

GSFC NEON Series-1 CMO

RELEASED

NS1-SMBA-CDRL-0001, Revision A
Near Earth Orbit Network (NEON) Series-1 Project, Code 473

**Near Earth Orbit Network
(NEON) Series-1 Project
Sounder for Microwave-Based
Applications
Contract Data Requirements List**



**Goddard Space Flight Center
Greenbelt, Maryland**

**Near Earth Orbit Network (NEON) Series-1 Project
Sounder for Microwave-Based Applications
Contract Data Requirements List
Review/Signature/Approval Page**

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Preface

This document is under NEON Series-1 (NS1) Project configuration control. Once this document is approved, NS1 approved changes are handled in accordance with Class I and Class II change control requirements as described in the Low Earth Orbit (LEO) Programs Division (LPD) Programs and Projects Configuration Management Procedure (CMP), and changes to this document shall be made by complete revision.

Any questions should be addressed to:

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Change History Log

Revision	Effective Date	Description of Changes
Revision –	December 20, 2024	This version incorporates 470-CCR-24-0404 which was approved by the SMBA ERB on March 26, 2024, and incorporates NS1-CCR-0004 by the NS1 SMBA CCB on the effective date shown.
Revision A	April 1, 2025	This version incorporates NS1-CCR-0007 which was approved by the NS1 SMBA CCB on the effective date shown. Removed duplicate language for UAR. Changed delivery date for MA-1.

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1 INTRODUCTION

This contract data requirements list (CDRL) document defines the requirements for deliverable documentation to be provided by the Sounder for Microwave-Based Applications (SMBA) Contractor. Section 1 includes definitions and instructions for distribution. Table 1 presents the CDRL item by item, with due dates, quantity, and media format. Section 2 contains the data item descriptions (DID), descriptions of each item which provide the use for the deliverables and the required preparation information. Except where specifically indicated to the contrary, the formats and drawing standards used shall be those normally used by the Contractor and by its subcontractors.

1.1 Export Control Limitations

Some of the content for documents provided by the National Aeronautics and Space Administration (NASA) (Goddard Space Flight Center [GSFC]) and United States (U.S.) industry will be subject to the U.S. International Traffic In Arms Regulations (ITAR), 22 CFR 120-130, by the U.S. Department of State and/or the Export Administration Regulations (EAR), 15 CFR 730-744, by the U.S. Department of Commerce. NASA may exchange technical information with the foreign partners and related entities (e.g., contractors and subcontractors) utilizing licensing exemptions to the ITAR and/or EAR, supported by the authority and content of the standing Letter of Agreement (LOA) and/or the Memorandum of Understanding (MOU) between NASA/the National Oceanic and Atmospheric Administration (NOAA) and that foreign partner, as appropriate. U.S. industry may exchange ITAR sensitive technical data under the authority and conditions of Technical Assistance Agreements (TAA), approved by the U.S. Department of State as appropriate.

NASA and U.S. industry will mark and thus notify foreign partners and related entities of the exchange of ITAR and/or EAR sensitive data with variations of the following language.

1.1.1 ITAR Technical Data

The unclassified technical information included is controlled under the U.S. ITAR, 22 CFR 120-130, by the U.S. Department of State. Transfer of this information to a foreign person or entity requires an export license issued by the U.S. Department of State or an ITAR exemption to the license requirement prior to the export of transfer.

These items are licensed by the United States for ultimate destination [foreign partners and related entities]. Diversion, retransfer, disclosure, or use contrary to applications specified in NASA/NOAA agreement without prior U.S. authorization are prohibited.

1.1.2 EAR Technical Data

The unclassified technical information included is controlled under the U.S. EAR, 15 CFR 730-744, Export Control Classification Number 9E515.a and 9E515.x by the U.S. Department of Commerce. Transfer of this information to a foreign person or entity requires an export license issued by the U.S. Department of Commerce or an EAR exemption to the license requirement prior to the export of transfer.

These items are licensed by the United States for ultimate destination [foreign partners and related entities]. Diversion, retransfer, disclosure, or use contrary to applications specified in NASA/NOAA agreement without prior U.S. authorization are prohibited.

1.2 Proprietary Data

Some of the content of documents may be of a proprietary nature to the document preparing/sending Party. In the event data is deemed to be proprietary, and for which protection is to be maintained, the sending Party shall mark the document with a notice to indicate that the data therein is proprietary and shall be used and disclosed by the receiving Party and its related entities (e.g., contractors and subcontractors) only for the purposes of fulfilling the receiving Party's responsibilities under the Near Earth Orbit Network (NEON) program, and that the identified and marked technical data shall not be disclosed or retransferred to any other entity without prior written permission of the document preparer. Data improperly marked as proprietary shall be rejected by the contracting officer's representative (COR) and returned for removal of proprietary markings.

1.3 Prior Submission Deliveries

If new hardware, software, or documentation is developed, procured, designed, fabricated, tested, or processed in a different manner from that of the first SMBA flight instrument effort (obsolescence, corrective actions, modifications, errors, etc.), the Contractor shall deliver the data item as required with the necessary updates for Government acceptance.

1.4 Due Dates – Related Definitions

The following definitions apply to the "DUE DATE, MATURITY" column in Table 1. Unless otherwise specified, deadlines are in calendar days.

(a) DUE DATE:

- (1) IOC: Initial Operating Capability
- (2) dCDR, PER, etc.: Delta Critical Design Review (dCDR), Pre-Environmental Review (PER), Pre-Ship Review (PSR), etc. (all major reviews). Electronic distribution to be delivered to the Government five working days prior to review, unless otherwise stated.
- (3) Monthly: Submitted on monthly basis
- (4) T: Launch Date
- (5) DACA/(D): (Calendar) Days After Contract Award/(Definitization)
- (6) JCT#: Joint Compatibility Test (JCT). Several JCTs are planned (reference the statement of work [SOW]), the number indicates for which JCT the deliverable is required (JCT#1 is the first JCT, JCT#2 is the second, etc.).

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- (7) MR: Mission Rehearsal (MR). Several MRs are planned (reference the SOW), the number indicates for which JCT the deliverable is required (MR#1 is the first MR, MR#2 is the second, etc.).
 - (8) FQT: Formal Qualification Test
 - (9) ATP: Authority to Proceed with the contractual effort (definitization or award of letter contract whichever is earlier).
 - (10) SCCDR: Spacecraft CDR
 - (11) SCPSR: Spacecraft PSR
- (b) MATURITY:
- (1) Initial: The first submission of an item, which shall be revised and resubmitted at a later date.
 - (2) Preliminary/Prelim: An early submission of an item. To be completed as is practicable at the time of preparation. Preliminary submittals are written with the best available current information and are resubmitted when further information becomes available.
 - (3) Final: The complete, thorough submission of an item for approval, review, or information that, to the best of the Contractor's knowledge and intention, will not require further revision or updates. However, this does not preclude updating if later found to be necessary. Any updates shall require the same "approval/review" process as was required for the original submissions.
 - (4) Current: The delivery is written with the best up-to-date information available at the time.
 - (5) Update: The delivery is revised with the best up-to-date information available at the time.

Other entries in the "DUE DATE" and "MATURITY" columns are self-explanatory.

1.5 Media – Related Definitions

The following definitions apply to the deliverables in Table 1.

H – Hardcopy(s) of this documentation shall be delivered to the Contracting Officer (CO), Data Manager (DM), and COR at GSFC Code 470.

E – Data items shall be delivered in electronic format to a GSFC LEO specified web portal unless otherwise noted in Table 1. Quantities refer to hardcopies, not electronic copies. The Contracting Officer shall be notified of electronic submission of the deliverable in writing.

Electronic deliverables shall be delivered in the following formats unless otherwise approved by the Government:

Text Documents:	PDF (searchable) or Microsoft Word
Presentations:	PDF (searchable) or PowerPoint
Spreadsheets:	Microsoft Excel
Database:	Delimited ASCII files accompanied with database schema document defining tables and entries.
Schedules:	PDF and Microsoft Project
Schematics and Drawings:	Design Web Format (DWF) and PDF
Photographs:	JPEG or current industry standard.
Video:	Any readily available open standard (e.g., AVI, MPEG)
	Test
Measurement Data	Delimited ASCII format or other acceptable format

R – For Reviews, hardcopies shall be made available at the review site for Government representatives. This will be in addition to electronic copies being made available prior to the review. The quantity specifies the number of hardcopies to be available at the review.

If separate distribution instructions are not specified for separate submission due date/maturity items, then the listed distribution applies to all submissions.

1.6 Category – Related Definitions

The following definitions apply to the "Submission Category (CAT)" column in Table 1. If separate approval instructions are not specified for separate submission due date/maturity items, then the listed approval instruction applies to all submissions.

- A Approval: CDRL items in this category require approval by the GSFC COR or Contracting Officer prior to use by the Contractor. Receipt by the Government shall occur within the time specified in the "Due Date" column of Table 1 of this document. Requirements for resubmission shall be as specified by the Contracting Officer. For most cases the Contractor shall be required to resubmit the document within 14 calendar days of receiving comments from the Government. If the Contractor has not received response from GSFC within 30 calendar days of delivery of a CDRL item, the Contractor may proceed as if the document has been approved.
- R Review: Documents in this category require delivery to the Government prior to use and within the time period specified in the "Due Date" column of Table 1 of this document. They are subject to evaluation by the Government or its designated representatives to determine Contractor effectiveness in meeting contract objectives. Upon submission, the Contractor may proceed with associated work while the Government reviews the submission. Requirements for resubmission shall be as specified by the CO.
- I Information: Data in this category shall be delivered to the Government within the time period specified in the "Due Date" column of Table 1 for the purpose of determining current program status, progress, and future planning requirements.

1.7 Applicability

The following definitions apply to "Applicability (APPLIC)" column in Table 1.

ETU – Applies to the Engineering Test Unit (ETU).

FM1 – Applies to the first flight model.

AFM – Applies to additional flight models (i.e., in addition to FM1).

1.8 Updates

All CDRLs shall be Update As Required (UAR). The Contractor shall notify the Government of any pending updates and re-deliver the CDRL on a schedule agreed to by the Government.

Table 1 does not specify this repetitious requirement, however UAR is required for each CDRL.

1.9 Documentation Change Procedures

Unless otherwise defined in the SOW or contract:

- a. When a minor correction/change is needed or has been made to a few parts of a “Final” version of a CDRL item that has already been delivered to the Government, the Contractor shall issue within 30 days a Documentation Change Notice starting with the number 001. Change bars or “change tracking” shall be used to indicate the locations of changes unless the particular file type does not have those features available. If not available, the details of the changes will be described in a text file.
- b. When a major correction/change is needed or has been made to all or multiple parts of a “Final” version of a CDRL item that has already been delivered to the Government, the Contractor shall deliver within 30 days a complete revision of the CDRL item. A description of all changes incorporated into the new revision shall be included in the new document or otherwise documented. No change bars are used when a document is updated by revision, and the Documentation Change Notice numbers for that document shall be automatically recycled to 001.

1.10 Delivery Instructions

- a. The Contractor shall deliver each CDRL item via the media indicated in accordance with the schedule and any special instructions indicated in Table 1 and/or the corresponding DID, and in accordance with the contract or SOW requirements and provisions.
- b. All CDRL items shall be delivered electronically (one electronic copy for each CDRL item) to an electronic library designated by the Government. If all or a portion of a CDRL item is not well-suited for electronic delivery, the item may be delivered via hardcopy upon mutual agreement between the Contractor and the Government.

With each prior delivery submission, the Contractor shall include a cover letter with the following information:

- CDRL #
- Title

- Due Date
- Flight Model
- Serial Number
- Original Submission Date
- Prior Delivery CDRL #
- Current Submission Revision
- Current Submission Revision Date

The Contractor may deliver a single letter for multiple submissions as long as the information above is contained the letter for each deliverable item.

Table 1. Contract Deliverable Items List

No.	CALIBRATION/VALIDATION	DUE DATE	CAT	APPLIC
CV-1	Calibration/Validation and Intercalibration Plan	- Draft: FM1 PDR - Final: FM1 CDR	I A	ETU, FM1
CV-2	Calibration/Validation Procedures	- Draft: 60 days prior to use - Final: 10 days prior to use <u>On-orbit Procedures</u> - Final: L-90 days	I A A	ETU, FM1
CV-3	Calibration/Validation Reports	- Electronic Data: 3 days post-test - Test Report: 10 days post-test - Analyses: At completion + 10 days, NLT PSR	I I I	ETU, FM1, AFM
CV-4	Radiometric Math Model (RMM)	- ETU: ETU Summary Review - Draft: CDR - Final: PSR	I I	ETU, FM1, AFM
CV-5	Calibration Data Book	- ETU: ETU Summary Review - Draft: CDR - Final: PSR	I I A	ETU, FM1, AFM
CV-6	Spectral Response Function (SRF) Measurements	- ETU: ETU Summary Review - Final: PSR	I A	ETU, FM1, AFM
CV-7	Intercalibration Analysis and Test Data	- Final: CDR - Update: PSR – 30 days	A A	ETU, FM1, AFM

No.	ETU	DUE DATE	CAT	APPLIC
ETU-1	ETU Development and Test Plan	• 90 DACA	I	ETU
ETU-2	ETU Requirements Alignment Matrix	• Draft: 90 DACA • Update: ETU Pre-Build Review -60 days • Final: ETU Summary Review -30days	I I I	ETU

No.	INTEGRATION AND TEST	DUE DATE	CAT	APPLIC
IT-1	Instrument Integration, Test and Verification Plan	- Draft: PDR - Final: CDR	I I	FM1
IT-2	Test Reports	Final: 10 days after test completion	I	ETU, FM1, AFM
IT-3	Detailed Test Procedures & As-Run Test Procedures	- Pre-Test: 10 days prior to test - As-Run: Conclusion of each test <u>Launch site-specific Test Procedures</u> - Draft: LRD – 12 months - Final: LRD – 3 months (i.e., SC PSR)	I for ETU A for Pre-Test I for As-Run I A	ETU, FM1
IT-4	Packaging, Handling, Storage, and Transportation (PHS&T) Plan and Procedures	- Plans Final: CDR + 30 Days - Procedures Final: PER – 21 Days	I I	ETU, FM1
IT-5	Instrument Post Delivery Support (PDS) Document	Final: SMBA Instrument PSR minus 30 days	I	FM1

No.	MISSION ASSURANCE	DUE DATE	CAT	APPLIC
MA-1	Mission Assurance Compliance Matrix	Final: 60 DACA	A	FM1, AFM
MA-2	Supplier List	- Initial: 60 DACA - Updates: 60 days after new suppliers are determined	I I	FM1
MA-3	Orbital Debris Assessment Report (ODAR) and End of Mission Plan (EOMP)	- Preliminary ODAR inputs 15 days prior to PDR (For Information) - Interim ODAR and Draft EOMP inputs 60 days prior to CDR (For Information) - Final/updated ODAR and EOMP inputs 90 days prior to PSR (For Information)	I I I	FM1, AFM
MA-4	System Safety Plan	- Prelim: SRR - Final: 45 days prior to PDR - Updates 30 days prior to implementation	I A A	ETU, FM1
MA-5	Safety Requirements Compliance Checklist	- Preliminary 45 days prior to SRR (For Approval) - Final 45 days prior to PDR (For Approval) - Updates 45 days prior to CDR (For Approval)	A A A	ETU, FM1AFM
MA-6	Operations Hazard Analysis and Hazard Verification Tracking Log	Submit OHA and hazard verification tracking log (VTL) 45 days prior to starting system level testing (For Approval)	A	FM1, AFM
MA-7	Instrument Safety Assessment Report	- Preliminary ISAR 30 days prior to PDR (For Approval) - Intermediate ISAR 30 days prior to CDR (For Approval)	A A A	FM1, AFM

No.	MISSION ASSURANCE	DUE DATE	CAT	APPLIC
		Final ISAR, with VTL, 45 days prior to instrument shipment (For Approval)		
MA-8	Hazardous Procedures for Payload I&T and Pre-Launch Processing	- I&T hazardous procedures to Project Office 7 days before first use (For Review) - Launch Range Hazardous Procedures 65 days prior to first use (For Review) - Note: Project Office approval of Launch Range Hazardous Procedures is required 10 days after delivery with subsequent Range Safety approval	I A	FM1, AFM
MA-9	Pre-Mishap Plan	- Preliminary plan 45 days prior to PDR (For Approval) - Final plan 45 days prior to CDR (For Approval)	A A	ETU, FM1
MA-10	Reliability Program Plan	- Preliminary: SRR – 30 days - Final: SRR + 30 days	I A	FM1
MA-11	FMECA and Critical Items List (CIL)	- Preliminary FMECA and CIL 30 days prior to PDR (For Review) - Updated FMECA and CIL 30 days prior to CDR and each subsequent milestone review - leading up to LRR (For Review)	I A	FM1
MA-12	Fault Tree Analysis	- Preliminary: PDR-30 days - Final: CDR-30 days - Updates: 30 days after identification of updates or changes	I I A	FM1
MA-13	Reliability Calculations	- Methodology 30 days prior to SRR (For Review) - Initial report 30 days prior to PDR (For Review) - Final report 30 days prior to CDR and subsequent milestone reviews (For Review)	I I I A	FM1

No.	MISSION ASSURANCE	DUE DATE	CAT	APPLIC
		Revisions 30 days after updates/changes (For Review)		
MA-14	Limited Life Items List	30 days prior to PDR (For Review) 30 days prior to CDR (For Review) - Updates 30 days of changes (For Review)	I I A	FM1
MA-15	Parts Stress Analysis	30 days prior to CDR (For Approval) - Revisions 30 days after identification (For Approval)	A A	FM1
MA-16	Worst-Case Analysis	30 days prior to CDR (For Approval) - Revisions 30 days after identification (For Approval)	A A	FM1
MA-17	Software Assurance Plan	- Preliminary plan 30 days prior to SRR (For Information) - Final plan 15 days prior to PDR (For Information) - Updates 30 days prior to implementation (For Information)	I I I	ETU, FM1
MA-18	Electrostatic Discharge (ESD) Control Plan	30 days prior to PDR (For Information)	I	ETU, FM1
MA-19	Printed Circuit Board Procurement Plan	- Preliminary information 30 days prior to CDR (For Approval) - Changes 14 days prior to manufacturing (For Approval)	A A	FM1
MA-20	Lead-Free Control Plan	60 days after contract award (For Information)	I	FM1
MA-21	Electrical, Electronic, and Electromechanical (EEE) Parts Control Plan	30 days after contract award (For Information)	I	FM1
MA-22	Master EEE Parts List	- EEE parts to the Parts Control Board prior to inclusion (For Approval) - Updates prior to inclusion (For Approval)	A A	FM1, AFM

No.	MISSION ASSURANCE	DUE DATE	CAT	APPLIC
MA-23	Radiation Effects Mitigation	Submission 60 days after contract award (For Approval)	A	FM1
MA-24	M&P Selection, Control, and Implementation Plan (MPSCIP)	14 days prior to SRR (For Information) 30 days prior to Mission PDR (For Approval) 30 days prior to CDR (For Approval)	I A A	FM1
MA-25	Materials Usage Agreement (MUA)	30 days prior to CDR (For Approval) - After CDR, submit new or revised MUAs, 30 days after their identification (For Approval)	A A	FM1, AFM
MA-26	Materials Identification and Usage List (MIUL)	30 days prior to PDR (For review) 30 days prior to CDR (For Approval) - Updates 30 days after identification (For Approval) - As-Built MIUL 30 days prior to PSR (For Approval)	I A A A	FM1, AFM
MA-27	Life Test Plan and Final Report for Lubricated Mechanisms	30 days prior to PDR (For Approval) 30 days after acceptance test completion (For Information)	A I	FM1, AFM
MA-28	Additive Manufacturing Control Plan (AMCP)	14 days prior to SRR (For Information) 30 days prior to Mission PDR (For Approval) 30 days prior to CDR (For Approval)	I A A	FM1
MA-29	AM Part Production Plan (PPP)	- Preliminary: 30 days prior to CDR (For Information) - Final: 30 days prior to an Additive Manufacturing Readiness Review (as defined by the AMCP (For Approval)	I A	FM1
MA-30	Contamination Control Plan	30 days before PDR (For Review) 30 days before the CDR (For Approval) - Final thermal vacuum bakeout results provided within 30 days of completion (For Review) - Contamination certificate of compliance with End Item Acceptance Data Package (DID 12-1) (For Review)	I A I I	FM1

No.	MISSION ASSURANCE	DUE DATE	CAT	APPLIC
MA-31	Foreign Object Damage Program	• Submit 30 days before PDR (For Review)	A	ETU, FM1
MA-32	End Item Acceptance Data Package	• Component/Subsystem Final: End Item Acceptance Data Package completion, or prior to consent to proceed to next level of integration, whichever is sooner. • 30 days prior to PSR (For Approval) • Final 30 days prior to end item delivery (For Approval) • Updates 30 days after identification (For Approval)	I (for comp/subsys) A A A	ETU, FM1, AFM

No.	ON ORBIT	DUE DATE	CAT	APPLIC
OO-1	Instrument On-Orbit Commissioning Plan	• Final: PSR	I	FM1
OO-2	Instrument User's Manual	• Final: PSR	I	ETU, FM1
OO-3	Commissioning Report	• Prelim: 10 Days after each test • Test Results: PLAR • Final: PLAR	I I I	FM1, AFM
OO-4	Command and Telemetry Handbook (see OO-8)	<u>Volumes 1 and 2</u> • Draft: PER - 30 days • Final: PSR – 6 months	I I	ETU, FM1
OO-5	Instrument Launch and On-Orbit Operations and Contingency Procedures	• Final: PSR	I	FM1
OO-6	Instrument and Simulator Training Materials	• Final: PSR	I	ETU, FM1
OO-7	Instrument Constraints, Restrictions, and Warnings/Alerts Document	• Draft: PER - 30 days • Final: PSR – 30 days	I I	ETU, FM1,
OO-8	Instrument Command and Telemetry Database	• Draft: PER - 30 days • Final: PSR – 6 months	I I	ETU, FM1
OO-9	Instrument Mission Data Definitions	• Final: PSR – 30 days	I	ETU, FM1
OO-10	Command and Telemetry Database (CTDB) Validation Plan	• Draft: PER - 30 days • Final: PSR – 30 days	I I	FM1
OO-11	SMBA Memory Table Baseline Content	• Final: PSR – 30 days	I	FM1

No.	MANAGEMENT	DUE DATE	CAT	APPLIC
PM-1	Monthly Project Management Reviews (PMR)	• Monthly at review	I	ETU, FM1, AFM
PM-2	Work Breakdown Structure (WBS) and Dictionary	• Final: 30 DACA	I	ETU, FM1
PM-3	Engineering Peer Review Plan	• Initial: with Proposal • Final: 60 DACA	I I	ETU, FM1
PM-4	Integrated Program Management Report	• Initial: Draft Integrated Master Schedule with Proposal • Draft Baseline: 45 DACA • Baseline: 12 th working day of the 3 rd fiscal month after contract definitization • Monthly financials: by the 12th working day following the close of the prior month accounting period • Monthly IMS snapshot no later than the 5 th working day of the month	I I A I I	ETU, FM1, AFM
PM-5	Manufacturing (Fabrication and Assembly) Flow Plan	• Initial: MRR-30days	A	ETU, FM1
PM-6	Hardware Configuration Management Plan	• Final: 90 DACA	I	ETU, FM1
PM-7	Project Management Plan	• Draft: 45 DACA • Final: IBR	I A	ETU, FM1
PM-8	Risk Management Plan	• Final: 60 DACA	I	ETU, FM1
PM-9	Giver/Receivers/Hand-offs/Receivables-Deliverables (Rec-Dels)	• Monthly with the IMS	I	ETU, FM1, AFM

No.	REVIEWS	DUE DATE	CAT	APPLIC
RE-1	System Requirements Review (SRR) / System Definition Review (SDR) Data Package	- Draft: 1 day prior to dry run - Final: 5 days prior to review / hardcopies at Review	I I	FM1
RE-2	Preliminary Design Review (PDR) Data Package	- Draft: 1 day prior to dry run - Final: 5 days prior to review / hardcopies at Review	I	FM1
RE-3	Critical Design Review (CDR) Data Package	- Draft: 1 day prior to dry run - Final: 5 days prior to review / hardcopies at Review	I I	FM1
RE-4	Pre-Environmental Review (PER) Data Package	- Draft: 1 day prior to dry run - Final: 5 days prior to review / hardcopies at Review	I I	FM1, AFM
RE-5	Pre-Ship Review (PSR) Data Package	- Draft: 1 day prior to dry run - Final: 5 days prior to review / hardcopies at Review	I I	ETU, FM1, AFM
RE-6	Integrated Baseline Review (IBR)	- Draft: 1 day prior to dry run - Final: 5 days prior to review / hardcopies at Review	I I	FM1
RE-7	Engineering Peer Review Data Packages	- Engineering Peer Review (EPR) Meeting Material : EPR – 3 days - Final: Hard copies at review	I I	ETU, FM1
RE-8	Manufacturing Readiness Review Package	- Draft: 5 days prior to dry run - Final: 5 days prior to review / hardcopies at Review	I I	ETU, FM1, AFM
RE-9	Design Modification Review (DMR)	As needed	I	AFM
RE-10	Firmware Design Data Package	- Final: CDR - Update: Firmware Review - 10 days	I I	ETU, FM1
RE-11	Test Readiness Review (TRR) Data Package	Final: 5 days prior to review / hardcopies at Review	I	ETU, FM1, AFM

No.	REVIEWS	DUE DATE	CAT	APPLIC
RE-12	ETU Pre-Build Review Package	ETU MRR – 30 days	I	ETU
RE-13	ETU Test Summary Review Package	- Prelim: CDR - Final: FM1 MRR	I I	ETU
RE-14	Inputs to Mission Level Reviews	- Draft: 1 day prior to dry run - Final: 5 days prior to review / hardcopies at Review	I I I	ETU, FM1, AFM

No.	SYSTEMS ENGINEERING	DUE DATE	CAT	APPLIC
SE-1	Engineering Change Requests	• Class I: As Generated • Class II: As Generated	A I	ETU, FM1, AFM
SE-2	Trend Plan/Analysis, Reports, and Operations Log	• Initial: PDR • Update: PER • Final: PSR	I I A	ETU, FM1, AFM
SE-3	Engineering Drawings	• Final: As generated	I	ETU, FM1, AFM
SE-4	System Performance Verification Plan and Requirement Verification Matrix	• Initial: SRR-60 days • Final: PER - 21 days • Update: PSR	I A A	ETU & FM1
SE-5	Verification Reports	• Final: Verification Test + 30 days	I	ETU, FM1, AFM
SE-6	Instrument Performance Specification	• Initial: SRR-60 days • Final: CDR	I I	ETU, FM1, AFM
SE-7	Analysis and Modeling			
SE-7a	Structural and Mechanical Analysis and Models	• Final: MRR • Correlated Model: PSR – 30 days	A A	ETU, FM1, AFM

No.	SYSTEMS ENGINEERING	DUE DATE	CAT	APPLIC
SE-7b	Thermal Analysis and Models	- Instrument DTM 1 & RTM 1: CDR – 60 days - Instrument DTM 2 + TVAC GSE: 90 days before PER - Correlated As-tested instrument DTM 3 + TVAC GSE: 90 days after instrument TVAC thermal balances - Correlated instrument DTM 4: PSR – 40 days Flight Cases Launch/Ascent/Deploy Science calibration maneuvers - Correlated instrument RTM 2: PSR-20 days If model deliveries per this schedule are not achievable due to the data analysis schedule, the Contractor shall coordinate with the COR to agree on acceptable delivery dates.	I I I I A	ETU, FM1, AFM
SE-7c	Stress Analysis	- Final: MRR - Update: PSR – 30 days	I I	ETU, FM1, AFM
SE-7d	Radiation Hardness & Survivability Analysis	- Final: MRR - Update: PSR – 30 days	I I	ETU, FM1, AFM
SE-7e	Shock Analysis	- Final: MRR - Update: PSR – 30 days with correlated model	I I	ETU, FM1, AFM
SE-7f	Contamination Analysis Report	- Initial: CDR - Final: PER - Update: PSR – 30 days, correlated model,	I I	ETU, FM1, AFM

No.	SYSTEMS ENGINEERING	DUE DATE	CAT	APPLIC
SE-7g	On-Orbit Pointing & Jitter Analysis	- Initial: PDR - Final: CDR - Update: PSR – 30 days, correlated model	I I	ETU, FM1, AFM
SE-7h	Electromagnetic Interference (EMI)/Electromagnetic Capability (EMC) Analysis Report	- Initial: CDR - Final: PSR – 30 days	I A	ETU, FM1, AFM
SE-7i	Hyperspectral Capability Analysis	- Initial: CDR - Final: PSR – 30 days	I A	ETU, FM1, AFM
SE-7j	Radio Frequency Interference Detection Capability Analysis	- Initial: CDR - Final: PSR – 30 days	I A	ETU, FM1, AFM
SE-8	Instrument Concept of Operations Document	- Initial: SRR - Final: PDR	I I	ETU, FM1
SE-9	Inputs to the Instrument to Spacecraft Interface Control Document (ICD)	- Initial: PDR - Final: CDR	I A	ETU, FM1
SE-10	Inputs to the Instrument to Spacecraft Mechanical Interface Control Document (MICD)	- Initial: PDR - Final: CDR	I A	ETU, FM1
SE-11	Electrical Systems Requirements Document	Final: SRR	I	ETU, FM1
SE-12	Spare Parts Plan and List	Final: with proposal	A	ETU, FM1
SE-13	Instrument Simulator Specification	Final: CDR	A	FM1
SE-14	Instrument Simulator Interface Verification Report	Final: Test + 30 days	A	FM1
SE-15	Field Programmable Gate Array (FPGA) Development Plan	Final : PDR – 30 days	I	ETU, FM1

No.	SYSTEMS ENGINEERING	DUE DATE	CAT	APPLIC
SE-16	Field Programmable Gate Array (FPGA) Design Data Package.	• Prelim: PDR- 30 days • Final: CDR – 45 days • As generated for each FPGA	I I	ETU, FM1
SE-17	Mass Properties Report	• Final: PSR	I	ETU & FM1, AFM
SE-18	Systems Engineering Management Plan (SEMP)	• Final: 90 DACA	I	ETU, FM1

No.	SOFTWARE	DUE DATE	CAT	APPLIC
SW-1	NPR 7150.2D Tailored Compliance Matrix	• Final: 30 DACA	A	ETU, FM1
SW-2	Instrument Software Development Plan	• Final: 30 DACA	I	ETU, FM1
SW-3	Software Configuration Management Plan	• Final: 30 DACA	I	ETU, FM1
SW-4	Software Requirements Specification	• ETU, FM1 Final: 30 DACA • FSTB Final: 60 DACA • Simulator Final: 90 DACA	A I I	ETU, FM1, FSTB, Simulator
SW-5	Software Detailed Design Document	• Final: 30 DACA • Simulator Final: 90 DACA	I I	ETU, FM1, Simulator
SW-6	Software Test Plan	• ETU, FM1 Final: 60 DACA • FSTB Final: FSTB TRR -30 days • Simulator Final: 120 DACA	I I I	ETU, FM1, FSTB, Simulator
SW-7	Software Test Procedures	• ETU, FM1 Initial: SWTRR - 15 days • ETU, FM1 Final: Software Acceptance Review (SWAR) - 15 days • Simulator Final: Simulator TRR – 3 weeks	I A A	ETU, FM1, Simulator
SW-8	Software Test Reports	• ETU, FM1, AFM Final: 30 days after test completion • FSTB Final: 30 days after test completion • Simulator Final: 30 days after test completion	I I I	ETU, FM1, AFM, FSTB, Simulator
SW-9	Software Version Description Document	• ETU, FM1, AFM Final: 30 days after acceptance test completion and with each FSW release/build • FSTB Final: FSTB TRR – 30 days • Simulator Final: Simulator Software Acceptance Review – 5 days	A I A	ETU, FM1, AFM, FSTB, Simulator
SW-10	Instrument Software Test Readiness Review Data Package	• Final: SWTRR – 5 days	I	ETU, FM1, AFM
SW-11	Software Acceptance Review Data Package	• ETU, FM1, AFM Final: 15 days after test completion • FSTB Final: 15 days after test completion • Simulator Final: 15 days after test completion	I I I	ETU, FM1, AFM, FSTB, Simulator

No.	SOFTWARE	DUE DATE	CAT	APPLIC
SW-12	Software User's Guide	- ETU, FM1 Prelim: each SW Release – 30 days - ETU, FM1 Final: PSR - Simulator Final: Simulator TRR	I I I	ETU, FM1, Simulator
SW-13	Instrument Software Development Facility Specification	Final: 60 DACA	I	ETU, FM1
SW-14	Instrument Software Maintenance Plan	Final: 30 DACA	I	ETU, FM1
SW-15	Instrument Data Sets	Final: PSR	I	ETU, FM1, AFM

2 DATA ITEM DESCRIPTIONS

2.1 Calibration/Validation Data Items

DATA ITEM DESCRIPTION

1. CDRL No.:

CV-1

2. Title:

CALIBRATION/VALIDATION AND
INTERCALIBRATION PLAN

3. Reference:

SOW 5.3.2

4. Use:

Document defines the planned calibration and validation activities to be performed by the Contractor. The scope is defined for a single ETU/FM and includes inter-calibration planning, RFI detection and hyperspectral capabilities.

5. Preparation Information:

The Contractor shall work with the Government's Sensor Data Records Team to develop a Calibration/Validation Plan that describes the approach for calibrating, characterizing, and validating the spectral, spatial, radiometric, and geometric performance of the SMBA. The Calibration and Validation Plan shall address all requirements and tests related to science performance and Cal/Val, including inter-calibration, RFI detection and hyperspectral capabilities. This document shall include On-Orbit calibration and validation requirements and references to the operations documents that describe these calibration and validation procedures. The Contractor's comprehensive radiometric/geometric calibration plan shall include preliminary work at subsystem levels of assembly as well as at the system level and component level (such as PRT sensor, Reflector emissivity).

The Calibration/Validation Plan shall incorporate the following information at a minimum:

- A. A description of planned tests and analyses including:
 - (a) what is being tested or analyzed and how it relates to instrument performance
 - (b) the expected test or analysis results
 - (c) integration level for test or analysis model, i.e., part, subassembly, assembly, instrument

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- (d) environmental conditions for test, e.g., ambient, thermal-vacuum, on-orbit
 - (e) sampling methods and their statistical validity
 - (f) operational phase of testing, i.e., pre-launch, or commissioning
 - (g) theoretical basis for the test or analysis, i.e., how the test is performed, how the data are reduced and the method rationale supported by fundamental physics principles and equations)
 - (h) the resolution, precision and accuracy of the results and relation to the expected results
 - (i) test configuration, i.e., equipment, test equipment calibration, and test setup
 - (j) description of test or analysis results usage, i.e., processing algorithms that use test or analysis results or calibration parameters generated by the test or analysis
 - B. A test and analysis schedule and flow chart
 - C. How test/analysis results are made available
 - D. Government access and participation in pre-launch testing including a Government-owned standards or Government-performed Cal/Val-related characterization activities
 - E. Description of commercial-off-the-shelf (COTS) and custom analysis tools
 - F. A description of the On-Orbit Calibration capabilities of the SMBA, including their design, characterization techniques, and operational implementation. This shall include a description of any necessary calibration data that must be collected during normal operations.
 - G. Reference Standards and their calibration traceability
 - H. Support data requirements
 - I. Tests to be conducted for calibrating the radiometric response of the Instrument
 - J. Inter-calibration
 - a. Tests to be conducted for intercalibrating the radiometric response of the Instrument to the NIST traceable standard.
 - b. The method of intercalibrating proxy standards to the primary NIST traceable standard.
 - c. The method of intercalibrating proxy standards to the primary NIST traceable standard.
 - d. Reference Standards and their calibration traceability

Given known issues with diffuser materials on other NASA missions being contaminated in manufacturing or in pre-launch testing, if these materials are used the Contractor shall ensure the Instrument Calibration/Validation Plan also includes a description of diffuser contamination management activities and characterizations including:

- a. Vacuum bakes
- b. Use of witness samples

The Committee on Earth Observation Satellites Working Group on Calibration & Validation provides the below definitions for calibration and validation. Reference <https://ceos.org/ourwork/workinggroups/wgcv/>.

- Calibration is the process of quantitatively defining a system's response to known and controlled signal inputs.
- Validation, on the other hand, is the process of assessing, by independent means, the quality of the data products derived from those system outputs.

Elaborating on these terms in the context of SMBA:

- Calibration inputs are electromagnetic (EM) energy from a scene at certain frequencies and the response (outputs) are the integrated energy under a channel spectral response function.
- The data products to be validated (i.e., assessed for quality) are:
- Raw Data Records (RDR)
- Temperature Data Records (TDR)
- Sensor Data Records (SDR)
- Environmental Data Records (EDR)

Note: Per NS1-SMBA-REQ-0001 section 1.4 and requirement SMBA-93, the instrument will produce RDR files. The NOAA Ground System (NGS) will use RDR data and instrument calibration data to produce higher-level data products such as TDRs, SDRs, EDRs. The CV-1 Cal/Val Plan only needs to describe Contractor activities for validation of RDR and validation of the SMBA calibration data used as inputs for TDRs, SDRs, EDRs. The document does not need to describe the entirety of TDR, SDR, and EDR validation, as those processes are controlled by NOAA.

DATA ITEM DESCRIPTION

1. CDRL No.:

CV-2

2. Title:

CALIBRATION/VALIDATION PROCEDURES

3. Reference:

SOW 5.3.2

4. Use:

To verify that the planned Calibration, Validation, Intercalibration, RFI detection and hyperspectral procedures meet the goals of the CV-1 Calibration/Validation and Intercalibration Plan.

5. Preparation Information:

1. For each test in the Calibration/Validation Plan, the Calibration, Validation or Intercalibration procedure for each test shall include:

- a. Test Objectives
- b. Test Methods
- c. Assumptions
- d. Applicable Documents and Software
- e. Associated algorithms
- f. Required Instrument Configuration, including any differences between pre-flight and flight configurations
- g. Mechanical and Electrical Test Equipment Configuration, including layout and interconnection of test equipment and articles including the grounding scheme. Location and identification of all measuring points on appropriate schematics and diagrams
- h. Test Equipment Identification
- i. Test Instrumentation
- j. Calibration reference requirements
- k. Safety Provisions and Cautions, including Identification of hazardous and potentially hazardous situations and operations and abort conditions
- l. Environmental and/or other conditions to be maintained, including contamination controls
- m. Personnel responsibilities and chain-of-command for test performance
- n. Expected results in telemetry and associated caution and warning levels.
- o. Program Quality Requirements
- p. Stimuli sources and associated and associated levels
- q. Test durations
- r. Test configurations

- s. Sequence of events
- t. Trending of performance characteristics during verification testing
- u. Step-by-step instructions
- v. Data Recording/Output Format Requirements
- w. Data Recording Forms and Tables
- x. Analysis techniques
- y. Expected results
- z. Pass/Fail Criteria
- aa. Test Termination Procedure

2. For each Analysis in the Calibration/Validation Plan, the procedure for each Analysis shall include:

- a. Analysis Objectives, i.e., short description of the item or items being validated by this analysis
- b. Analysis Methods including the analysis level of depth (i.e., 1st order, 2nd order, etc.)
- c. Assumptions
- d. Applicable Documents and Software
- e. Associated algorithms and software tools
- f. Required analysis mode/model configuration(s)
- g. List of input data that will be used and related test IDs that were used to acquire the data
- h. List of external reference sources of input data and/or model used in analysis
- i. List of fixed analysis parameters and set points
- j. List of adjustable analysis parameters and set points that may vary between runs.
- k. Equipment requirements
- l. Program Quality Requirements
- m. Sequence of events
- n. Step-by-step description of the data analysis procedure/algorithm
- o. Description of any statistical sampling method being used
- p. Data Recording/Output Format Requirements
- q. Data Recording Forms and Tables
- r. Expected results
- s. Pass/Fail Criteria

DATA ITEM DESCRIPTION

1. CDRL No.:

CV-3

2. Title:CALIBRATION/VALIDATION REPORTS

3. Reference:SOW 5.3.2

4. Use:

To provide results of individual calibration/validation, intercalibration, RFI detection, and hyperspectral tests and analyses for Government review.

5. Preparation Information:

The Contractor shall provide a Calibration/Validation Report for each calibration, validation or intercalibration test or analysis. All supporting data shall be provided in electronic format, where applicable.

The Calibration/Validation Reports shall include at a minimum:

- A. Identification of article or component tested or being validated with date of test or related analysis and the relevant Test/Analysis ID and run number
- B. Performance trends during and between each planned test
- C. List of uncertainty levels for each input data in analysis or test measurements
- D. If a given test or analysis has more than one configuration state the test or analysis configuration selected.
- E. Problems or failures with tests, procedures, or analyses
- F. Anomalies and deviations from plans or procedures and their resolution/status
- G. Test or analysis results, including:
 - a. any calibration parameters to be used for on-orbit processing
 - b. comparison of results with expectations and requirements
 - c. interpretation of the results
- H. Instrument Data Sets
 - a. General Notes
 - 1. These data sets shall be provided on standard commercially available digital media.
 - 2. These data shall be provided in a format consistent with the Instrument-to-Spacecraft ICD accompanied by corresponding descriptive documentation.
 - 3. For Instrument and Observatory level testing where test pass/fail status reporting is required in a timely manner during the execution of the test (for

example in thermal vacuum (TVAC) before the consent to proceed to end a plateau), quick look data sufficient to evaluate the tests pass/fail status shall be provided in near real time.

- b. For instrument-level test output these data sets shall consist of:
 - 1. Unprocessed detector sample data with associated time indexing information.
 - 2. Instrument ancillary data that would be included in the SMBA mission data stream and associated time indexing information.
 - 3. Instrument housekeeping information that would be included in the observatory telemetry stream and associated time indexing information.
- c. For observatory-level test output these data sets shall consist of:
 - 1. Unprocessed detector sample data with associated time indexing information.
 - 2. Instrument ancillary data that would be included in the SMBA mission data stream and associated time indexing information.
 - 3. Observatory ancillary data that would be included in the SMBA mission data stream and associated time indexing information. Ancillary data fields that are not available or are not meaningful in the prelaunch environment (e.g., ephemeris) may be simulated.
 - 4. Instrument housekeeping information that would be included in the observatory telemetry stream and associated time indexing information.
 - 5. Observatory housekeeping information that would be included in the observatory telemetry stream and associated time indexing information.

Characterization measurements of noise power spectral density shall be made for all continuity channels and provided in the Calibration/Validation reports.

DATA ITEM DESCRIPTION

1. CDRL No.:

CV-4

2. Title:RADIOMETRIC MATH MODEL (RMM)

3. Reference:SOW 2.4.1

4. Use:

For modeling, evaluating, and characterizing the end-to-end radiometric performance of the instrument.

5. Preparation Information:

- A. The Radiometric Math Model shall be used to: evaluate the end-to-end radiometric performance of the Instrument (Photons to calibrated brightness temperature); conduct sensitivity analyses; determine absolute and relative calibration accuracies; identify error contributors which can be eliminated or mitigated during the design phase; identify impact of error budget trades; assess instrument performance in terms of Noise Figure (NF), Noise Equivalent Delta Temperature (NE Δ T), stability in orbit, etc., including radiometric sensitivity, calibration accuracy, nonlinearity, and noise.
- B. The model shall incorporate actual test and calibration data; the model shall be updated and refined during the course of the Instrument development program until it simulates instrument performance within the accuracy required by the specifications.
- C. The model shall include calibration algorithms. The estimated data shall be replaced with measured data when it is available. The post-launch calibration algorithms are used along with pre-launch calibration data to demonstrate that the absolute and relative radiometric accuracies are being met. The calibration data shall also be provided in a mutually agreed upon electronic format. Algorithms to be provided shall include:
 - 1. Radiometric Temperature Computation
 - 2. Target Temperature Computation
 - a. Physical Temperature Computation
 - b. Brightness Temperature Computation
 - 3. Calibration Accuracy/Uncertainty Measurement
 - 4. Noise Power Spectral Density (NPSD) and NE Δ T Measurements
 - 5. non-Linearity
 - 6. Dynamic Range

7. Gain Stability ($\Delta G/G$)
8. Hot and Cold Calibration References
 - a. Hot Target calibration reference uncertainty and corrections
 - b. Cold Target calibration reference uncertainty and corrections

The algorithm delivery shall consist of:

- a. An overview of each algorithm including the objective or purpose of the algorithm
 - b. The rationale for each algorithm and identification of any specifications it supports
 - c. A description of each algorithm's inputs and outputs including any external ancillary data (i.e., from the spacecraft) used by the algorithm
 - d. A mathematical description of each algorithm
 - e. An implementation of the algorithm in source code and identification of the Operating System that the source code was built on
 - f. An error analysis showing the expected accuracy of the algorithm results
 - g. A description of the methods used to validate each algorithm and description of the validation results
 - h. A list of the instrument data components used by each algorithm
 - i. A list of the calibration parameters used by each algorithm
 - j. A list of references
- D. The Sensor Performance Math Model (SPMM) shall also propagate uncertainties in model input parameters to uncertainties in model output parameters, including error correlation effects. The basis for uncertainties in SPMM inputs shall be documented. The delivered SPMM shall include full documentation of the model, including the error propagation methodology, any software code/scripts to run the model, and instructions for Government customer use of the model.
- E. System performance predictions shall include uncertainty estimates for calibrated radiance, reflectance, and equivalent blackbody temperatures based on propagation of uncertainties in SDR algorithm look-up table values and other SDR algorithm parameters.

DATA ITEM DESCRIPTION

1. CDRL No.:

CV-5

2. Title:

CALIBRATION DATA BOOK

3. Reference:

SOW 5.3.2

4. Use:

For demonstrating specification compliance and for full interpretation of the orbital data.

5. Preparation Information:

The calibration data book is to be prepared and processed for public release prior to delivery to NASA.

The calibration data book, which is to be delivered at the end of pre-launch instrument calibration and intercalibration tests and as part of Pre-Ship Review (PSR) Data Package, are to be a record of all pertinent test, calibration and intercalibration data, including all raw data from which graphs, curves, etc., are prepared, and whose primary functions include 1) providing NOAA calibration parameters for RDR, TDR, SDR generation and 2) providing data users info needed to process data.

The calibration data book shall be updated after commissioning Cal/Val activities.

The Cal Data Book needs to provide calibration parameters needed by NOAA (or NGS) to produce TDRs and SDRs.

A summary section shall be prepared, which contains functional equations and charts pertinent to the data contained therein. The summary shall depict, as a minimum, all information pertinent to the performance requirements, the final system level bench test data, the final calibration and intercalibration of the instrument and every instrument subsystem as determined from all tests and calibrations performed prior to delivery of the instrument to the Government, and a set of trend charts of all critical parameters.

The calibration book data shall be submitted in accordance with Table 1 of this document.

Data shall be provided for each tested instrument temperature

The results of all tests and calibrations shall be summarized by the contractor in the calibration log book(s).

The following information shall be derived from the calibration data.

- (a) Temperature sensitivity ($NE\Delta T$) of each channel
- (b) Standard deviation of individual calibration points from the best-fit calibration curves for each channel and each calibration
- (c) Calibration of in-flight calibration blackbodies in terms of effective brightness temperature versus temperature monitor bit outputs.
- (d) Calibration curves of all temperature sensors.
- (e) Calibration curves of all voltage monitors.
- (f) Temperature stability data for the instrument.
- (g) Beam profile in the far field for each channel

Spectral Response Function (SRF) frequency data shall be in the radiofrequency (RF). This ensures that data is delivered at RF frequencies, not the intermediate frequency (IF).

SRF data shall be delivered in digital format (such as .csv or MATLAB) that includes numerical values for gain vs. frequency and graphical plots of the SRFs. The data format is needed so that Government can process and analyze SRF data.

SRFs, and associated text describing the SRFs, shall be prepared for public release and included in CV-5 Calibration Data Book. SRFs are planned for public release to be made available to data users.

DATA ITEM DESCRIPTION

1. CDRL No.:

CV-6

2. Title:Spectral Response Function (SRF) Measurements

3. Reference:PRD Sec 4.10; SOW Sec 5

4. Use:To provide the SRF data of each channel.

5. Preparation Information:

How the data was collected, the raw measurement data, analysis of data and any corrections to the data needed prior to use by data users, final data set after all corrections have been applied, text to be associated with data for use in CV-5.

SRF frequency data shall be in the radiofrequency (RF).

SRF data shall characterize SRF changes over the operational lifetime of the instrument.

SRF data shall be delivered in digital format (such as .csv or MATLAB) that includes numerical values for gain vs. frequency and graphical plots of the SRFs.

SRFs, and associated text describing the SRFs, shall be prepared for public release and included in CV-5 Calibration Data Book.

DATA ITEM DESCRIPTION

1. CDRL No.:

CV-7

2. Title:INTERCALIBRATION ANALYSIS AND TEST
DATA3. Reference:

SOW 2.4.1

4. Use:

To provide results from the intercalibration activities of the SMBA units for Government review. CV-7 documents the measurements and analyses from the tests performed with the SMBA units that are needed to complete intercalibration across those units. Of key importance is data to enable quantification of the main sources of uncertainty, and any applicable limitations on the test data, plus key assumptions made during the analysis.

5. Preparation Information:

How the data was collected, the raw measurement data, analysis of data and any corrections to the data needed prior to use by data users, final data set after all corrections have been applied, text to be associated with data for use in CV-3.

2.2 Engineering Test Unit (ETU) Data Items

DATA ITEM DESCRIPTION

1. CDRL No.:

ETU-1

2. Title:

ETU DEVELOPMENT AND TEST PLAN

3. Reference:

SOW 5.3.1

4. Use:

To provide information on the development, integration, testing and verification of the instrument Engineering Test Unit.

An overview of the development plans and I&T flow and its relation to performance verification.

To describe the methods, policies, and organization, and qualification test program.

The Contractor may submit separate plans for each phase of the test program as appendices or one plan covering all phases. To define the test requirements and the tests to be performed, instrumentation and locations, the tests methods, stimuli/measurements for acceptance criteria, and a time sequence for the testing.

5. Preparation Information:

The Contractor shall provide an Instrument ETU Development, Integration and Test Plan which describes the series of activities required to develop the Instrument ETU, to integrate the various instrument ETU components into the final flight configuration. The Contractor shall prepare a test plan describing the methods, policies, and organization of the ETU test program. The plan shall also define the test requirements and the tests to be performed, instrumentation and locations, the tests methods, stimuli/measurements for acceptance criteria, and a time sequence for the testing.

Limitations in the ETU instrument verification program which preclude the verification by test of any system requirement shall be documented. Examples of limitations in the ability to demonstrate requirements include replacement parts used because the flight parts were not available or different materials used for the structure to reduce cost.

DATA ITEM DESCRIPTION

1. CDRL No.:

ETU-2

2. Title:

ETU REQUIREMENTS ALIGNMENT MATRIX

3. Reference:

SOW 5.3.1

4. Use:

To provide information on the integration, testing and verification of the instrument ETU. To identify the flight requirements that can and cannot be demonstrated by the instrument ETU due to difference in the fidelity of the ETU in comparison to the flight model or due to decisions on verifications that can be performed in the timeframe during which the ETU testing sufficiently precedes the flight development so as to be influential.

5. Preparation Information:

The Contractor shall provide a verification matrix of all instrument-level flight requirements (including all performance and environments requirements) and indicate which requirements can be met, tested, and verified by the ETU and which tests/verifications are planned to be performed during the ETU verification phase.

The matrix shall identify all of the Product Requirements Document (PRD) requirements that can be verified by the ETU and identify which of those that are planned to be verified with the ETU.

The matrix shall identify all of the Instrument Mission Assurance Requirements (IMAR) that can be verified by the ETU and identify which of those that are planned to be verified with the ETU.

The matrix shall identify all of the SOW requirements that can be verified by the ETU and identify which of those that are planned to be verified with the ETU.

2.3 Integration and Test Data Items

DATA ITEM DESCRIPTION

1. CDRL No.:

IT-1

2. Title:

INSTRUMENT INTEGRATION, TEST AND
VERIFICATION PLAN

3. Reference:

SOW 5.3.1
ICD and PRD

4. Use:

To provide information on the integration, testing and verification of the instrument. An overview of the entire I&T flow and its relation to performance verification. To describe the methods, policies, and organization, proto-flight and flight unit test program. The Contractor may submit separate plans for each phase of the test program as appendices or one plan covering all phases. To define the test requirements and the tests to be performed, instrumentation and locations, the tests methods, stimuli/measurements for acceptance criteria, and a time sequence for the testing.

This DID includes all the Contractor generated documents necessary to perform the integration and testing of the instrument (e.g., the Environmental Verification Plan (EVP), the Environmental Test Matrix, ICD/PRD verification, etc.). The EVP documents the Contractor's approach for environmental qualification and acceptance tests. The Environmental Test Matrix summarizes the tests performed. This DID also satisfies the environmental verification requirements of the launch services provider.

5. Preparation Information:

The Contractor shall provide an Instrument Integration and Test Plan which describes the series of activities required to integrate the various instrument components into the final flight configuration. The Contractor shall prepare a test plan describing the methods, policies, and organization of the flight test program. The plan shall also define the test requirements and the tests to be performed, instrumentation and locations, the tests methods, stimuli/measurements for acceptance criteria, and a time sequence for the testing.

Limitations in the instrument verification program which preclude the verification by test of any system requirement shall be documented. Examples of limitations in the ability to demonstrate requirements include:

- Inability to deploy hardware in a 1-g environment.
- Facility limitations which do not allow testing at system level of assembly.
- Inability to perform certain tests because of contamination control requirements.
- Inability to perform powered-on testing because of voltage breakdown concerns.

Alternative tests and analyses shall be evaluated and implemented as appropriate, and an assessment of program risk shall be included in the System Performance Verification Plan.

At a minimum, the plan shall contain the following information:

- (1) Deliverables to Instrument Integration and Test
- (2) Flow of instrument Integration and Test sequence, including identification of calibration tests and characterizations. Identify instrument level requirements to be verified at each stage, cross referenced to the System Performance Verification Plan (SPVP). Identify dependencies within the I&T sequence.
- (3) Description of each instrument activity or test including the general test configuration
- (4) Description of instrument procedures required to support the activity or test
- (5) Description of simulated data/ simulator requirements
- (6) Description of instrument special handling requirements to support the activity or test
- (7) Description of instrument ground support equipment requirements to support the activity or test
- (8) Description of instrument safety requirements to support the activity or test
- (9) Description of instrument personnel required to support the activity or test
- (10) Requirement to take photographs during the integration process for reference
- (11) Description of the process to be used for handling anomalies as they occur during I&T
- (12) Requirement of log books to be maintained throughout the I&T process

Proposed tailoring of tests shall include a comparison (gap analysis) of the planned test program to the ICD/PRD. Note: the comparison to ICD/PRD will be used by the Government team to support future reviews with Independent Review Teams. Justification shall be provided for any requested deviations to ICD/PRD.

This CDRL shall have the following as an appendix:

As an appendix to the instrument test and verification plan, a test matrix shall be prepared that summarizes all tests that will be performed on each component and each subsystem. The purpose is to provide a ready reference to the contents of the test program in order to prevent the deletion of a portion thereof without an alternative means of accomplishing the objectives; it has the additional purpose of ensuring that all flight hardware has been subjected to environmental exposures that are sufficient to demonstrate acceptable workmanship. In addition, the matrix shall provide traceability of the qualification heritage of hardware. All flight hardware, spares, and prototypes (when appropriate) shall be included in the matrix. Details of each test shall be provided (e.g., number of thermal cycles, temperature extremes, vibration levels). It shall also relate the design environments to the test environments and to the anticipated mission

environments. The matrix shall be prepared in conjunction with the initial environmental verification plan and shall be updated as changes occur.

A complementary matrix shall be included showing the tests that have been performed on each component or subsystem (or applicable level of assembly). This should include tests performed on prototypes or engineering units used in the qualification program, and should indicate test results (pass/fail or malfunctions).

DATA ITEM DESCRIPTION

1. CDRL No.:2. Title:

IT-2

TEST REPORTS

3. Reference:SOW 5.3.6

4. Use:

Report the results of all tests identified in the Test Plans, including test procedures used, test results, configuration status of all items under test, all measured data, a summary of observed results, and a discussion of compliance.

5. Preparation Information:

The Contractor shall provide test reports that summarize the results of verification tests on the instrument such as:

- a) Hardness and Survivability Test Reports
 - (1) Part Lot Acceptance and Latch-up Screening Test Report
- b) Quality Related Reports
 - (1) DPA Reports/Coupons
- c) Reliability Related Reports
 - (1) Life Test Report(s)
- d. Test Data
 - Results of Engineering, Performance, Calibration or Acceptance testing; Raw data and Test Data Analysis (TDA) System Engineering Reports only.
- e. Hardware and Software Qualification and Acceptance Test Reports

The following shall be included in test reports at a minimum:

- a. Test identification and hardware configuration – for specific test
- b. Facility description
- c. Reference - applicable test plan, test procedures, and test requirements, test log including the dates of the testing, photographs of test setup, any malfunction reports written during the test
- d. Test results, to include:

1. Identification of test results which confirmed the expected results as specified in the test plan / procedures or for which variations between actual and expected results were within specified tolerance. For the latter case, actual test results shall be shown.
 2. Identification of test results which differ from expected results beyond expected or acceptable limits.
 3. Identification of any planned test objective or requirement for which actual results were not obtained. Reasons for not meeting the objective/requirement shall be stated.
 4. Identification of any false or aberrant results noted during the test or subsequent analyses. Note that any such behavior that can prevent the instrument from accomplishing its mission objectives can be a basis for rejection.
- d. Recommendations for subsequent actions shall be stated, based on the test results, to include:
1. Redesign of a particular component to enable the instrument to meet a specific requirement which was not fulfilled.
 2. Revision of a system / subsystem specification in cases where the test results disclose ambiguity or conflicting requirements.
 3. The performance of additional tests where results were not acceptable to fulfill objectives.

DATA ITEM DESCRIPTION

1. CDRL No.:

IT-3

2. Title:DETAILED TEST PROCEDURES AND AS-RUN
TEST PROCEDURES3. Reference:SOW 5.3.4
IMAR4. Use:

To provide information on how the Instrument and associated hardware such as cables and Ground Support Equipment (GSE)/System Test Equipment (STE) will be tested and how their performance will be verified.

5. Preparation Information:

The Contractor shall provide Quality Assurance Related Procedures (printed wiring board coupon analysis, wire bond pull test, die shear). The Contractor shall provide Reliability Related Procedures (Accelerated Testing, Reliability Improvement Testing). The Contractor shall provide Hardness and Survivability Test Plans/Procedures (Part Lot Acceptance and Latch-up Screening Test Plans/Procedures (the plan and procedure (may be combined as one document) shall identify the parts requiring radiation lot acceptance testing and latch-up screening and describe the plans for the testing including parameters to be tested, radiation levels, and acceptance criteria.) The Contractor shall deliver existing, as-run, functional test procedures. The Contractor shall provide detailed test procedures to be used during Instrument Level Testing. Test procedures shall provide information sufficient to understand the purpose and methodology of all tests, and to provide the required environmental and configuration controls necessary for successful completion of the test. These procedures shall be prepared for each test activity defined in the Performance Verification Plan (CDRL SE-4) and shall cover all Instrument test operations, interfaces, and Instrument performance requirements (i.e., electrical, structural and mechanical, EMC), and shall cover specialized tests such as mechanical function and deployments, environmental exposure tests (i.e., vacuum, vibration), Instrument calibration, GSE/STE calibration and checkout, and pre-launch end-to-end tests. If tests are conducted in conjunction as part of a “group” test, for example, Limited Performance Test (LPT) or Comprehensive Performance Test (CPT), one test procedure may encompass this group. At a minimum, the procedures shall contain the following information:

- a. Test Objectives including hardware to be tested

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- b. Test Methods and Sequence, including Test methods documented within a test procedure prior to start of test
 - c. Applicable Documents and Software
 - d. Required Instrument configuration, including any differences from flight configuration
 - e. Test Equipment Configuration, including layout and interconnection of test equipment and articles including the grounding scheme. Location and identification of all measuring points on appropriate schematics and diagrams
 - f. Test Equipment and Facility Identification (Test equipment & tooling lists including type, range, and accuracy for each item)
 - g. Test Instrumentation
 - h. A confirmation that each piece of test equipment is within its calibration period.
 - i. Safety Provisions and Cautions, including Identification of hazardous and potentially hazardous situations and operations and abort conditions
 - j. Environmental and/or other conditions to be maintained, including contamination controls
 - k. Responsibilities and chain-of-command for test performance
 - l. Expected results in telemetry and associated caution and warning levels.
 - m. Data Recording Requirements
 - n. Data Recording Forms and Tables
 - o. Accept/Reject Criteria
 - p. Note any test phases and profiles
 - q. List the requirements for the test procedure and test report development
 - r. Description of any necessary functional operations required during the test (i.e., a CPT performed at hot and cold during thermal vacuum testing)
 - s. Test parameters/specific data to be measured
 - t. Test limits and tolerances
 - u. Inspections points as applicable

As-Run Test Procedures shall be all as-run, red-lined, Contractor-format test procedures used during instrument development, integration, and test at instrument and at observatory levels. With the as-run procedures, include all deviations from the test procedures with a rationale for each deviation. The Contractor shall provide Automated Test Scripts.

DATA ITEM DESCRIPTION

1. CDRL No.:

IT-4

2. Title:PACKAGING, HANDLING, STORAGE, AND
TRANSPORTATION (PHS&T) PLAN AND
PROCEDURES3. Reference:SOW 6.1
IMAR4. Use:

Provide the requirements, plans, instructions and procedures for safe and effective packaging, handling, storage, and transporting of flight hardware, spares, and associated GSE/STE throughout the contract. For As-Run Procedures, to provide an archive of test procedures used during system development to aid in on-orbit anomaly resolution.

5. Preparation Information:

This documentation shall discuss the plan and all of the step-by-step procedures for the packaging, handling, storage, and transporting of the instrument, spares, and GSE/STE. This plan shall be prepared in accordance with the IMAR.

The documentation shall include:

For Transporting:

- A. Nomenclature of all supportive equipment
- B. Calibration and load-tested data
- C. Identification of special environmental conditions, such as cleanliness, temperature, humidity, etc., and the controls to be implemented to maintain those conditions
- D. Format for recording QA stamp, deviations, and approval columns
- E. Requirements for special personnel, tools, equipment, special handling fixture and containers, including:
 - a. Specific procedures for use of instrument protective covers.
- F. Method of transportation and carrier
- G. Cargo manifest including aircraft layout diagrams, if applicable
- H. Ground and/or air shipment cargo loading and unloading procedures.
- I. Staging area plans and diagrams.
- J. Trip planning schedule of events, required support, route, contingency plans, permits.

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- K. Procedures to comply with local, state, and Federal safety requirements
 - L. Procedures for maintaining contact with the transported item.
 - M. Procedures and precautions to be taken (against shock, contamination, electrostatic discharge (ESD), etc.) for handling the instrument
 - N. Transportation methods, for the instruments and its subassemblies, within the manufacturing plant, from the plant to the spacecraft contractor's facilities, and if applicable, precautions to be taken for transportation to the launch site while integrated to the spacecraft.
 - O. Shipping containers to be used for transporting the instrument and its deliverable GSE/STE; environmental control, including ESD, during transportation
 - P. Marking of the shipping containers

For Storage:

The Contractor shall provide a detailed plan for ground storage of the instrument in case it is required at the direction of the Government. This shall include the plans, requirements, and procedures for maintaining and monitoring the instruments while in storage at the spacecraft contractor's facilities, and the actions to be taken in case of anomalies. The plan shall describe the following:

- a. Preparation for storage
- b. Ground storage facilities
- c. Environmental control
- d. Monitoring of critical functions during storage
- e. GSE/STE and testing requirements during storage
- f. Impact of prolonged storage on instrument operational lifetime including expendables.
- g. Removal from storage, including retesting requirements
- h. Storage, before and after integration to the spacecraft; corresponding environmental control, including ESD; required monitoring during storage

DATA ITEM DESCRIPTION

1. CDRL No.:

IT-5

2. Title:

Instrument Post Delivery Support (PDS) Document

3. Reference:

SOW 6.6

4. Use:

This document is intended to define the required instrument operations from receipt at the spacecraft contractor's facility through encapsulation. This document will include the details pertaining to equipment, personnel, testing, safety, and contamination providing the baseline integration plan, description, and procedure for conducting post-delivery activities from instrument delivery at the Spacecraft Contractor to encapsulation for SMBA. This document will be used to guide the spacecraft and instrument team to complete all operations and activities the instrument requires to be ready for flight. This document will be used by the Government and Spacecraft Contractor to prepare a successful satellite integration and test campaign.

5. Preparation Information:

The Contractor shall provide the stated information in the format provided in the document outline below (or as close as possible to this intended format/outline). Sufficient detail shall be provided to meet the intended uses of this document as stated above.

Equipment details shall include all equipment required for every test event at the spacecraft contractor's facility and at the launch site. The information should be delineated by type of equipment, where and when the equipment is used, if it is temporary, what events the equipment is used for, operating restrictions for the equipment, equipment safety concerns, and equipment cleaning/maintenance required.

Personnel details shall include all personnel required for all types of test events and all operations along with their roles and responsibilities. Each test event or operational should clearly describe what instrument personnel are required, what their roles are, and whether they are responsible for the operation or whether they are there to support the spacecraft contractor personnel.

Testing details shall include a mapping of all instrument test activities mapped to each satellite level test event from Bench Acceptance Test (BAT) to launch site testing. The testing details should also include any required spacecraft configurations, any personnel required for the

activity, all equipment or GSE needed for the activity, all applicable constraints, and any safety considerations.

Safety details shall include what type of safety concern the operation presents, what personnel are required, constraints to be referenced, and if any special equipment is needed.

Contamination details shall define the cleaning and operational processes throughout all events at the spacecraft contractor's facility and launch site. These details should include measurements needed, any purge plans, any cleanings required, any cleaning supplies required along with which organization is supplying them, and any spacecraft/instrument configuration requirements (bags/covers/etc.).

The contractor shall provide the instrument scripts or inputs to SC scripts necessary to run the instrument after delivery.

2.4 Mission Assurance Data Items

Mission Assurance Data Item Descriptions are included in the Souder for Microwave-Based Applications (SMBA) IMAR document: NEON SMBA IMAR 470-SMBA-00436 Appendix A.

CDRL	Title	IMAR DID
MA-1	Mission Assurance Compliance Matrix	1-1
MA-2	Supplier List	1-2
MA-3	Orbital Debris Assessment Report (ODAR) and End of Mission Plan (EOMP)	2-1
MA-4	System Safety Plan	3-1
MA-5	Safety Requirements Compliance Checklist	3-2
MA-6	Operations Hazard Analysis and Hazard Verification Tracking Log	3-3
MA-7	Instrument Safety Assessment Report	3-4
MA-8	Hazardous Procedures for Payload I&T and Pre-Launch Processing	3-5
MA-9	Pre-Mishap Plan	3-6
MA-10	Reliability Program Plan	4-1
MA-11	FMECA and Critical Items List	4-2
MA-12	Fault Tree Analysis	4-3
MA-13	Reliability Calculations	4-4
MA-14	Limited Life Items List	4-5
MA-15	Parts Stress Analysis	4-6
MA-16	Worst-Case Analysis	4-7
MA-17	Software Assurance Plan	5-1
MA-18	ESD Control Plan	6-1
MA-19	Printed Circuit Board Procurement Plan	6-2
MA-20	Lead-Free Control Plan	6-5
MA-21	EEE Parts Control Plan	7-1
MA-22	Master EEE Parts List	7-2
MA-23	Radiation Effects Mitigation	7-3
MA-24	M&P Selection, Control, and Implementation Plan (MPSCIP)	8-1
MA-25	Materials Usage Agreement (MUA)	8-2
MA-26	Materials Identification and Usage List (MIUL)	8-3
MA-27	Life Test Plan and Final Report for Lubricated Mechanisms	8-4
MA-28	Additive Manufacturing Control Plan (AMCP)	8-5
MA-29	AM Part Production Plan (PPP)	8-6
MA-30	Contamination Control Plan	9-1
MA-31	Foreign Object Damage Program	9-2
MA-32	End Item Acceptance Data Package	12-1

2.5 On Orbit Data Items

DATA ITEM DESCRIPTION

1. CDRL No.:

OO-1

2. Title:

INSTRUMENT ON-ORBIT COMMISSIONING
PLAN

3. Reference:

SOW 6.3.1.1

4. Use:

To define how the Instrument will be activated, calibrated, and tested (functionally and performance); and how its on-orbit performance will be verified. The Plan should include test activities, test equipment and plans for inclusion of the Government test team.

5. Preparation Information:

This CDRL item may be combined with CDRL OO-8.

The On Orbit Commissioning Plan shall be the governing document for initialization and validation of the Instrument during the pre-operational check-out period. The plan shall include:

- a. A summary of the initialization and verification methodology
- b. A matrix or list of any Instrument requirements, with associated acceptance criteria, to be verified on-orbit which is cross-referenced to the appropriate Instrument On-Orbit Test Procedures or Calibration/Validation Procedures
- d. The schedule of initialization and verification activities, including start times and durations.
- e. Procedure numbers of the Instrument On-Orbit Test Procedures to be used during initialization and verification
- f. Constraints to operations

DATA ITEM DESCRIPTION

1. CDRL No.:

OO-2

2. Title:INSTRUMENT USERS MANUAL

3. Reference:SOW 6.3.1.1

4. Use:

The report shall be a self-contained document in that a reader not familiar with the Instrument can obtain a reasonably complete understanding of the Instrument without recourse to another document or drawing. The document is not meant to be an engineering working document but a reference document for SMBA Instrument data users including: Observatory operators, Government personnel, scientists, Observatory contractor personnel, and the general public.

5. Preparation Information:

The Instrument User's manual shall provide a non-proprietary description of the system, subsystems, functions, and operations (with emphasis on how the system operates), with illustrations, block diagrams and circuitry descriptions. The Instrument to spacecraft interface shall be described. Physical details (mechanical, electrical, thermal, optical train & detectors, etc.), information on the software design, interfaces, instrument modes and states, commands and telemetry, and performance characteristics shall be provided by inclusion or by reference. The report shall be a self-contained document in that a reader not familiar with the Instrument can obtain a reasonably complete understanding of the Instrument and its operation without recourse to another document or drawing. The document is not meant to be an engineering working document but a reference document for Government personnel, scientists, Observatory contractor personnel, and the general public of SMBA Instrument data users.

The SMBA Instrument User's Manual shall characterize Instrument performance with respect to: relative spectral response; radiometric accuracy, sensitivity, and stability; and line-of-sight accuracy. Each Manual shall provide pre-flight test results characterizing performance and a flight performance evaluation with updates at major milestones. The document shall describe the Instrument modes of operation and the equipment, methods, accuracies, and concepts of operation for in-flight calibration of radiometric response and line-of-sight. The document shall include a description of the data format and packet structure.

The User's Manual shall stand alone. There shall be no references to training packages or other materials (e.g., internal vendor reports/memos) that will not be readily available. Note that what

is ‘readily available’ is not the same at beginning of life (BOL) as at end of life (EOL), and the On-Orbit User’s Manual must apply to EOL.

In general, if there is any doubt whether an operator may need to know the information, the Contractor should include it.

Constraints shall be discussed as appropriate and in context; e.g., if critical commands must be enabled prior to sending, this should be discussed in the command and data handling (C&DH) section.

The User’s Manual shall cover the life of the instrument from launch pad through decommissioning.

Note the following is not intended to be a table of contents. The information below does not have to be in the order or groupings given here, but are provided to ensure expected content is included in the deliverable.

Information should include, but is not limited to, the items below:

- Sensor overview including science mission
 - o Channel Wavelengths / Frequencies and Resolutions
 - o Scan pattern
 - o Field of View (FOV)/footprint on the Earth
 - o Scan period
 - o Scan patterns
 - o A description of the hyperspectral capabilities
 - o For hyperspectral include:
 - a. Identification of which bands and frequency ranges have hyperspectral capability
 - b. The number of channels for each hyperspectral band
 - c. The start and stop frequencies, the center frequency and bandwidth of each hyperspectral channel
 - d. Performance trades and studies used to select the hyperspectral configuration, including rationale on the selected configuration
 - e. The SRF of each channel, including a description of which passbands are combined to form the channel (a reference to CV-10 SRFs is acceptable).
 - f. The raw data used to derive the SRF along with the frequencies associated
 - g. The description of the SRF measurement test that provides details about the test procedure (set up and instrumentation), the environmental conditions, the data generated, the data analysis and the results obtained.
 - h. A description of the algorithms that combine the channels including rationale; limitations and specifications; mathematical description; a performance analysis; and assumptions such as the data sets used for simulating the environmental conditions

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- i. The center frequency stability, calibration accuracy, and NEDT of each channel
 - j. Flexibility in the frequency plan, including capabilities to update the frequencies pre- and post-launch
 - k. Power consumption impacts
 - l. Thermal impacts
 - m. Data rate impacts on downlink
 - n. Data storage impacts
 - o. Description of the signal flow and digitization, including Fast Fourier Transforms, binning, windowing, and similar functions as applicable
 - p. A description of the key contributing hardware elements and their functions
 - q. A description of the software
 - r. Digitization errors, including how closely a digitized signal matches the analog signal
 - o A description of the RFI capabilities
 - o For RFI include:
 - a. The Instrument bands that have RFI capability
 - b. The number of channels for each band
 - c. The bandwidth of each channel
 - d. The Radio Frequency Interference (RFI) detection algorithms chosen for each band
 - e. The implementation of the RFI detection algorithms
 - f. An overview of each algorithm including the objective or purpose of the algorithm
 - g. The rationale for each algorithm and identification of any specifications and limitations
 - h. A description of each algorithm's inputs and outputs including any external ancillary data (i.e., thresholds) used by the algorithm
 - i. A mathematical description of each algorithm
 - j. An implementation of the algorithm in source code and identification of the Operating System that the source code was built on.
 - k. A performance analysis of the algorithm that provides metrics on the algorithm including its detection sensitivity, its probability of detection and its false alarm rate as function of RFI characteristics
 - l. A description of the methods used to validate each algorithm and description of the validation results
 - Subsystem/Component Overview (The purpose of this section is to inform the reader of overall subsystem information; however it should cover in detail those aspects that are monitored or changeable by the operator.)
 - o Mechanical
 - Include diagrams of motors and information about bearings and lubricants, because electro-mechanical devices are notorious troublemakers
 - o Electrical
 - Nominal operating power ranges
 - o Thermal

-
- Locations of all heaters and their relationship (location relative) to thermistors
 - Heater set points
 - Nominal operating temperature ranges
 - o Optical / RF / energy path through the system
 - Detector layout (if appropriate)
 - Calibration targets/sources
 - o Flight Software
 - Functional description
 - Simplified flowchart if practical
 - Fault Protection
 - What it monitors
 - What it does in response to ‘violations’ of monitored information
 - Description and discussion of uploadable and downloadable tables including how to know when an update is needed
 - Fault codes (or where to find)
 - Response to fault codes (or where to find)
 - Sensor Modes
 - o Description of and how to recognize each mode
 - o Allowable Transitions
 - o Recommended Transitions
 - o Indicate what transitions can occur autonomously versus by command
 - o Table of major components and their status (e.g., ON/OFF) for each mode
 - o Table of telemetry available in each mode
 - o Activities allowable in each mode (e.g., loads and/or dumps)
 - Flight model-specific signatures
 - o ‘Features’ of the specific flight model being documented
 - o Differences with existing/operating sensors should be explicitly pointed out
 - Interfaces
 - o Sensor Internal
 - o Spacecraft External
 - Point-to-point telemetry and commands
 - Survival heater locations, set points and relationship to thermistors (if not covered under sensor thermal subsystem)
 - o Databus interaction
 - o Timing, clocks, and synchronization
 - o Average and peak data rate
 - o Average and peak power consumption
 - Command and Data Handling
 - o APIDs
 - o Telemetry Rates
 - o Packet descriptions
 - o Description on on-board stored commanding and macro capabilities
 - o Critical commands
 - o Database

-
- Flight Telemetry Limits ‘philosophy’: i.e., How did the Contractor ‘choose’ the flight limits? E.g., Did the Contractor take the ‘tested to’ maximum and add margin? Did the Contractor take the ‘Qual’ limits and add margin? If a limits philosophy requirement is defined elsewhere, it supersedes this guidance.
 - Science
 - o Algorithms
 - o On-board data processing
 - Overview of Recommended Operating Procedures (ROP)
 - o Purpose and short summary of functionality
 - Normal Operations
 - o On-Orbit Activation Sequence (could be a ROP or series of ROPs) including timing as determined from testing, e.g., cool-down is expected to take x minutes or x hours or x days
 - o Cal/Val Activities
 - o On-Orbit Maintenance Activities
 - Constraints/configurations for Maneuvers DELTA-V and DELTA-I maneuvers
 - o Trending Recommendations (health and safety and science performance parameters)
 - Nominal values
 - Frequency of trending
 - Limits
 - o Special Activities (e.g., Cal/Val maneuvers, mid-mission out-gassing)
 - o Table and/or Flight Software (FSW) Loads
 - Contingency Operations and Troubleshooting
 - o Red/yellow limit responses
 - o Document common or ‘expected’ anomalies and their response
 - o Redundancy and redundancy management
 - o Diagnostic modes/activities/data and when appropriate to use

DATA ITEM DESCRIPTION

1. CDRL No.:

OO-3

2. Title:

COMMISSIONING REPORT

3. Reference:

SOW 6.5.6

4. Use:

The purpose of the Commissioning Report is to show that the on-orbit performance will meet the specifications.

5. Preparation Information:

For each activity required to validate on-orbit performance, the following information shall be reported:

- a. Activity information including name, short description, and analysis plan
- b. Test IMT number
- c. Success criteria
- d. Test data used for validation including test date(s) and time
- e. Activity results including activity date and time, data used for validation on the same level as the TVAC reports and where possible show how on-orbit performance compares to ground characterization data
- f. Any test anomalies or idiosyncrasies and their impacts
- g. Health, functionality, and performance

DATA ITEM DESCRIPTION

1. CDRL No.:

OO-4

2. Title:TELEMETRY AND COMMAND HANDBOOK

3. Reference:SOW 6.3.1.1

4. Use:Establishes the command and telemetry definitions for the SMBA.

5. Preparation Information:

The Telemetry and Command Handbook shall be delivered in Contractor format. The Government prefers two volumes for clarity and ease of use.

The Handbook shall contain the telemetry information. The Government prefers this be Volume

1. This section of the Handbook shall provide at least, but not be limited to, the following:

- a. Detailed listing of all telemetry assignments and Parameter IDs.
- b. Key parameters necessary for description of the telemetry requirements as a part of the list.
- c. Summary quantifying the number and type of telemetry assignments for each subsystem and the number of spares remaining.
- d. Description of telemetry interfaces, telemetry format, telemetry requirements data, and caution and warning levels. Note: Modal Limits are based on sensor configuration (e.g., subsystem X is not on, therefore subsystem X limits are not checked. Heater Y is on, so limits are ABC. Heater Y is off, so limits are CDE, etc.)
- e. Descriptive information necessary for interpretation of the telemetry requirements.
- f. Listing of telemetry assignments that confirm commands.
- g. Schematic reference for each telemetry assignment.
- h. Data format, to include: detailed bit definitions, command, meta and ancillary data definitions for all telemetry and command data (command and telemetry list), detailed data format and content of the specific Instrument data and ancillary data files as stored on the flight data recorder and specific detailed data formats and outputs of the flight data recorder for Instrument and ancillary data transferred via the Instrument mission data downlink interfaces.

The Handbook shall contain the command information. The Government prefers this be Volume

2. This section of the Handbook shall provide at least, but not be limited to, the following:

- a. Detailed listing of all serial and digital commands

- b. Key parameters necessary for description of the commands as part of this listing
- c. Summary quantifying the number of types of commands used by each subsystem and by each unit and the number of spares remaining
- d. Description of command input, command verification, command rate and filler (no-op) commands.
- e. Description of command requirements data and information necessary for interpretation.
- f. Listing of commands verified by telemetry and the telemetry verifiers.
- g. Schematic reference for each command.

DATA ITEM DESCRIPTION

1. CDRL No.:

OO-5

2. Title:INSTRUMENT LAUNCH AND ON-ORBIT
OPERATIONS AND CONTINGENCY
PROCEDURES3. Reference:

SOW 6.3.1.1

4. Use:

The Launch and Early Orbit Procedure section is required to coordinate the launch operation.

The Instrument On-Orbit Procedures section contains the complete set of procedures required for operating the Instrument following completion of commissioning activities. It shall cover all mission phases: activation & checkout, calibration/validation, routine operations, and contingencies.

The Instrument On-Orbit Test and Cal/Val Procedures section of the document contains the set of procedures required to conduct activation, testing, and calibration/validation during the commissioning phase of the mission. These procedures are unique to the activation and commissioning phase and do not duplicate the procedures in the Instrument On-Orbit Operations and Contingency Procedures section. This document shall include the plans, requirements, and procedures for on-orbit sensor calibrations.

5. Preparation Information:

The Instrument Launch and Early Orbit Procedure section of the document shall contain Instrument sequences required to support the Instrument state of health monitoring and operational events during the launch phase of the mission. These procedures are applicable from launch vehicle-Observatory separation until the SMBA Observatory is configured for on-orbit operation prior to the start of the on-orbit system test activities.

The procedure shall include:

- a) Detailed flight timeline
- b) Survival heater configurations (when they need to be turned on), set points, etc.
- c) Script of each communication event including required actions and responses

The Instrument On-Orbit Procedures section of the document shall provide a detailed set of operations procedures for operating the Instrument. It shall provide detailed operating

procedures for each instrument mode, including overview/purpose, prerequisites, applicable mission phases, affected systems/subsystems, flowcharts, commands/mnemonics/parameter values, telemetry mnemonics/verification values, constraints (e.g., time-spacing between commands) and cautions, associated contingencies, duration (if known) and start time (if known). These procedures shall include:

- A. Normal on-orbit command and control operations
- B. Instrument State-of-Health Monitoring and management
- C. Instrument mode transition and mode operations
- D. Contingency and recovery procedures
- E. Calibration procedures
- F. Instrument procedures for satellite reconfigurations

The Instrument On-Orbit Test and Cal/Val Procedures section of the document shall provide a detailed set of operations procedures for activating, testing, and calibration/validation of the instrument. These procedures shall include:

- A. Instrument command and control operations
- B. Instrument State-of-Health Monitoring and management
- C. Instrument mode transition and mode operations
- D. Contingency and recovery procedures
- E. Calibration procedures
- F. Instrument procedures for satellite reconfigurations

Detailed Calibration/Validation Procedures delivered under CV-2 may be referenced from this document instead of duplicated.

All sections, all procedures shall include:

- a. Test/Procedure Purpose
- b. Any Instrument requirement(s) being verified
- c. Procedure Methodology
- d. Support Resources Required
- e. Observatory and Instrument configuration before and after the procedure is executed
- f. Step-by-step commands to be issued and expected Instrument response after each step; including possible contingencies and responses
- g. Cautions and warnings

DATA ITEM DESCRIPTION

1. CDRL No.:

OO-6

2. Title:INSTRUMENT AND SIMULATOR
TRAINING MATERIALS3. Reference:

SOW 6.4.1.2, 4.2.11

4. Use:

The Instrument Training Materials are used to train the Operations Team in the operation and functionality of the Instrument and the Simulator. This document shall support the training to be provided by the Mission Operations Team (MOT). The Training Materials shall include a Simulator Description and Training Manual.

5. Preparation Information:

The training plan shall provide course descriptions and objectives, subjects to be covered, time estimates, facilities & resources needed, classroom materials required and completion criteria. The training materials shall cover unit description, operation during activation/checkout, calibration/validation, routine operations, and contingency phases; and operational constraints.

The Instrument Training Material shall include training in the following items:

- a. Instrument Operations concepts
- b. Functional description of the SMBA
- c. All Instrument modes of operation and functional description of the SMBA
- d. Instrument Procedures
- e. Instrument Normal Imaging activities
- f. Describe and include execution of timelines for instrument-related launch, Observatory deployment, on-orbit checkout, and Observatory commissioning activities
- g. Instrument operating constraints and rules
- h. Instrument Anomaly scenarios and procedures

The Simulator Training Manual shall provide detailed information on how to configure and operate the simulator including power on/initialization, mode changes, anomaly injection, running/modification of scenarios (if applicable), administrative options, and maintenance; simulator capabilities and limitations; material to train the MOT (Mission Operations Team) to operate the simulator, including power on/off, initialization, restarting (in event of fault or operator error), scenario generation & usage, anomaly injection, and maintenance.

In order for the operations team to perform properly, the training should include how to do table uploads and downloads, required instrument operations during maneuvers; and understand the Command and Telemetry Handbook (OO-4), The Instrument Launch and On-Orbit Operations and Contingency Procedures (OO-5), the Constraints, Restrictions and Warnings/Alerts Document (OO-7) and the Instrument Command and Telemetry Database (OO-8).

Materials could be text, PowerPoint files and videos.

DATA ITEM DESCRIPTION

1. CDRL No.:

OO-7

2. Title:CONSTRAINTS, RESTRICTIONS, AND
WARNINGS/ALERTS DOCUMENT3. Reference:

SOW 2.6

4. Use:

The Instrument Constraints, Restrictions, and Warnings/Alerts Document provides the operating constraints and restrictions for the SMBA, including conditions and cautions associated with sensor/satellite activities and commands.

5. Preparation Information:

The Instrument Constraints, Restrictions, and Warnings/Alerts Document shall include:

- a) Operating constraints for each Instrument operational mode and activity
 - b) Operating restrictions for each Instrument operational mode and activity
 - c) Any warning or alert conditions for Instrument operational activities
- 1) Constraints shall be categorized into the following four types:
- i. Constraint – consequence of violation is Damage/Loss of Mission.
 - ii. Restriction – consequence of violation is Reduced Capability.
 - iii. Limitation – consequence of violation is Loss of Mission Data.
 - iv. Warning/Alert – procedural notes.
- 2) Constraints shall be grouped into the following three types:
- i. Test Only
 - ii. On-Orbit Only
 - iii. Shared (applies to test and on-orbit)

Each constraint, restriction, limitation, and alert should contain:

- 1. Title
- 2. Description (Concise statement of the constraint, i.e., ‘Do not do XYZ,’ ‘Always do XYZ,’ ‘Y must always follow X’)
- 3. Source (usually ROPs, C&T handbook, etc. – where the reader can go for more information)
- 4. Impact of Violation (If X, then Y.)
- 5. Recommended Action

DATA ITEM DESCRIPTION

1. CDRL No.:

OO-8

2. Title:INSTRUMENT COMMAND AND TELEMETRY
DATABASE3. Reference:

SOW 6.5.1 and 6.6.2.3

Sounder for Microwave-Based Applications (SMBA) Command & Telemetry Database (CTDB)
Schema per revision specified in Contract TBD, Attachment TBD4. Use:

The Instrument Command and Telemetry Database will be used to operate, control, and monitor the SMBA.

5. Preparation Information:

The Instrument Command and Telemetry Database shall be in a format defined by (SMBA C&T Database Schema) per revision specified in Contract TBD, Attachment TBD.

The telemetry and command database shall contain

- a) All commands required to operate, monitor, and control the SMBA
- b) All telemetry parameters required to operate, monitor, and control the SMBA
- c) Instrument command and telemetry unique identifiers
- d) Database validation results (i.e., the results of the Validation Plan defined in CDRL OO-14), provide the results of the CTDB validation with every CTDB delivery
- e) Database validation results shall continue to be provided for each Government-provided satellite-level database

DATA ITEM DESCRIPTION

1. CDRL No.:

OO-9

2. Title:

INSTRUMENT MISSION DATA DEFINITIONS

3. Reference:

SOW 6.5

470-00TBD SMBA Application Packet Schema, revision in Contract TBD, Attachment TBD.

4. Use:

The SMBA Mission Data Definitions will be used within the SMBA Missions Operation Element to identify and process data products produced by the sensor.

5. Preparation Information:

The SMBA Mission Data Definitions shall be provided via an XML database submitted to the Government per the format defined in 472-00371 (SMBA Application Packet Schema). The Application Packet Schema is the XML schema that is used to generate the application packet structure XMLs for each instrument's mission data packet type. These XML files will be called out in the Mission Data Format Control Book and be considered the official configuration managed data format definitions for all mission data packets.

DATA ITEM DESCRIPTION

1. CDRL No.:

OO-10

2. Title:COMMAND AND TELEMETRY DATABASE
(CTDB) VALIDATION PLAN3. Reference:

SOW 6.5

4. Use:

The CTDB Validation Plan documents the processes used to verify the integrity and accuracy of the instrument command and telemetry database.

5. Preparation Information:

1. The CTDB Validation Plan shall include, but not be limited to, the following information:
 - a. Difference report (e.g., “Was-Is Reports”)
 - b. How difference reports are generated
 - c. Verification of content
 - i. the instrument level database (CDRL OO-8)
 - ii. the system level database (provided by the Government)
 - d. How updates are tracked
 - e. Methods of in-flight or ground-test verification
2. Note: Results from executing the CTDB Validation Plan are provided via CDRL OO-8.

DATA ITEM DESCRIPTION

1. CDRL No.:

OO-11

2. Title:SMBA MEMORY TABLE BASELINE
CONTENT3. Reference:

SOW 6.5

472-00370 SMBA FSW Table Schema, revision in Contract TBD, Attachment TBD.

4. Use:

Instrument memory table content deliveries will be used to modify onboard FSW and/or tables as well as maintain the ground reference image of on-board memory.

5. Preparation Information:

The contractor shall provide baseline content for the loadable and dumpable Instrument Memory Tables per the XML formatting defined in 472-00370 (SMBA FSW Table Schema) Appendix B.

The full package of information that accompanies any FSW or tables delivered to the Mission Operations Element shall be

1. FSW/Table Files (XML file referenced above)
2. Back-Out Load Files – or instructions for using previous load files to back out bad load
3. Version Description Document including
 - ☐ FSW/Table Version Number.
 - ☐ Database Version Number (as appropriate)
 - ☐ Brief description of the purpose for the changes
 - ☐ Was/is list (not needed for FSW updates)
 - ☐ Installation procedure for the FSW/Table and Back-out on both Observatory and for the simulator (if the software or installation process is different)
 - ☐ MD5 checksum for the delivered bundle (tools available on Secure File Transfer Protocol (SFTP)/eRooms servers)
 - ☐ Checksums of loads
 - i. A checksum or hash value for the SFTP/eRoom delivered bundle (for Mission Support Team (MST) CM to unbundle). A tool/algorithm is provided, typically in eRooms, to create the checksum/hash value.
 - ii. Expected value for each table's (or packet's) checksum or Cyclic Redundancy Check (CRC)

4. Load Certification Packet including:
- ☐ Approval – personal, written (e.g., e-mail or signature) approval from a supplier
 - ☐ identified stakeholder demonstrating the load is released by the supplier.
 - ☐ Test Information (or request to test on simulator). For any load that cannot be functionally validated at either the supplier facility or on the simulator, a thorough manual verification must be completed and documented.
- D. Platform tested
- ii. Nominal and Contingency Path (including back-out test)
- D. Entry and exit conditions

The full package (md5 bundle) shall be delivered by one of these methods (in order of preference):

1. eRooms
2. sftp
3. hard media

2.6 Project Management Data Items

DATA ITEM DESCRIPTION

1. CDRL No.:

PM-1

2. Title:

MONTHLY PROJECT MANAGEMENT
REVIEWS (PMR)

3. Reference:

SOW 1.4, 1.5.1.2, 2.4.1, 4.2.10

Reference is made to NASA Procedural Requirement (NPR) 7120.5 (Current Revision), NASA Space Flight Program and Project Management Requirements.

4. Use:

To evaluate contract status. These reports will be used to provide an opportunity for face-to-face discussions between the Contractor and the Government regarding project status, plans and issues. The PMR shall be designated to provide a management level review of technical, schedule and programmatic activities. A Quarterly Program Management Review (QPMR) will typically consist of a 4 hour meeting to review formal status and an additional 4 hours to address special topics to be mutually determined prior to each review. These reviews will typically address the topics listed below as required. Addition of special topics will be coordinated between Government and Contractor prior to each meeting.

5. Preparation Information:

Scope: The PMR shall include all aspects of the contract effort.

The Monthly Project Management Review (PMR) shall be presented at a face-to-face meeting with the Government. These meetings shall occur at the Contractor's facility, unless modified by mutual consent.

The Monthly Project Management Review shall include the following:

- A. Program Manager's Assessment and Program Summary
- B. Report of Key Technical Parameters; including mass and power budgets, peak and average power budgets, software metrics, pointing knowledge and control error budgets, their current best estimated, calculated, and/or measured values for all parameters, and trends for each Technical Performance Metric (TPM). The list of reported Key Technical Parameters shall be agreed upon with the Government and may be revised by the Government as the situation dictates. The values to be presented shall be at least to the major component level of the subsystems, e.g., Mechanisms, Power supplies, heaters, cabling, etc. The accuracy of the values and units shall be identified. The status of

margins and contingency based on maturity shall be presented. Report Monthly TPM (Current Best Estimate (CBE) and Margin) against the specification for:

- Mass (Kg)
 - Operational Peak Power (W)
 - Operational Average Power (W)
 - Survival Peak Power (W)
 - Average Data Rate (Mbps)
 - Peak Data Rate (Mbps)
 - Memory Utilization
 - CPU Utilization
 - NedT
 - Calibration Accuracy for each channel
 - Non-linearity
 - Gain Stability
 - Center Frequency Stability
 - Bandwidth
 - Out-of-band Rejection
 - Passband Ripple
 - Pointing Accuracy
 - Beamwidth
 - Beam Efficiency
 - Science metrics that are selected to show the satisfaction and stability of the science performance requirements
- C. Technical status for system and subsystem design and development activities, including subcontract technical performance.
- D. Design Status (when design work is being accomplished); covering the design activities for the previous month, including: schedule and technical status of development, manufacturing status, qualification and test status, issues, and concerns
- E. (1) List of Class I in process and approved changes; (2) List of Class II approved Changes available upon request
- F. Summary of Integrated Master Schedule Status, including a brief description of the current status of each subsystem or subassembly along with descriptions of any existing or potential problems areas. The critical path and near critical paths shall be explained along with possible work-arounds being considered to maintain the schedule. A slack metric with at least 1 month of aging (prior month slack) shall be maintained and provided for major subassemblies and activities (the Government and Contractor managers shall determine which items are to be tracked at least quarterly).
- G. Integration & Test Status such as Test Issues, Test Results, Performance Trends
- H. Verification & Validation status and issues (as required)
- I. Mission Assurance status such as summary of MRB actions and status; summary of Failure Review Board (FRB) actions and status; failure investigations and corrective actions
- J. Lower Tier Suppliers Status (as required)
- K. A detailed 12-month “rolling-wave” schedule (3 months of actual, plus 9 months of forecast)

-
- L. Metrics summarizing all milestones depicting planned versus actual accomplishments and explanations for variances, e.g., graphic display of planned versus actual cumulative milestone status, xx of 1,000 milestones are behind plan and why, etc.
 - M. Summary explanation of key Earned Value Management (EVM) parameter performance and trends
 - N. Metrics by Integrated Product Team (IPT) and Total Project, summarizing Baseline, Forecast and Actual Headcounts where a head equals one equivalent work-month of effort. Include explanation of variance from plan.
 - O. Problems (problems that are significant in nature, questionable “significance” should be discussed with and determined by the Instrument Manager or COR) encountered during the reporting period, and anticipated approaches for resolution (including, as appropriate, technical issues, manpower and staffing, and supplier and subcontractor issues, etc.)
 - P. Status of open issues and problems from prior reporting periods
 - Q. Risk Status for top 10 risks and ability to discuss mitigations plans of each.
 - R. Review of MR Liens or other unfunded risk mitigations including estimated cost and impact
 - S. Status of action items with focus brought to those actions which are not closing in a timely manner
 - T. PM&P Issues including identification and status of long-lead purchases/acquisitions planned.
 - U. Prohibited Materials Compliance Status (as required)
 - SMBA Product Alert evaluation status
 - Production Readiness Issues (as required)
 - V. Significant plans and activities for the following month, including upcoming milestone events depicting critical items of project status.
 - W. CDRL Status Report that includes the following information for each document delivered in accordance with the CDRL or overdue from previous reporting periods:
 - a. Document Number
 - b. Document Title
 - c. Scheduled Due Date
 - d. Actual Submittal Date
 - e. Current Status
 - f. List of documents planned for delivery during the next reporting period, listed by document number, title, and scheduled submittal date.
 - X. Summary of financial status and Cost Performance per SOW
 - Y. Business issues, including personnel changes
 - Z. Status of Giver/Receiver items

The Contractor shall place PMR material on the Government-access electronic database by the day of the PMR. Presentation material may be in Contractor format.

DATA ITEM DESCRIPTION

1. CDRL No.:

PM-2

2. Title:WORK BREAKDOWN STRUCTURE (WBS)
AND DICTIONARY

3. Reference:

SOW 1.2.3

Reference is made to NPR 7120.5 (Current Revision), *NASA Space Flight Program and Project Management Requirements*, and MIL-HDBK-881, *Department of Defense Handbook Work Breakdown Structure*. These documents shall be used as guides in the preparation of the WBS and the WBS dictionary.

4. Use:

The WBS establishes a product-oriented framework for reporting program cost, schedule, and technical performance and provides a basis for uniform planning, reporting status, program visibility, and assignment of responsibilities.

Additionally, the WBS provides a product-oriented logical subdivision of hardware, software, services, facilities, etc., that contribute to and/or make up the total project scope of effort and work.

The WBS Dictionary provides a narrative description of the work tasks and effort to be performed in each WBS element.

5. Preparation Information:

The WBS and WBS Dictionary are two distinct project documents used for defining the approved project scope of work. The contents of each document are detailed in the following paragraphs:

- a. WBS - A logical, hierarchical display of the subdivision of all project work to be completed. The WBS shall include the approved element title and element number.
- b. WBS Dictionary - The WBS dictionary shall describe and document the work content of every WBS element and relevant efforts associated with each element (e.g., design, development, manufacturing).

The WBS dictionary shall be arranged in the same order as the contract WBS. The WBS dictionary shall include the following for each WBS element:

1. WBS element title.
2. WBS element code.

3. WBS element content description (including quantities, relevant associated work, and contract end items where applicable).
4. WBS Index.
5. SOW paragraph number.
6. Specification (number and title) associated with the WBS element (if applicable).
7. Contract line item associated with the WBS element.
8. Date, revision number, revision authorization and approved changes.
9. Contract Identification Number.

The WBS is reflected in an electronic chart format that depicts element relationships arranged in the same order as the WBS provided in the proposal. The WBS Dictionary is ordered in consonance with the WBS index and references each WBS element by its identifier and name.

Changes shall be incorporated by change page or complete reissue.

DATA ITEM DESCRIPTION

1. CDRL No.:

PM-3

2. Title:

ENGINEERING PEER REVIEW PLAN

3. Reference:

SOW 1.5.1.8, 5.4

GPR 8700.6A, Engineering Peer Reviews

4. Use:

Early in project/product formulation, the Contractor shall identify subsystems, components, software and crosscutting functional elements to be subject to the Engineering Peer Review (EPR) process. The Peer Review Plan is used to identify the methodology and scope of the Contractor's peer review process.

5. Preparation Information:

The Engineering Peer Review Plan shall:

- a) Describe the methodology used to determine the scope of the EPR process, including rationale for specific components or subsystems that will not be peer reviewed.
- b) Describe the peer review process, including personnel, nominal agenda, and Request for Action (RFA) generation, tracking, and closure process.
- c) Identify the concepts, designs, plans, processes, subsystems, components, software, etc. that will be Peer Reviewed.
- d) Identify a schedule or associated milestones for the EPRs; peer reviews shall be planned and scheduled in the Contractor integrated master schedule (IMS).

In addition to other standard discipline EPRs, the Contractor shall, at a minimum, conduct EPRs to cover the following specific items as appropriate based on item maturity and risk:

- a) Sensor focal plane assembly
- b) RF mixers, local oscillators
- c) Data system, including the allocation of functions between the Instrument and the spacecraft solid state recorder
- d) Algorithm development
- e) Mechanism design and test procedures
- f) Opto-mechanical design and alignment processes
- g) Antenna and receiver design and alignment processes
- h) Sensor calibration design and test
- i) Instrument integration and test (I&T)
- j) Spacecraft-Level I&T

- k) Fault detection and correction
- l) Flight software development
- m) Command and control procedures
- n) System engineering and interfaces

The Contractor shall track action items from EPRs and maintain EPR presentation and closure documentation for the duration of the contract.

DATA ITEM DESCRIPTION

1. CDRL No.:

PM-4

2. Title:

INTEGRATED PROGRAM MANAGEMENT
REPORT (IPMR)

3. Reference:

SOW 1.2, 1.2.2

NPR 7120.5 (Current Revision), NASA Space Flight Program and Project Management
Requirements

NFS 1852.242-75 Earned Value Management System (Deviation)

FAR 52.234-4 Earned Value Management System

DI-MGMT-81861 (06/20/12)

Department of Defense Earned Value Management Implementation Guide (EVMIG) (website:
<http://guidebook.dcmamc.mil/79/EVMIG.doc> October 2006)

NASA Schedule Management Handbook (Current Revision)

4. Use:

To provide information for: (1) integrating cost and schedule performance data with technical performance measures, (2) assessing the magnitude and impact of actual and potential problem areas causing significant cost and schedule variances, and (3) providing valid, timely project status information to higher management.

5. Preparation Information:

Contract Performance Report (CPR)

The CPR shall include data pertaining to all authorized contract work, including both priced and un-priced effort that has been authorized at a not-to-exceed amount in accordance with the Contracting Officer's direction. The CPR shall include the following reporting levels:

- Format 1 – Work Breakdown Structure
- Format 3 – Baseline
- Format 4 – Staffing
- Format 5 – Explanations and Problem Analyses
- Format 6 – Integrated Master Schedule
- Format 7 – Electronic History and Forecast File

To ensure an integrated approach to risk management, the data provided by this CPR DID shall be in consonance with the WBS, Integrated Master Schedule (IMS), Risk Management Processes, Plans and Reports (where required), Probabilistic Risk Assessment Processes and

Reports (where required), the Cost Analysis Data Requirement (CADRe) and the Monthly/Quarterly Contractor Financial Management Reports (533/Q).

CPR formats shall be completed according to the instructions outlined in DI-MGMT-81861 and the following forms: Format 1 (DD Form 2734/1); Format 3 (DD Form 2734/3) and Format 4 (DD Form 2734/4)

Format 1, Work Breakdown Structure (WBS): Format 1 shall provide data to measure cost and schedule performance by WBS elements.

Format 3, Baseline: Format 3 provides the budget baseline plan against which performance is measured. It is the baseline report used to track all changes to the Performance Measurement Baseline (PMB). In addition to the six month forecast Format 3 shall contain the cost to complete estimate for the current year (Column 10), and the following three out years (Columns 11-13).

Format 4, Staffing: Defines staffing forecasts.

Format 5, Explanations and Problem Analyses: The variance analysis thresholds, which if exceeded, require problem analysis, narrative explanations, and corrective action plan descriptions at the Control Account level. Variance analysis thresholds shall be +/- \$100K and +/- 10% for the current month, +/- \$250K and +/- 10% cumulative, and +/- \$500K and +/- 10% at complete cost and schedule variance to date. The variance analysis thresholds may change once the Project personnel evaluate the Contractor's schedule and cost performance, and risk. Special emphasis should be placed in the variance analysis on cost and schedule growth linked to technical risks (e.g., technology development efforts; design engineering; integration; complexity; project management; systems engineering; duration constraints; etc.) identified by both the Government and contractor.

Format 6, Integrated Master Schedule: Provides the discrete tasks/activities, consistent with all authorized work, and relationships necessary for successful contract completion. Subcontractor work shall also be incorporated as tasks with the prime IMS at a level necessary for a realistic critical path. The IMS shall contain the contract milestones, accomplishments, criteria, discrete tasks/activities, work packages and planning packages, as applicable, from contract award or formal contract Definitization to contract completion. The IMS shall be statused according to the contractor's accounting month and aligned with Format 1 reporting month, as applicable.

Following Integrated Baseline Review (IBR) approval, the IMS shall become the Contract Baseline Schedule and the basis for monitoring Contractor performance. Changes shall be incorporated by a baseline change request (BCR) or complete reissue. BCR data shall be provided to the Government upon request. A BCR shall be developed for all rolling wave planning.

IMS Submission shall be submitted per CDRL PM-8 except as interrupted by Contractor holidays. Submission of the deliverables shall be by standard electronic media. Electronic media submittals shall be in native file format utilizing schedule management software approved

by the project office. A legend identifying the Contractor's schedules symbols used and their meanings shall be provided.

The IMS shall contain:

- a. Clearly identified "need dates" for Government Furnished Property.
- b. A month end status date (i.e., "data date")
- c. A custom table, view and fields will be provided to the contractor for inclusion in the IMS to designate deliverables, (Del) and receivables (Rec). iRec and iDel will be used to designate internal deliverable/receivables to I&T.
- d. A text field will be chosen by the Contractor to designate the CDRL number.
- e. A text field will be chosen by the Contractor to designate a traceability reference to the SOW

Format 7, Electronic History and Forecast File: Provides supplemental historical and time phased information in the approved electronic XML format by WBS, provided at level 7.

Contractor formats may be substituted for CPR formats whenever they contain all the required data elements at the specified reporting levels in a form suitable for use by NASA management. The CPR shall be submitted electronically using the American National Standards Institute (ANSI) X12 standards (transaction sets 839), or XML using WINST.DTD format (format can be found at <http://evm.nasa.gov/reports.html>) or the UN/EDIFACT standard (PROCST message).

DATA ITEM DESCRIPTION

1. CDRL No.:

PM-5

2. Title:MANUFACTURING (FABRICATION AND
ASSEMBLY) FLOW PLAN3. Reference:

SOW 4.0

4. Use:

Define the fabrication, assembly, inspection and test sequences, locations, operations, and controls for hardware articles at all levels of assembly to be manufactured by the prime contractor on this contract.

5. Preparation Information:

Provide narrative material and flow charts, which define the contractor's fabrication and assembly plan. This plan shall include, as a minimum:

- (a) Description and the sequence of fabrication and assembly operations and processes
- (b) Identification of the site(s) of the operation(s)
- (c) Identification of tooling and facilities required
- (d) Description of the cognizant organizational structure(s)
- (e) Requirements for inspection and test stations
- (f) Description of system of controls on fabrication and assembly, including documented procedures.
- (g) Integrated manufacturing and inspection flow chart, which depicts all fabrication, processing, and assembly operations; testing at the subcomponent levels; associated inspection points and documentation. The sequence of this flow chart shall start with the point of entry of purchased materials, parts, components, or Government Furnished Property and the initiation of manufacturing and conclusion with as assembled electrical or mechanical component being delivered for acceptance test, or an item of structure being delivered for integration. This shall be submitted in accordance with Table 1 of this document.

DATA ITEM DESCRIPTION

1. CDRL No.:

PM-6

2. Title:HARDWARE CONFIGURATION
MANAGEMENT PLAN3. Reference:

SOW 1.4

4. Use:

Defines the Contractor's configuration management system (including policies and procedures) that will be implemented for the instrument flight hardware.

5. Preparation Information:

Contractor format is acceptable. The Contractor's hardware configuration management plan shall be prepared in accordance with the Contractor's standards. This plan shall describe in detail all flight hardware configuration management processes, methods, and procedures the Contractor intends to use on the Instrument. This plan shall describe how hardware configuration management is accomplished and how consistency between product definition, the product's configuration, and the configuration management records is achieved and maintained throughout the applicable phases of the product's life cycle by the Contractor.

The configuration management plan shall describe the Contractor's approach, methodology, and application of configuration management principles and practices and shall include the following:

1. General product definition and scope
2. Description of configuration management activities and procedures for each of the following configuration management functions:
 - a. Configuration planning and management
 - b. Configuration identification
 - c. Configuration Change management
 - d. Configuration status accounting
 - e. Configuration verification and audit
 - f. Configuration management of digital data
 - g. Configuration management of software
3. Organization, roles, responsibilities, and resources
4. Definition of terms
5. Programmatic and organizational interfaces
6. Subcontract flow down of configuration management

DATA ITEM DESCRIPTION

1. CDRL No.:

PM-7

2. Title:

PROJECT MANAGEMENT PLAN

3. Reference:

SOW 1.2.7, 1.5.1.14

4. Use:

Describes how the project is organized and managed by the Contractor. It provides the management structure, its system of operation, responsible lines of communications, and key personnel assignments. The organization chart identifies the Contractor's project organization with names, functions, lines of authority, coordination, etc. This PMP can be in Contractor's format, however accomplishments and criteria must be defined.

5. Preparation Information:

Contractor format is acceptable.

This plan shall address the overall organization, management approach, and structure of the Contractor's project plus its interrelationships with the parent company and the subcontractors.

Describe how and where the project will operate during all phases of the contract. This plan shall identify and describe interfaces with the Government. This plan shall include graphical displays such as flow diagrams, WBS, logic networks, etc., to reduce verbal descriptive material where appropriate. As attachments or Appendices, the Contractor shall prepare and deliver a WBS Index and Dictionary describing each of its elements to level 4 in the Contractor's preferred format. Note: Revisions of the WBS index or dictionary must be submitted prior to incorporating the proposed changes into the contract.

This plan shall provide an organizational chart(s) and sufficient supplemental narrative to briefly describe the following:

- (a) Organization proposed for carrying out the project showing interrelationships of technical management, business management, and subcontract management, from lower level through intermediate management to top-level management with explanation of:
 1. The authority of the Project Manager relative to other ongoing projects and applicable support organizations within the company structure. Discuss the Project Manager's control over essential resources and functions necessary to accomplish the work.
 2. How and by whom interdepartmental work will be monitored and the authority of the Project Manager over interdepartmental work.

-
3. Process to be followed by the Project Manager in obtaining decisions beyond his/her authority and in resolving priority conflicts for resources and functions not under the Project Manager's direct control such as personnel, finances, and facilities.
 4. The project team members with names and functions.
- (b) Implementation approach for the project. Describe in general how the requirements of the Statement of Work (SOW) will be achieved. Identify potential problems related to this work, and your approach to problem avoidance and/or solution. Briefly, identify how your risk management system and processes are integrated into the daily management, decision making, and strategic direction of the project and how you foresee Government participation in that process. Address the degree to which your proposed personnel and overall management procedures are addressing any lessons learned from previous efforts. For new procurements or changes from the previous instrument development process, briefly describe such things as make/buy strategies, acquisition plans, sparing philosophy, project dependencies, facility requirements, internal review strategies and plans, significant work elements on critical paths, long-lead items, and significant milestones down to at least the lowest level of the WBS.
- (c) Contractual procedures proposed for the project to effect administrative and engineering changes, describing any differences from existing procedures.
- (d) Management technique lessons learned from prior efforts or otherwise to be employed to minimize: 1) project costs and schedule overruns, and 2) risks of violating interface requirements and agreements. Describe associated controls to be exercised over subcontractors and suppliers. Describe how issues will be surfaced in a timely manner and at the proper levels. Identify initial proposed Technical Performance Metrics (TPM) the project will use to monitor and report on interface compliance and resource status.
- (e) The proposed Safety and Mission Assurance organizational structure, including staffing plans, reporting channels, authority and responsibilities, and management visibility. Discuss whether the technical, test, manufacturing and system safety/quality assurance/reliability/ configuration management personnel required for this project (as indicated in your proposed labor hours) are presently on your payroll and immediately available for this work. State the number and disciplines/skills of persons who would have to be hired, and plans to obtain them.

DATA ITEM DESCRIPTION

1. CDRL No.:

PM-8

2. Title:

RISK MANAGEMENT PLAN (RMP)

3. Reference:

SOW 1.3

GPR 7120.4A, Risk Management

NPR 8000.4 Risk Management Procedural Requirements Rev A

4. Use:

The Risk Management Plan is the basis for identifying and managing all performance, reliability, schedule, cost, and safety risks on the Contractor's project.

5. Preparation Information:

The Contractor shall prepare and update a Risk Management Plan that meets the requirements of Goddard Procedural Requirement (GPR) 7120.4A and NPR 8000.4. The Risk Management Plan shall describe Contractor's process for Risk Identification, Risk Analysis, Risk Handling and Risk Monitoring. This RMP shall include a process for tracking and reporting risk status and changes. The risk management plan shall clearly describe:

- Overview of the risk management process
- Organizational responsibilities
- Risk identification approach
- Risk mitigation planning
- Interface of risk management to schedule and financial/cost management
- Risk tracking/documentation
- Risk management list reporting

The plan shall include risks associated with hardware and software (e.g., technical challenges, new technology qualification, etc.), COTS, system safety, performance, cost, and schedule (i.e., programmatic risks). The plan shall identify which tools and techniques will be used to manage the risks and the plan shall use the GSFC risk 5x5 definitions as described in Figure 1. The plan shall also express risk statements in the condition-consequence format as expressed in the following:

- Risk Statement: For a risk to be understandable, it must be expressed clearly and include:
 - Condition: A description of the current condition that may lead to the loss. It is a single phrase that identifies possible future problems, and describes current key circumstances, situations, etc. that are causing concern, doubt, anxiety, or uneasiness.

- **Consequence:** A description of the loss or consequence. It is a single phrase or sentence that describes the key, negative outcome(s) of the current conditions.

GSFC 5x5 Risk Scorecard

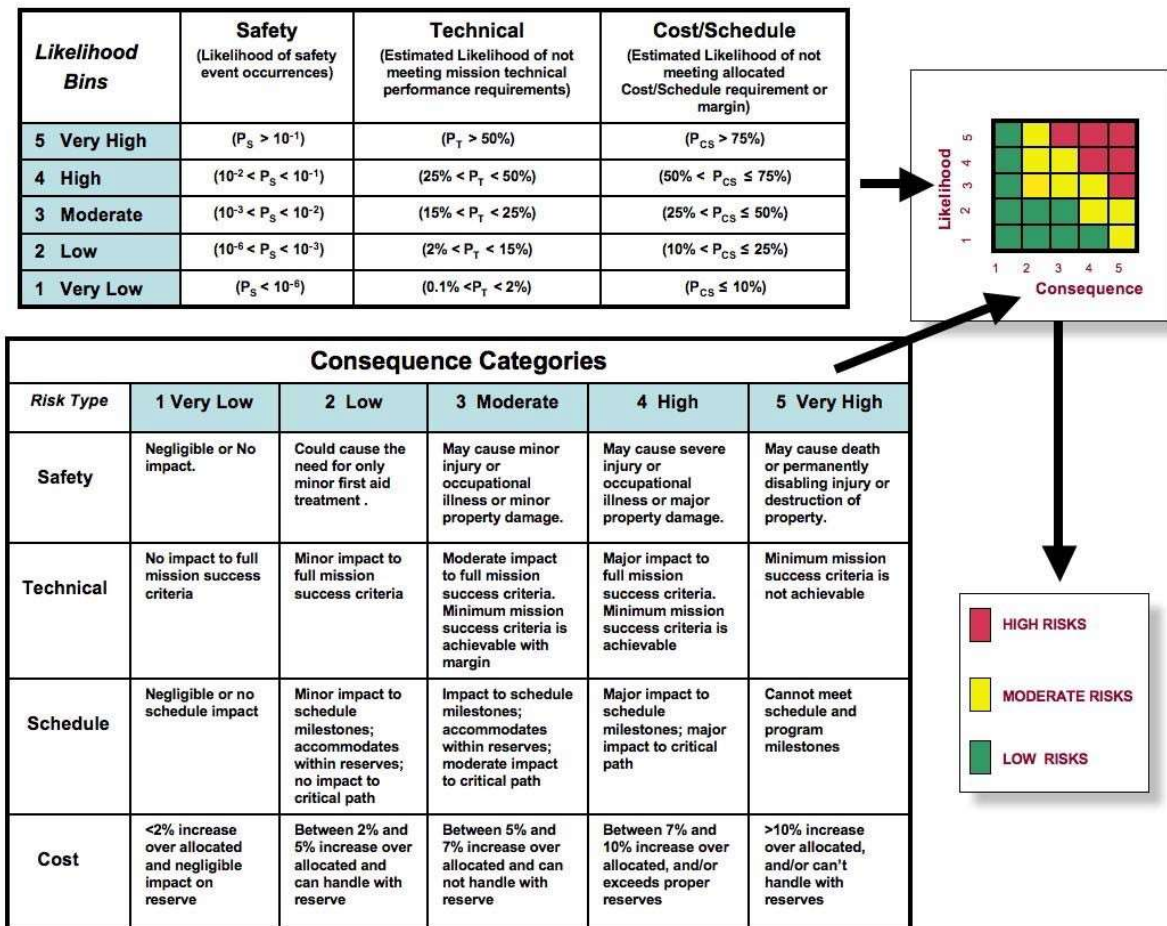


Figure 1. GSFC 5 x 5 Risk Definitions

DATA ITEM DESCRIPTION1. CDRL No.:

PM-9

2. Title:GIVER/RECEIVER/HAND-OFFS/RECEIVES-
DELIVERABLES (REC-DELS)3. Reference:

SOW 2.4.2.2

4. Use:

To provide the project a description of each Giver-Receiver (G-R) indicated in the IMS.

5. Preparation Information:

Contractor format is acceptable.

The Contractor shall deliver a table of Giver-Receivers with descriptions of each giver or receiver item, whether it is a giver (deliverable from the Contractor) or receiver (delivery to the Contractor), UID for this G-R the Contractor IMS, baseline date, forecast date, and actual date as well as any comments regarding the status of the Giver or change to need date.

2.7 Review Data Items

DATA ITEM DESCRIPTION

1. CDRL No.:

RE-1

2. Title:

SYSTEM REQUIREMENTS REVIEW (SRR)
DATA PACKAGE

3. Reference:

SOW 2.4
GSFC-STD-1001A

4. Use:

To evaluate the requirements, requirements flow-down, and the operational concepts and to validate the realism of the functional and performance requirements and their congruence with the system configuration selected to conduct the mission. A “delta” review is used to achieve the same, though typically, but not always, on a subset of the design and requirements.

5. Preparation Information:

The Systems Requirements Review (SRR) shall contain the flow-down of requirements from the Instrument Performance Requirements Document to the Instrument Performance Specification and first-order allocations.

The SRR Data Package shall contain all relevant instrument information required to satisfy paragraph 4.2 (Criteria for Successful Completion) of GSFC-STD-1001A, Criteria for Project Flight Critical Milestone Reviews.

The SRR Data Package shall discuss Contractor system level requirements, rationale, and flow-down plans to lower level requirements.

The SRR Data Package shall show the allocation and traceability of requirements to major subsystems.

The SRR Data Package shall include the specification tree illustrating the Contractor’s specifications starting at the top-level Instrument Performance Specification, showing all lower-level specifications to the sub-assembly (box/board) level, and indicating the relationships and interfaces between the documents.

The SRR Data Package shall address any identified Single Point Failures.

The SRR Data Package shall show how the current concept meets all Government specified requirements including interface requirements.

The SRR Data Package shall discuss the preliminary operations concept of the instrument.

Where the Government determines a partial Instrument System-level review or a second review is required based on changes in the instrument design baseline, a “delta” review shall be conducted. This “delta” review shall meet the requirements of the appropriate Instrument System-level review (Preliminary Design Review (PDR), Critical Design Review (CDR), etc.) as defined herein and the applicable CDRL. For a “delta” review, the CDRL shall be tailored to cover those items identified for review by the Government.

The Contractor shall prepare minutes of design reviews that shall consist of a list of meeting attendees, copies of presentation material provided at the review not in the presentation charts package, action items assigned, and agreements made. The reviews will be considered complete when all outstanding action items are closed out to the Government's satisfaction.

Results of Review – As a result of successful completion of the SRR, the system and its operation are well enough understood to warrant design and acquisition of the end items. Approved specifications for the system, its segments, and preliminary specifications for the design of appropriate functional elements may be released.

DATA ITEM DESCRIPTION

1. CDRL No.:

RE-2

2. Title:PRELIMINARY DESIGN REVIEW (PDR) DATA
PACKAGE OR DELTA PDR3. Reference:SOW 1.5.2.2
GSFC-STD-1001A4. Use:

To demonstrate the Flight Equipment and GSE/STE design meet the documented requirements. A “delta” review is used to achieve the same, though typically, but not always, on a subset of the design and requirements.

5. Preparation Information:

The PDR Data Package shall address all relevant instrument information required to satisfy paragraph 5.2 (Criteria for Successful Completion) of GSFC-STD-1001A, Criteria for Project Flight Critical Milestone Reviews.

The Instrument PDR data package shall contain information to cover the Instrument data processing algorithm design.

The PDR data package shall address the instrument compliance with the Instrument Performance Requirements Document.

The PDR data package shall include responses to action items from previous reviews, including subsystem PDRs/Engineering Peer Reviews.

The PDR data package shall include changes since the last review.

The PDR data package shall address performance requirements and their flow-down to the card or equivalent level.

The PDR data package shall address instrument performance budgets including Beginning of Life to End of Life performance degradation estimates.

The PDR data package shall address error budget determination.

The PDR data package shall address mass, power, data rate, coding and format, commands, EMI/EMC.

The PDR data package shall contain a detailed report of Key Technical Parameters down to a level below the one reported in the PMR.

The PDR data package shall address interface requirements, including the following information:

- a. Preliminary analysis that will allow the Contractor to ensure that the "swept" or deployed volume is verified, accounting for all distortions and misalignments.
- b. Preliminary estimates of gimbaled masses, inertias, and to permit sizing the spacecraft control components to meet pointing and stability requirements.
- c. Preliminary analysis of disturbances of the sum of any periodic disturbance torques, in order to produce the corresponding magnitude spectrum.

The PDR data package shall address mechanical/structural design, analyses, and life tests.

The PDR data package shall address electrical, thermal, optical, radiometric, and calibration design and analyses.

The PDR data package shall address software requirements, design, and development environment.

The PDR data package shall address Ground Support Equipment design and workflow, and describe how each item will be fabricated, tested, and certified when needed.

The PDR data package shall address design verification, test flow and calibration/test plans.

The PDR data package shall address the instrument operations concept.

The PDR data package shall address parts selection, and qualification.

The PDR data package shall address preliminary Failure Modes and Effects Analysis (FMEA); Fault Tree Analysis; and reliability analysis and results.

The PDR data package shall address redundancy and redundancy management.

The PDR data package shall address single point failures.

The PDR Data Package shall address the list of long lead items, and of items that may become obsolete prior to completion of all flight instruments, identify those items that must be procured prior to CDR (including a list of those that were ordered prior to PDR and SRR), and provide a plan for procuring these items and all parts.

The PDR data package shall address contamination requirements and control plan.

The PDR data package shall delineate the status of each document required at PDR as to its acceptability for use as is.

The PDR Data Package shall present all instrument risks and address their mitigation.

The PDR Data Package shall provide the status of all sub-contracts and discuss the preliminary design status of critical assemblies and sub-assemblies.

The PDR Data Package shall present a summary of all breadboard and brassboard testing and present the available test results.

The PDR Data Package shall present the development status of sub-assembly engineering units, and available test data.

The PDR Data Package shall present the planned and/or expected level of functionality of the components of the Engineering Development Unit.

The PDR Data Package shall address the producibility of the design solution.

The PDR Data Package shall address mission assurance to be imposed including parts and materials usage as well and workmanship standards imposed.

The PDR Data Package shall address software assurance process.

The PDR Data Package shall include the specification tree illustrating the Contractor's specifications starting at the top-level Instrument Performance Specification, showing all lower-level specifications to the sub-assembly (box/board) level, and indicating the relationships and interfaces between the documents.

Where the Government determines a partial Instrument System-level review or a second review is required based on changes in the instrument design baseline, a "delta" review shall be conducted. This "delta" review shall meet the requirements of the appropriate Instrument System-level review (PDR, CDR, etc..) as defined herein and the applicable CDRL. For a "delta" review, the CDRL shall be tailored to cover those items identified for review by the Government.

The Contractor shall prepare minutes of design reviews that shall consist of a list of meeting attendees, copies of presentation material provided at the review not in the presentation charts package, action items assigned, and agreements made. The reviews will be considered complete when all outstanding action items are closed out to the Government's satisfaction.

The Contractor shall complete the GSFC Opening Learning Design (GOLD) Rules Alignment Matrix (Appendix A) for review and discussion prior to and at the PDR at the Government's discretion.

DATA ITEM DESCRIPTION

1. CDRL No.:

RE-3

2. Title:CRITICAL DESIGN REVIEW (CDR) DATA
PACKAGE3. Reference:SOW 1.5.2.3
GSFC-STD-1001A4. Use:

To present the Instrument Flight Equipment and GSE/STE design and operation, and to demonstrate that all related manufacturing documentation, processes, and fixtures are in place before hardware manufacture of new or improved hardware begins, and to demonstrate that the design meets all performance requirements.

5. Preparation Information:

The CDR Data Package shall contain all relevant instrument information required to satisfy paragraph 6.4 (Criteria for Successful Completion) of GSFC-STD-1001A, Criteria for Project Flight Critical Milestone Reviews.

The CDR data package shall include responses to action items from previous reviews, including subsystem CDRs/Peer Reviews, and all subsequent subsystem peer reviews held before the CDR.

The CDR data package shall include changes and improvements cited in Section 1.2.7 of the SOW.

The CDR data package shall address instrument compliance with the most recent revision of the Instrument Performance Requirements Document under contract.

The CDR data package shall address instrument performance budgets including Beginning of Life to End of Life performance degradation estimates.

The CDR Data Package shall address the procurement status of long lead items and Electrical, Electronic, and Electromechanical (EEE) parts.

The CDR Data Package shall address manufacturing flow, and the status of the improved manufacturing and assembly drawings, bill of materials, etc.

The CDR Data Package shall address the improved manufacturing procedures.

The CDR Data Package shall address mission assurance product checkpoints and evaluation criteria.

The CDR Data Package shall address standard applicable in-house processes.

The CDR Data Package shall address special/unique to SMBA/fixtures

The CDR Data Package shall address facilities required for manufacturing. The CDR Data Package shall address personnel resources (time phased).

The CDR Data Package shall address the delivery schedules for flight hardware and GSE/STE.

The CDR data package shall address detailed analysis from FMEA, fault tree analysis, and reliability analysis.

The CDR Data Package shall address worst case analyses of:

- (a) Electrical circuits
- (b) Scanning drive system
- (c) Lubrication and lubrication loss
- (d) Tolerance and tolerance sensitivity analysis (including thermal and mechanical considerations)

The CDR Data Package shall address stress analyses using NASTRAN with hand verification

The Instrument CDR Data Package shall address thermal analysis of:

- All subsystems (Antenna, Scan Drive, Electronics and Receiver)

The CDR Data Package shall address weight, power, and volume.

The CDR data package shall contain a detailed report of Key Technical Parameters down to a level below the one reported in the PMR.

The CDR Data Package shall address test plans (including all environmental tests)

The CDR Data Package shall address manufacturing considerations

The CDR Data Package shall address maintainability considerations, including storage.

The CDR Data Package shall address materials and processes lists

The CDR Data Package shall provide a summary of deviations/waivers

The CDR Data Package shall address contamination control and monitoring considerations

The CDR Data Package shall address spares program

The CDR Data Package shall address system safety hazards analyses:

- (a) Hazards identification matrix
- (b) Single point failure summaries
- (c) Risk assessment rationale

The CDR Data Package shall delineate the status of each incomplete deliverable document, due at CDR, as to its acceptability for use as is.

The CDR package shall address the operations concept.

The CDR Data Package shall present any additional test results from breadboard and brassboard testing

The CDR Data Package shall present the test data from sub-assembly engineering development units and the status of ETU development and testing results, if any.

The CDR Data Package shall address the development status of all GSE/STE, including test and calibration procedures, and the software/firmware design and operation and interface aspects as evaluated since the PDR

The CDR Data Package shall address the status of all program risks and their mitigation

The CDR Data Package shall address the status of all sub-contract design activities and schedule for delivery of flight hardware, as appropriate, and demonstrate that designs are complete and have been adequately reviewed.

The CDR Data Package shall include the qualification history\status and details for each component for each instrument including data.

The CDR Data Package shall include the specification tree illustrating the Contractor's specifications starting at the top-level Instrument Performance Specification, showing all lower-level specifications to the sub-assembly (box/board) level, and indicating the relationships and interfaces between the documents.

The Contractor shall prepare minutes of design reviews that shall consist of a list of meeting attendees, copies of presentation material provided at the review not in the presentation charts package, action items assigned, and agreements made. The reviews will be considered complete when all outstanding action items are closed out to the Government's satisfaction.

Additional Details:

- Implement coax cable and instrument design changes to eliminate blind mates in order to reduce/eliminate risk during cable mate/de-mate operations

- Simplify troubleshooting efforts and reduce risk of hardware damage by eliminating de-integration of instrument components
 1. Implement design changes to allow external measurement of RF ground at the fully assembled instrument level instrument
 2. Provide an assessment on routing the signals for Signal Processor Assembly (SPA), Scan Drive Electronics (SDE) and Receiver Power Supply (RPS) diagnostics to an accessible location at the fully assembled instrument level instrument
- Implement design changes to improve access to W, G, V feed horns as well as improve ease of assembly
- Implement integration fixtures that will be used during receiver shelf integration to eliminate handling errors
- Implement design changes to improve receiver shelf wiring to simplify manufacturing of receiver shelves, implement single point ground, and improve repeatability of design execution
- Automate cable functional tests subsequent to cable builds to reduce handling

DATA ITEM DESCRIPTION

1. CDRL No.:

RE-4

2. Title:PRE-ENVIRONMENTAL REVIEW (PER) DATA
PACKAGE OR DELTA PER3. Reference:SOW 1.5.2.4, 5.5
GSFC-STD-1001A4. Use:

To present the description and results of the Instrument Pre-Environmental Test program, and demonstrates readiness for environmental testing.

5. Preparation Information:

The PER Data Package shall contain all relevant instrument information required to satisfy paragraph 9.2 (Criteria for Successful Completion) of GSFC-STD-1001A, Criteria for Project Flight Critical Milestone Reviews.

The PER Data Package shall include status of action items generated at prior reviews.

The PER Data Package shall include analyses and reports required at the review.

The PER Data Package shall contain the results of any analyses updated or revised since the CDR.

The PER Data Package shall include test and integration program descriptions and results.

The PER Data Package shall include failure report summaries including status of action and rationale for closure.

The PER Data Package shall include as-built documentation summary.

The PER Data Package shall include results of the functional and interface tests.

The PER Data Package shall include descriptions of any malfunctions and corrective actions.

The PER Data Package shall include comparison of measured performance with requirements and discussion of the effect of any variance and waivers.

The PER Data Package shall include mission operation constraints.

The PER Data Package shall include contamination avoidance requirements.

The PER Data Package shall include safety requirements.

The PER Data Package shall include list of spares for flight equipment and GSE/STE.

The PER Data Package shall include review of flight hardware handling procedures.

The PER Data Package shall include spacecraft interface concerns, problems, and solutions, including:

- a. Analysis and test data that demonstrates that the "swept" or deployed volume is verified, accounting for all distortions and misalignments.
- b. In electronic format, time/magnitude plots of their disturbances, for decomposition, of the sum of the instrument-induced periodic disturbance torques, in order to produce the corresponding magnitude spectrum.

The PER Data Package shall include orbital operations plans and status of documentation and databases.

The PER Data Package shall include end-item data packages (submit a summary of the content prior to review and have package available for inspection at review)

1. As-built configuration list
2. Hardware parts lists
3. Hardware materials and processes lists
4. Test Log Book (including total operating time and cycle records)
5. Open item lists (including reasons for being open)
6. Safety compliance data package
7. Limited life items listings and status
8. Critical parameters trend data
9. Final comprehensive performance test results
10. Failure report summaries including status of action and rationale for closure

The Instrument PER Data Package shall discuss the compatibility of instrument with spacecraft flight support equipment, ground support equipment and operational ground equipment.

The PER Data Package shall address the availability and readiness of facilities and GSE/STE required for environmental testing.

The PER Data Package shall address the status of all program risks and their mitigation plans.

The PER Data Package shall address the readiness of environmental test plans and procedures.

Where the Government determines a partial Instrument System-level review or a second review is required based on changes in the instrument design baseline, a “delta” review shall be conducted. This “delta” review shall meet the requirements of the appropriate Instrument System-level review (PDR, CDR, etc.) as defined herein and the applicable CDRL. For a “delta” review, the CDRL shall be tailored to cover those items identified for review by the Government.

The Contractor shall prepare minutes of design reviews that shall consist of a list of meeting attendees, copies of presentation material provided at the review not in the presentation charts package, action items assigned, and agreements made. The reviews will be considered complete when all outstanding action items are closed out to the Government's satisfaction.

DATA ITEM DESCRIPTION

1. CDRL No.:

RE-5

2. Title:

PRE-SHIP REVIEW (PSR) DATA PACKAGE

3. Reference:SOW 1.5.2.5, 6.3
GSFC-STD-1001A4. Use:

To evaluate instrument performance during qualification or acceptance testing, and evaluate readiness to ship.

5. Preparation Information:

This data package shall address, as a minimum:

The PSR Data Package shall contain all relevant instrument information required to satisfy paragraph 11.2 (Criteria for Successful Completion) of GSFC-STD-1001A, Criteria for Project Flight Critical Milestone Reviews.

The PSR Data Package shall address responses to action items generated at prior reviews.

The PSR Data Package shall address the solutions to all problems encountered during the environmental test and validation program and the solution rationale.

The PSR Data Package shall address any rework/replacement of hardware, regression testing, and test plan changes.

The PSR Data Package shall address compliance with the test verification matrix.

The PSR Data Package shall address measured test margins versus requirements.

The PSR Data Package shall address instrument performance budgets including Beginning of Life to End of Life performance degradation estimates.

The PSR Data Package shall address qualification/acceptance temperature margins.

The PSR Data Package shall address any data that has been trended to identify compliance with specification, indicating a change or drift to the trend.

The PSR Data Package shall summarize and analyze the monthly trend reports (CDRL SE-2) of trended instrument data, focusing on trends that have changed or drifted anomalously (see SOW paragraph 2.4.2).

The PSR Data Package shall address total failure-free operating time of the item.

The PSR Data Package shall address the number of cycles during testing of parts with finite lifespans.

The PSR Data Package shall address the results of the final audit of any remaining drawing changes.

The PSR Data Package shall address “could-not-duplicate failures” along with assessment of the problem and the residual risk that may be inherent in the item.

The PSR Data Package shall address project assessment of any residual risk.

The PSR Data Package shall provide an update from CDR on shipping containers, monitoring/transportation/control plans.

The PSR Data Package shall address ground support equipment status.

The PSR Data Package shall address post shipment plans.

The PSR Data Package shall address spacecraft integration plan.

The PSR shall address the plans for storage of the instrument, if required.

The PSR Data Package shall include documentation verifying that the instrument meets contamination budgets and requirements specified in the Contamination Control Plan.

The PSR Data Package shall include alignment data relative to the Instrument Boresight and the mounting surface datum(s) associated with the instrument.

DATA ITEM DESCRIPTION

1. CDRL No.:

RE-6

2. Title:

INTEGRATED BASELINE REVIEW (IBR)

3. Reference:

SOW 1.5.1.3

4. Use:

To assess and validate the Contractor's Performance Measurement Baseline (PMB) by conducting an IBR.

5. Preparation Information:

The Contractor shall support a formal on-site Integrated Baseline Review (IBR) conducted by the Government with Government participation. This review will be established by the Government as early as practicable and conducted in accordance with NASA requirements within 180 days of contract award or formal contract definitization. Additional baseline reviews may be required when substantial changes are made to the baseline (e.g., subsequent detailed planning; including routine rolling wave planning and program re-planning resulting from incorporation of significant contract modifications). The IBR will be tailored to meet the needs of the SMBA project.

The Government will coordinate the timeframe and scope of the IBR with the Contractor. The IBR will ensure that the PMB captures the entire scope of work, is consistent with the contract schedule requirements and has an adequate resource plan. The IBR team shall include Government personnel. Baseline Reviews held during the execution of the Contract will be less comprehensive in nature and specific to the interests of GSFC and the SMBA customer.

The IBR shall cover plans related to the SOW.

DATA ITEM DESCRIPTION

1. CDRL No.:

RE-7

2. Title:ENGINEERING PEER REVIEW DATA
PACKAGES3. Reference:SOW 1.5.1.8, 2.6
IMAR4. Use:

Engineering Peer Reviews (EPR) focus on the design and implementation details at levels that system-level reviews cannot address. They provide a resource for Design Teams to identify potential engineering design and implementation flaws, and increase the probability of success. Applying the EPR process early and throughout the product life cycle affords the maximum advantage in terms of resource efficiency as well as design confirmation and ultimate mission success. Peer review documentation represents knowledge that may prove invaluable later.

5. Preparation Information:

Engineering Peer Review documentation shall be in Contractor format.

The Contractor shall prepare minutes of design reviews that shall consist of a list of meeting attendees, copies of presentation material provided at the review not in the presentation charts package, action items assigned, and agreements made. The reviews will be considered complete when all outstanding action items are closed out to the Government's satisfaction.

DATA ITEM DESCRIPTION

1. CDRL No.:

RE-8

2. Title:MANUFACTURING READINESS REVIEW
PACKAGE3. Reference:

SOW 1.4, 1.5.1.9, 5.4.1

4. Use:

The purpose of the Manufacturing Readiness Review (MRR) is to verify the engineering baseline by a design, drawing, and engineering requirement review to assure that the release to manufacturing represents the final and correct flight equipment and that the manufacturing planning is complete and reflects the SMBA instrument configuration.

5. Preparation Information:

The content of this review package shall be as follows:

- a) Process flowchart
- b) Schedule
- c) Product description
- d) Configuration Item Identification List (CIIL)
- e) Computer Software Configuration Item (CSCI)
- d) Summary of any changes to proto-flight configuration baseline, including, parts, materials, and processes.
- e) Summary of any updates to the CDR data submittals, design analysis and SMBA plans and procedures provided under the NASA SMBA Contract.
- f) Detailed manufacturing schedules
- g) Manufacturing flow diagram including inspection, MIPs, and test points
- h) Drawing readiness
- i) Process readiness
 - i. List of drawings
 - ii. Procedures
 - iii. Machine availability and certification
 - iv. Process qualification
 - v. Training and personnel qualification
 - vi. Workmanship standards
- j) Product FMEA (Manufacturing risk mitigation)
- k) Special process list
- l) ESD control plan and cleanliness control
- m) Material availability

- n) Personnel availability
- o) Tooling readiness
- p) Facility readiness
- q) Parts status and qualification status on piece parts
- r) Summary of all nonconformance and test failure reports initiated for the protoflight unit.
The summary shall include the status of all corrective/preventive action.
- s) Status of lower-tier suppliers
- t) Status of prohibited material screening
- u) a table listing all in-process and end item procedures where Consent to Proceed (CTP), Consent to Break (CTB) and/or Consent to Integrate (CTI) decision points require Government approval.

The MRR shall be held as part of a scheduled program review meeting mutually agreed to by the Subcontractor and the Buyer in the baseline schedule; however, where schedule requirements dictate, the MRR may be held at an earlier date. The MRR will be considered approved when all outstanding action items are closed to the Buyer's satisfaction. In the event of disapproval or conditional approval, the required correction(s) shall be submitted to the Buyer for approval. Design review approvals in no way relieve the Subcontractor of responsibility for meeting the subcontract requirements.

DATA ITEM DESCRIPTION

1. CDRL No.:

RE-9

2. Title:

DESIGN MODIFICATION REVIEW

3. Reference:

SOW 1.5.1.6

4. Use:

To identify changes from the previous flight model and assess cost/schedule/technical risk.
Verify:

- Integration, Test, and Verification approach is adequate
- Testability items identified that support implementation of the development plan
- Critical issues and/or activities are identified
- Risk Items are properly identified with adequate mitigation activities
- A cohesive plan from the conduct of the review through launch
- Identify any disconnects, missing items, etc.
- Identify sub-tier vendor viability, parts availability, issues with parts obsolescence, facility conflicts, lessons learned.

5. Preparation Information:

The Data Package shall include at a minimum:

- Instrument Overview
- Compliance Documents
- By Subsystem/component (hardware and software)
 - o Description
 - o Documentation Status
 - o Sub-tier vendors – viability, changes in process or product
 - o Modifications from previous unit – design, process, qualification approach
 - o Test approach
 - o Discrepancies
 - o Waivers
 - o Staffing
 - o Risk assessment
- Test Program
 - o Plans
 - o Procedures
 - o Facilities
 - o GSE/STE
 - o Risk assessment
- Operations products

DATA ITEM DESCRIPTION

1. CDRL No.:

RE-10

2. Title:FIRMWARE DESIGN DATA PACKAGE

3. Reference:SOW 2.0

4. Use:

A Firmware Design Data Package is required for each firmware design to allow independent Government review and assessment of Contractor firmware designs.

5. Preparation Information:

The final, flight Firmware Design Data Package shall be submitted for each Firmware developed by the Contractor. If the flight version is not ready for review when requested by the Government, a Design Data Package for the current engineering/test/prototype Firmware shall be submitted as well.

A Firmware Design Data Package shall include the:

1. Firmware design (place and route) database and any constraint file(s)
 - a. For Actel SX and AX FPGAs, the database file is the .adb file
2. Firmware synthesis constraint and report files
 - a. For designs synthesized with Synplify, the synthesis report is the .srr file
3. Firmware timing analyses for external inputs and outputs, internal domain(s), etc.
4. Firmware internal clock domain crossings disposition table/spreadsheet
 - a. For each clock domain crossing, briefly describe how the source and destination domain logic work without error
5. Firmware simulation coverage report and testbenches/scripts
6. Firmware source code (ex: Very High-Speed Integrated Circuits Hardware Description Language (VHDL), Verilog), schematics and/or state machines/tables, including any 3rd party intellectual property code and/or cores
7. System, box, board & Firmware requirements, specifications presentations and/or verification document(s) relevant to the Firmware and its role in the system, box, board
8. Board schematics (PDF and native format) that contain this Firmware
9. Board netlist (any ASCII format)
10. Board part list (any ASCII or common spreadsheet format)
11. Board layout (PDF and native format)
12. Signal integrity analyses relevant to this Firmware

DATA ITEM DESCRIPTION

1. CDRL No.:

RE-11

2. Title:

TEST READINESS REVIEW (TRR)

3. Reference:SOW 1.5.1.10
GSFC-STD-1001A4. Use:

A TRR ensures that the test article (hardware/software), test facility, support personnel, and test procedures are ready for testing and data acquisition, reduction, and control. A TRR is held prior to commencement of verification or validation testing. The objectives of the TRR are to:

- Ensure a thorough review of the products supporting the review.
- Ensure the products meet the entrance criteria and success criteria.
- Ensure issues raised during the review are appropriately documented and a plan for resolution is prepared.

A successful TRR signifies that test and safety engineers have certified that preparations are complete, and that the project manager has authorized formal test initiation.

5. Preparation Information:

Prior to holding the TRR the Contractor shall present data to the Government that verifies:

- The objectives of the testing have been clearly defined and documented and all of the test plans, procedures, environment, and the configuration of the test item(s) support those objectives
- Configuration of the system under test has been defined and agreed to
- All interfaces have been placed under configuration management or have been defined in accordance with an agreed-to plan, and a version description document has been made available to TRR participants prior to the review
- All applicable functional, unit-level, subsystem, system, and qualification testing has been conducted successfully
- All TRR-specific materials such as test plans, test cases, and procedures have been made available to all participants prior to conducting the review
- All known system discrepancies have been identified and disposed in accordance with an agreed-upon plan
- All previous design review success criteria and key issues have been satisfied in accordance with an agreed-upon plan

- All required test resources people (including a designated test director), facilities, test articles, test instrumentation, and other enabling products have been identified and are available to support required tests
- Roles and responsibilities of all test participants are defined and agreed to
- Test contingency planning has been accomplished, and all personnel have been trained.

At the TRR and within the TRR data package, the Contract shall show that:

- Adequate test plans are completed and approved for the system under test
- Adequate identification and coordination of required test resources are complete
- Previous component, subsystem, and system test results form a satisfactory basis for proceeding into planned tests
- Risk level is identified and accepted by program/competency leadership as required
- Plans to capture any lessons learned from the test program are documented
- The objectives of the testing have been clearly defined and documented, and the review of all the test plans, as well as the procedures, environment, and the configuration of the test item, provide a reasonable expectation that the objectives will be met
- The test cases have been reviewed and analyzed for expected results, and the results are consistent with the test plans and objectives; test personnel have received appropriate training in test operation and safety procedures.

DATA ITEM DESCRIPTION

1. CDRL No.:

RE-12

2. Title:

ETU PRE-BUILD REVIEW

3. Reference:

SOW 1.5.1.4

4. Use:

To get Government agreement for the ETU Program – the scope and fidelity of the ETU, what testing will be done, and an attempt to define priorities ahead of time to aid in quicker decision making should there be cost, schedule, procurement, or facilities challenges that require an adjustment to the scope or fidelity of the ETU.

5. Preparation Information:

The content of this review package shall be as follows:

- a) Process flowchart
- b) Schedule
- c) Product description
- d) Configuration Item Identification List (CIIL)
- e) Computer Software Configuration Item (CSCI)
- d) Summary of any changes between ETU and flight configuration baseline, including, parts, materials, and processes.
- f) Detailed manufacturing schedules
- g) Manufacturing flow diagram including inspection, MIPs, and test points
- h) Drawing readiness
- i) Process readiness
 - i. List of drawings
 - ii. Procedures
 - iii. Machine availability and certification
 - iv. Process qualification
 - v. Training and personnel qualification
 - vi. Workmanship standards
- j) Product FMEA (Manufacturing risk mitigation)
- k) Special process list
- l) ESD control plan and cleanliness control
- m) Material availability
- n) Personnel availability
- o) Tooling readiness
- p) Facility readiness

- q) Parts status and qualification status on piece parts
- s) Status of lower-tier suppliers
- t) Status of prohibited material screening
- u) a table listing all in-process and end item procedures where Consent to Proceed (CTP), Consent to Break (CTB) and/or Consent to Integrate (CTI) decision points require Government approval.

The ETU Pre-Build Review shall be held as part of a scheduled program review meeting mutually agreed to by the Subcontractor and the Buyer in the baseline schedule; however, where schedule requirements dictate, the Pre-Build Review may be held at an earlier date. The Pre-Build Review will be considered approved when all outstanding action items are closed to the Buyer's satisfaction. In the event of disapproval or conditional approval, the required correction(s) shall be submitted to the Buyer for approval. Design review approvals in no way relieve the Subcontractor of responsibility for meeting the subcontract requirements.

DATA ITEM DESCRIPTION

1. CDRL No.:

RE-13

2. Title:

ETU TEST SUMMARY REVIEW

3. Reference:

SOW 1.5.1.5

4. Use:

To present the actual testing conducted compared to the plan, the results and any lessons of that testing, and any subsequent changes, either to the ETU and/or the flight design.

5. Preparation Information:

This data package shall address, as a minimum:

The ETU Test Summary Review Data Package shall address responses to action items generated at prior reviews.

The ETU Test Summary Review Data Package shall address the solutions to all problems encountered during the environmental test and validation program and the solution rationale.

The ETU Test Summary Review Data Package shall address any rework/replacement of hardware, regression testing, and test plan changes.

The ETU Test Summary Review Data Package shall address compliance with the test verification matrix.

The ETU Test Summary Review Data Package shall address measured test margins versus requirements.

The ETU Test Summary Review Data Package shall address instrument performance budgets including Beginning of Life to End of Life performance degradation estimates.

The ETU Test Summary Review Data Package shall address qualification/acceptance temperature margins.

The ETU Test Summary Review Data Package shall address any data that has been trended to identify compliance with specification, indicating a change or drift to the trend.

The ETU Test Summary Review Data Package shall summarize and analyze the monthly trend reports (CDRL SE-2) of trended instrument data, focusing on trends that have changed or drifted anomalously (see SOW paragraph 2.4.2).

The ETU Test Summary Review Data Package shall address total failure-free operating time of the item.

The ETU Test Summary Review Data Package shall address the number of cycles during testing of parts with finite lifespans.

The ETU Test Summary Review Data Package shall address the results of the final audit of any remaining drawing changes.

The ETU Test Summary Review Data Package shall address “could-not-duplicate failures” along with assessment of the problem and the residual risk that may be inherent in the item.

The ETU Test Summary Review Data Package shall address project assessment of any residual risk.

The ETU Test Summary Review Data Package shall provide an update from the ETU Pre-Build Review on shipping containers, monitoring/transportation/control plans

The ETU Test Summary Review Data Package shall address ground support equipment status.

The ETU Test Summary Review Data Package shall address post-test plans.

The ETU Test Summary Review shall address the plans for storage of the instrument, if required.

The ETU Test Summary Review Data Package shall include documentation verifying that the instrument meets contamination budgets and requirements specified in the Contamination Control Plan.

The ETU Test Summary Review Data Package shall include alignment data relative to the Instrument Boresight and the mounting surface datum(s) associated with the instrument.

DATA ITEM DESCRIPTION

1. CDRL No.:

RE-14

2. Title:INPUTS TO MISSION LEVEL REVIEWS

3. Reference:SOW 1.5.3.1, 1.5.3.2, 1.5.3.3

4. Use:

Inputs to mission level reviews. To evaluate the requirements, requirements flow-down, and the operational concepts and to validate the realism of the functional and performance requirements and their congruence with the system configuration selected to conduct the mission.

5. Preparation Information:

Inputs to the Mission Definition Review (MDR) Data Package shall contain all instrument information required to satisfy paragraph 3.2 (Criteria for Successful Completion) of GSFC-STD-1001A, Criteria for Project Flight Critical Milestone Reviews. The MDR Data Package shall discuss contractor system level requirements, rationale, and flow-down plans to lower level requirements.

The MDR Data Package shall show the allocation and traceability of requirements to major subsystems.

The MDR Data Package shall address any identified Single Point Failures.

The MDR Data Package shall cover requirements for data processing algorithms.

The MDR Data Package shall show how the current concept meets all Government-specified requirements including interface requirements.

The MDR Data Package shall discuss the preliminary operations concept of the instrument.

Results of Review – As a result of successful completion of the MDR, the system and its operation are well enough understood to warrant design and acquisition of the end items. Approved specifications for the system, its segments, and preliminary specifications for the design of appropriate functional elements may be released. Inputs to the Mission-Level

Reviews – Inputs to the mission-level reviews (PDR, CDR, PER, PSR) shall be a subset of the corresponding Instrument level reviews, with appropriate updates. Inputs to the System Integration Review (SIR) shall be a subset of the instrument PSR, with appropriate updates.

2.8 Software Data Items

DATA ITEM DESCRIPTION

1. CDRL No.:

SW-1

2. Title:

NPR 7150.2D TAILORED COMPLIANCE
MATRIX

3. Reference:

SOW 4.2

4. Use:

The NPR 7150.2D Tailored Compliance Matrix specifies compliance of the Contractor's processes with each Software Engineering Requirement (SWE) from the NPR that is levied by NASA on the Contractor, as indicated in the appendix to the SOW.

5. Related Documents:

NPR 7150.2D

6. Preparation Information:

The Tailored Compliance Matrix (template contained in SOW Appendix A) shall be completed based on the Contractor's institutional and program processes applied to the SMBA development work. This information will be used by NASA to assess the level of process compliance and the need for any waivers. The requirements levied in the matrix should not instigate new work or CDRL updates unless directed by NASA.

DATA ITEM DESCRIPTION

1. CDRL No.:SW-2
PLAN2. Title:

INSTRUMENT SOFTWARE DEVELOPMENT

3. Reference:SOW 1.5.1.12, 4.2.1
NPR 7150.24. Use:

Defines Contractor activities required to develop and manage all software, whether developed by the Contractor or by a third party.

5. Related Documents:

NPR 7150.2D

6. Preparation Information:

The Software Management and Development Plan shall describe processes and activities used in the development and testing of the various types of software being acquired, acknowledging the fact that not all software must have the same criticality level or process requirements. Topics to be included in the Software Development Plan are (as per NPR 7150.2D, SWE-102):

- a. Project organizational structure showing authority and responsibility of each organizational unit, including external organizations (e.g., Safety and Mission Assurance, Independent Verification and Validation (IV&V), Technical Authority, NASA Engineering and Safety Center, NASA Safety Center).
- b. The safety criticality and classification of each of the systems and subsystems containing software.
- c. Tailoring compliance matrix for approval by the designated Engineering Technical Authority, if the project has any waivers or deviations to this NPR.
- d. Engineering environment (for development, operation, or maintenance, as applicable), including test environment, library, equipment, facilities, standards, procedures, and tools.
- e. Work breakdown structure of the life-cycle processes and activities, including the software products, software services, non-deliverable items to be performed, budgets, staffing, acquisition approach, physical resources, software size, and schedules associated with the tasks.

-
- f. Management of the quality characteristics of the software products or services.
 - g. Management of safety, security, privacy, and other critical requirements of the software products or services.
 - h. Subcontractor management, including subcontractor selection and involvement between the subcontractor and the acquirer, if any.
 - i. Verification and validation.
 - j. Acquirer involvement.
 - k. User involvement.
 - l. Risk management.
 - m. Security policy.
 - n. Approval required by such means as regulations, required certifications, proprietary, usage, ownership,
 - o. Process for scheduling, tracking, and reporting.
 - p. Training of personnel, including project unique software training needs.
 - q. Software life-cycle model, including description of software integration and hardware/software integration processes, software delivery, and maintenance.
 - r. Configuration management.
 - s. Software documentation tree.
 - t. Software peer review/inspection process of software work products.
 - u. Process for early identification of testing requirements that drive software design decisions (e.g., special system level timing requirements/checkpoint restart).
 - v. Software metrics.
 - w. Content of software documentation to be developed on the project.
 - x. Management, development, and testing approach for handling any COTS, Government-off-the-shelf (GOTS), modified-off-the-shelf (MOTS), reused, or open source software component(s) that are included within a NASA system or subsystem.

Note: Verification includes:

- a. Identification of selected software verification methods and criteria across the life cycle (e.g., software peer review/inspections procedures, re-review/inspection criteria, testing procedures).
- b. Identification of selected work products to be verified.
- c. Description of software verification environments that are to be established for the project (e.g., software testing environment, system testing environment, regression testing environment).
- d. Identification of where actual software verification records and analysis of the results will be documented (e.g., test records, software peer review/inspection records) and where software verification corrective action will be documented.

Note: Validation includes:

- a. Identification of selected software validation methods and criteria across the life cycle (e.g., prototyping, user groups, simulation, analysis, acceptance testing, operational demonstrations).
- b. Identification of selected work products to be validated.
- c. Description of software validation environments that are to be established for the project (e.g., simulators for operational environment).

- d. Identification of where actual software validation records and analysis of the results will be documented (e.g., user group records, prototyping records, and acceptance testing records) and where software validation corrective action will be documented.

Additionally, the Contractor shall evaluate all flight software using software metrics. The metrics collected, trended, and presented monthly. Metrics shall include at a minimum:

- A. Number of flight software requirements and their change status
- B. Design/Code complexity index at CSU, Computer Software Components (CSC), and Computer Software Configuration Item (CSCI) levels
- C. Source code production rate estimates versus actuals
- D. Number of Software Change Requests/Problem Reports and their status
- E. Resource margins for Utilization of memory, CPU, I/O Bandwidth and Bus traffic
- F. Effort data (staffing profile) estimates versus actuals

Include an alphabetized list of definitions for abbreviations and acronyms used in this document. Include an alphabetized list of definitions for special terms used in the document, i.e., terms used in a sense that differs from or is more specific than the common usage for such terms.

Material that is too detailed or sensitive to be placed in the main body of text may be placed in an appendix or included as a reference. Include the appropriate reference in the main body of the text. Appendices may be bound separately, but are considered to be part of the document and shall be placed under configuration control as such.

DATA ITEM DESCRIPTION

1. CDRL No.:

SW-3

2. Title:SOFTWARE CONFIGURATION
MANAGEMENT PLAN3. Reference:

SOW 4.2.2

4. Use:

The Software Configuration Management Plan (CMP) describes the functions, responsibilities, and authority for the accomplishment and implementation of software configuration management to be performed during the software life cycle. This plan identifies the required coordination of software configuration management activities with other activities of the project. This information may be combined into an instrument-level CM Plan.

5. Related Documents:

NPR 7150.2

6. Preparation Information:

The Software CM Plan shall be prepared in accordance with the NPR 7150.2D, SWE-103, and shall contain, at minimum:

- a. The project organization(s).
- b. Responsibilities of the software configuration management organization.
- c. References to the software configuration management policies and directives that apply to the project.
- d. All functions and tasks required to manage the configuration of the software, including configuration identification, configuration control, status accounting, and configuration audits and reviews.
- e. Schedule information, which establishes the sequence and coordination for the identified activities and for all events affecting the plan's implementation.
- f. Resource information, which identifies the software tools, techniques, and equipment necessary for the implementation of the activities.
- g. Plan maintenance information, which identifies the activities and responsibilities necessary to ensure continued planning during the life cycle of the project.
- h. Release management and delivery.

DATA ITEM DESCRIPTION

1. CDRL No.:

SW-4

2. Title:SOFTWARE REQUIREMENTS
SPECIFICATION3. Reference:

SOW 4.2.3

4. Use:

The Software Requirements Specification specifies in detail each Software Element's requirements for a particular computer Software Configuration Item (CSCI), including functional and performance requirements, interface requirements, testing requirements, and security and safety requirements.

5. Related Documents:

NPR 7150.2

6. Preparation Information:

The Software Requirements Specification (SRS) shall be prepared in accordance with the NPR 7150.2D, SWE-109 (listed below). Alternatively, the Contractor may, with agreement from the Government, use an alternative industry standard SRS approach such as MIL-STD-498 or Institute of Electrical and Electronics Engineers (IEEE) standards.

SWE-109 specifies that the SRS shall contain:

- a. System overview.
- b. CSCI requirements:
 - (1) Functional requirements.
 - (2) Required states and modes.
 - (3) External interface requirements.
 - (4) Internal interface requirements.
 - (5) Internal data requirements.
 - (6) Adaptation requirements (data used to adapt a program to a given installation site or to given conditions in its operational environment).
 - (7) Safety requirements.
 - (8) Performance and timing requirements.
 - (9) Security and privacy requirements.
 - (10) Environment requirements.

- (11) Computer resource requirements:
 - (a) Computer hardware resource requirements, including utilization requirements.
 - (b) Computer software requirements.
 - (c) Computer communications requirements.
- (12) Software quality characteristics.
- (13) Design and implementation constraints.
- (14) Personnel-related requirements.
- (15) Training-related requirements.
- (16) Logistics-related requirements.
- (17) Packaging requirements.
- (18) Precedence and criticality of requirements.
- c. Qualification provisions (e.g., demonstration, test, analysis, inspection).
- d. Bidirectional requirements traceability.
- e. Requirements partitioning for phased delivery.
- f. Testing requirements that drive software design decisions (e.g., special system level timing requirements/checkpoint restart).
- g. Supporting requirements rationale.

DATA ITEM DESCRIPTION

1. CDRL No.:

SW-5

2. Title:

SOFTWARE DETAILED DESIGN DOCUMENT

3. Reference:

SOW 2.6, 4.2.4

4. Use:

The Software Detailed Design (SDD) Document describes the software design and operation for use by software maintenance team. It includes details of the architecture, structure, and organization of a particular Computer Software Configuration Item (CSCI), decomposing the top-level CSCI into Computer Software Components (CSC) and lower levels of units as appropriate. The SDD describes each unit of software in terms of its interfaces (input/output), data architectures, and processing (e.g., logic, algorithms).

5. Related Documents:

NPR 7150.2

6. Preparation Information:

Provide a system level design overview that contains:

- (a) Design Methodology
- (b) Design Overview
- (c) Design Studies
- (d) Design Issues
- (e) Hardware Interface

Provide a system design description that contains, at a minimum:

- (a) Subsystem Description for each subsystem
- (b) Software Description for each software component, including (as per NPR 7150.2D SWE-111):
 - (1) Description of how the software item satisfies the software requirements, including algorithms, data structures, and functional decomposition.
 - (2) Software item I/O description.
 - (3) Static/architectural relationship of the software units.
 - (4) Concept of execution, including data flow, control flow, and timing.
 - (5) Requirements, design, and code traceability.
 - (6) CSCI's planned utilization of computer hardware resources.

- (c) Software Interface Control Description for both software-to-software and software-to-hardware interfaces

Describe the system operations design, including:

- (a) Operations Scenarios
- (b) User-System Interface
- (c) Operations Environment and Facilities

DATA ITEM DESCRIPTION

1. CDRL No.:

SW-6

2. Title:

SOFTWARE TEST PLAN

3. Reference:

SOW 4.2.6

4. Use:

Provide overall view of the software acceptance test program detailing test philosophy objectives and rationale for all software testing and hardware/software integration activities planned for the program, this includes new software capabilities for SMBA such as RFI detection and hyperspectral. Provide descriptions for planned flight software tests.

5. Related Documents:

NPR 7150.2

6. Preparation Information:

The Software Test Plan shall include, at a minimum (as per NPR 7150.2D, SWE-104):

- a. Test Levels
- b. Test Types
- c. Test classes
- d. General test conditions
- e. Test progression
- f. Data recording, reduction, and analysis.
- g. Test coverage (breadth and depth) or other methods for ensuring sufficiency of testing
- h. Test personnel and participating organizations.
- i. Test descriptions

The test descriptions shall include, as a minimum (as per NPR 7150.2D, SWE-114):

- (1) Test identifier
- (2) System or CSCI requirements addressed by the test case
- (3) Prerequisite conditions
- (4) Test input
- (5) Instructions for conducting procedure
- (6) Expected test results, including assumptions and constraints
- (7) Criteria for evaluating results.

DATA ITEM DESCRIPTION

1. CDRL No.:

SW-7

2. Title:

SOFTWARE TEST PROCEDURES

3. Reference:SOW 4.2.6
CDRL SW-64. Use:

Provide listing of the software test procedures/scripts for review prior to the Test Readiness Review.

5. Related Documents:

NPR 7150.2

6. Preparation Information:

The test procedures shall include, as a minimum (as per NPR 7150.2D, SWE-114):

- a. Test preparations, including hardware and software
- b. Test descriptions (see separate CDRL item)
- c. Requirements traceability
- d. Identification of test configuration.

DATA ITEM DESCRIPTION

1. CDRL No.:

SW-8

2. Title:

SOFTWARE TEST REPORTS

3. Reference:

SOW 4.2.6

4. Use:

The Software Test Report is a record of the qualification testing performed on a CSCI, a software system or subsystem, or other software-related item. These test procedure results are provided for review following the test completion.

5. Related Documents:

NPR 7150.2

6. Preparation Information:

The test reports shall include, as a minimum (as per NPR 7150.2D, SWE-118):

- a. Overview of the test results:
 - (1) Overall evaluation of the software as shown by the test results.
 - (2) Remaining deficiencies, limitations, or constraints detected by testing (e.g., including description of the impact on software and system performance, the impact a correction would have on software and system design, and recommendations for correcting the deficiency, limitation, or constraint).
 - (3) Impact of test environment.
- b. Detailed test results:
 - (1) Project-unique identifier of a test and test procedure(s).
 - (2) Summary of test results (e.g., including requirements verified).
 - (3) Problems encountered.
 - (4) Deviations from test cases/procedures.
- c. Test log:
 - (1) Date(s), time(s), and location(s) of tests performed.
 - (2) Test environment, hardware, and software configurations used for each test.
 - (3) Date and time of each test-related activity, the identity of the individual(s) who performed the activity, and the identities of witnesses, as applicable.
- d. Rationale for decisions.

DATA ITEM DESCRIPTION

1. CDRL No.:

SW-9

2. Title:SOFTWARE VERSION DESCRIPTION
DOCUMENT3. Reference:

SOW 4.2.7

4. Use:

The Software Version Description Document (VDD) contains a letter describing what is being delivered, the actual executable software, and accompanying documentation. For all firmware media, formatted and numbered drawings are required with assembly numbers assigned to all loaded chips and drawing notes used to identify specific loading and test procedures, markings, etc.

5. Related Documents:

NPR 7150.2

6. Preparation Information:

The VDD shall include, as a minimum (as per NPR 7150.2D, SWE-116):

- a. Full identification of the system and software (e.g., numbers, titles, abbreviations, version numbers, and release numbers).
- b. Executable software (e.g., batch files, command files, data files, or other software needed to install the software on its target computer).
- c. Software life-cycle data that defines the software product.
- d. Archive and release data.
- e. Instructions for building the executable software, including, for example, the instructions and data for compiling and linking and the procedures used for software recovery, software regeneration, testing, or modification.
- f. Data integrity checks for the executable object code and source code.
- g. Software product files (any files needed to install, build, operate, and maintain the software).
- h. Open change requests and or problem reports, including any workarounds.
- i. Change requests and/or problem reports implemented in the current software version since the last Software Version Description was published.

DATA ITEM DESCRIPTION

1. CDRL No.:

SW-10

2. Title:INSTRUMENT SOFTWARE TEST READINESS
REVIEW (SWTRR) DATA PACKAGE3. Reference:

SOW 1.5.1.12, 4.2.9

4. Use:

Presents the description and results for the Software and System Integration/Test program.

5. Preparation Information:

The SWTRR shall show that the Contractor has adequately prepared for formal software acceptance testing to include, at a minimum, the check-out of test procedures, test cases, and requirements traceability.

This design review package shall address, as a minimum:

- A. All documentation as called for in the Software Development and Management Plan
- B. Test and Integration program descriptions and results
- C. Software test results
- D. Failure report summaries including status of action and rationale for closure
- E. As-built documentation summary

DATA ITEM DESCRIPTION

1. CDRL No.:

SW-11

2. Title:SOFTWARE ACCEPTANCE REVIEW (SWAR)
DATA PACKAGE3. Reference:

SOW 4.2.9

4. Use:

For review of all test data and designs for compliance against specification requirements, variances, mission operations requirements, etc.

5. Preparation Information:

The software acceptance review data package CDRL SW-11 may be provided together with Version Description Document CDRL SW-9

This data package shall address, as a minimum:

- A. Results of the functional and interface tests
- B. Malfunctions and corrective actions
- C. Reliability predictions
- D. Comparison of measured performance with requirements and discussion of the effect of any variance and waivers
- E. Mission operation constraints
- F. Safety requirements
- G. Maintenance and operation manuals
- H. Interface concerns, problems, and solutions
- I. Compatibility of instrument with spacecraft flight support equipment, ground support equipment and operational ground equipment

DATA ITEM DESCRIPTION

1. CDRL No.:

SW-12

2. Title:

SOFTWARE USER'S GUIDE

3. Reference:

SOW 4.2.8, 6.4.1.2

4. Use:

The Software User's Guide shall contain the information required to use the software, including detailed procedures and functionalities. The Guide shall describe the installation, build, and operating procedures for the flight software. This Guide shall provide flight software description, and the necessary on running and updating the instrument flight software.

5. Related Documents:

NPR 7150.2

6. Preparation Information:

The User's Guide shall include, as a minimum (as per NPR 7150.2D, SWE-115):

- a. Software summary, including: application, inventory, environment, organization, overview of operation, contingencies, alternate states, and modes of operation, security, privacy, assistance, and problem reporting.
- b. Access to the software: first-time user of the software, initiating a session, and stopping and suspending work.
- c. Processing reference guide: capabilities, conventions, processing procedures, related processing, data backup, recovery from errors, malfunctions, emergencies, and messages.
- d. Assumptions, limitations, and safety-related items/concerns or constraints.
- e. Information that is unique or specific for each version of the software (e.g., new and modified features, new and modified interfaces).

DATA ITEM DESCRIPTION

1. CDRL No.:

SW-13

2. Title:INSTRUMENT SOFTWARE DEVELOPMENT
FACILITY SPECIFICATION3. Reference:

SOW 4.2.5

4. Use:

Documentation of the development environment components shall be provided in a Sensor Software Development Facility Specification.

5. Related Documents:6. Preparation Information:

The Sensor Software Development Facility Specification shall be available for review by the Government, and shall include: functional and performance requirements for Instrument Ground Support Equipment/Special Test Equipment (GSE/STE) Software; GSE/STE Software verification and validation program details, including testing approach and methodology for ensuring that GSE/STE Software being developed or maintained satisfies functional and performance requirements defined in the GSE/STE requirements; the development facility integration and test program, including GSE/STE Software Test Descriptions, detailed descriptions of all qualification and acceptance tests, acceptance criteria for GSE/STE Software prior to and after loading, and test reports capturing the results of all qualification and acceptance tests for the GSE/STE Software; specifications for simulators, test equipment, and other facility components; and a GSE/STE User's Guide describing the installation, build, and operating procedures for the GSE/STE Software.

The Contractor shall include in the specification host development computer operating systems, high-level language compilers and debuggers, auto-code generator software systems, machine language emulators, and test scenarios and procedures. It includes the software in the instrument test environment simulators that model the detectors and mechanisms. It also includes development support software such as configuration management systems for tracking software and document versions.

DATA ITEM DESCRIPTION

1. CDRL No.:SW-14
PLAN2. Title:

INSTRUMENT SOFTWARE MAINTENANCE

3. Reference:

SOW 4.2.13

4. Use:

The Software Maintenance Plan shall contain the information required to sustain the software, including detailed procedures and functionalities. The Plan shall provide insight into the method, approach, responsibility, and processes to be followed for maintenance of software and its associated documentation.

5. Related Documents:

NPR 7150.2

6. Preparation Information:

The Software Maintenance Plan shall include, as a minimum (as per NPR 7150.2D, SWE-105):

- a. Plan information for the following activities:
 - (1) Maintenance process implementation.
 - (2) Problem and modification analysis.
 - (3) Modification implementation.
 - (4) Maintenance review/acceptance.
 - (5) Migration.
 - (6) Software Retirement.
 - (7) Software Assurance.
 - (8) Software Risk Assessment for all changes made during maintenance and operations.
- b. Specific standards, methods, tools, actions, procedures, and responsibilities associated with the maintenance process. In addition, the following elements are included:
 - (1) Development and tracking of required upgrade intervals, including implementation plan.
 - (2) Approach for the scheduling, implementation, and tracking of software upgrades.
 - (3) Equipment and laboratories required for software verification and implementation.
 - (4) Updates to documentation for modified software components.
 - (5) Licensing agreements for software components.

- (6) Plan for and tracking of operational backup software (e.g., backup flight software, backup to the primary operational software).
- (7) Approach for the implementation of modifications to operational software (e.g., testing of software in development laboratory prior to operational use).
- (8) Approach for software delivery process, including distribution to facilities and users of the software products and installation of the software in the target environment (including, but not limited to, spacecraft, simulators, Mission Control Center, and ground operations facilities).
- (9) Approach for providing NASA access to the software version description data (e.g., revision number, licensing agreement).

DATA ITEM DESCRIPTION

1. CDRL No.:

SW-15

2. Title:

INSTRUMENT DATA SETS

3. Reference:

SOW 4.2.14

4. Use:

This data will be used to support ground system development and end-to-end mission rehearsals.

5. Preparation Information:

The Contractor shall prepare and deliver sensor test data sets simulating 72-hour continuous science data.

2.9 Systems Engineering Data Items

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-1

2. Title:

ENGINEERING CHANGE REQUESTS,
DEVIATIONS, AND WAIVERS

3. Reference:

SOW 1.4.1.1, 2.0, 3.1

4. Use:

To facilitate the orderly processing of engineering change requests to specifications or requirements, changes to plans or procedures, specification deviations, waivers, to appropriate level of review, classification, and approval authority for disposition.

5. Preparation Information:

Consistent with the Contractor's Configuration Management Plan (CMP), the Contractor shall prepare, and submit Configuration Change Requests (CCR) for Government review. CCRs shall be either Class I or Class II. In addition to the change description, the CCR shall contain sufficient information in the form of attachments, drawings, test results, etc., to allow the Government to evaluate the total impact of the proposed change. The Government Contracting Officer may direct the Contractor to prepare CCRs under the "Changes" clause of the contract. For the purposes of this DID, a Class I CCR is a change that:

- a. Affects any Government Contract requirement, specification, or interface requirement.
- b. Affects schedules of end item deliverables to the Government.
- c. Impacts Government Furnished Equipment.
- d. Affects configuration to the extent that changes would be required to prior deliverables in order to maintain specified performance.
- e. Creates a new Single Point Failure possibility.

A change may be classified Class II when it does not fall within the definition of a Class I change as given above. Examples of Class II changes are:

- a. A change in documentation only (for example, correction of errors, addition or clarifying notes or views).
- b. A minor change in hardware (for example, substitution with an approved alternative material) which does not affect any item listed under Class I changes.
- c. Drawing changes that do not affect a baseline, interface, etc.

Class II changes normally do not require Government Configuration Control Board (CCB) approval unless they are written against Government CM- controlled documents.

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-2

2. Title:TREND PLAN/ANALYSIS, REPORTS, AND
OPERATIONS LOG3. Reference:SOW 2.7.2
IMAR4. Use:

Provides a list and a means of tracking critical engineering and performance parameters for the Instrument and across Instruments. Provides a log of Instrument operating hours. Starting at component acceptance testing and continuing during the system integration and test phases through the on-orbit commissioning phase, trend parameters are to be monitored for trends leading toward degradation of performance or reliability of the Instrument and out of family with other Instruments.

5. Preparation Information:

The Contractor shall develop a trending plan, analysis and log that includes at a minimum:

Trend List:

The list of parameters to be trended shall be subdivided by assembly or subassembly. A brief rationale for including the parameter shall be included. The list shall be coordinated with the Government prior to implementation. The trend list shall be revised with the on-orbit operational trend parameters prior to launch with Government Flight Operations Team input.

Trend Reports:

The trend data shall be graphed with clearly marked axes. The scale of the graphs shall be set such that trends can be clearly identified. The scale of each graph shall be tailored for each parameter for the best clarity. The scale of the graphs shall be readjusted with the range of the data in order to continue identification of trends.

Operations Log:

A log shall be maintained of the accumulated operating time of the Instrument. The log shall include the following information, as a minimum:

- A. Identification of hardware item
- B. Serial number
- C. Total operating time since assembly as a unit

- D. Total operating time since last failure
- E. Total additional operating time projected for the unit prior to launch
- F. Identification of key parameters being monitored
- G. Upper/lower spec tolerance limit for each parameter being monitored
- H. Observed value (in sequence) for the reporting interval
- I. Flight Software image/patch version or firmware version used in the subsystem

Contractor format is acceptable.

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-3

2. Title:

ENGINEERING DRAWINGS

3. Reference:

SOW 1.4

4. Use:

To provide layouts and engineering drawings to serve as the basis for technical discussions, evaluations, manufacturing, fabrication, assembly, test, operations, and maintenance. To satisfy some of the Observatory drawings requirements of the launch services provider.

5. Preparation Information:

The Contractor shall submit, upon request, engineering drawings used to procure, manufacture, assemble, integrate, test and control interfaces. Included in this engineering drawing package shall be all reference type drawings such as layouts, schematics, diagrams, mechanical drawings, electrical schematics, logic diagrams, and block diagrams. The logic diagrams shall cover the system, subsystem and component electronics and shall identify the signal inputs and outputs, internal signal flow, and the next level external connections.

Sketch type drawings shall not be used. Interface control drawings and applicable Instrument layouts shall include the stowed, extended, and critical intermediate positions of the moving mechanical assemblies and deployables with respect to fields of view and surrounding structure, components, or other hardware. All drawing changes and change notices are included under this requirement.

This delivery includes wiring diagrams. These wiring diagrams shall cover the system, subsystem, component electronics, and interface with electrical/mechanical ground support equipment. It shall identify each wire by its classification:

- Ground (differentiate between power return, shield, and chassis grounds)
- Signal
- Power
- Wire type, ratings, material, etc.
- Connector/Backshells
- Harness bundle braids and termination w/ backshell
- Harness between subsystems and GSE/STE

The diagrams shall trace each wire's runs identifying all path connections (by connector/pin number). Wire designators shall be clearly delineated for legibility.

An indentured drawing list (including drawings from subcontractors) shall be provided to the Government. An explanation of company procedures for locating drawings in this package shall be provided with this list.

All engineering drawings shall be delivered in the Contractor's designated format.

Drawing Trees shall be provided to the Government to show quick-reference relationships between drawings and next level of assembly. One set for the flight hardware, and one set for the GSE/STE is sufficient. The drawing trees shall also include parts lists, firmware, and PROM versions.

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-4

2. Title:SYSTEM PERFORMANCE VERIFICATION
PLAN AND REQUIREMENT VERIFICATION
MATRIX3. Reference:

SOW 2.4.2.1, 5.7.1

4. Use:

Provides the overall approach for accomplishing the verification program. Defines the specific tests, analyses, calibrations, alignments, hardware models, etc. that will demonstrate that the flight hardware complies with the mission requirements.

5. Preparation Information:

The SPVP shall:

- a. Provide an overview of the entire Instrument verification program,
- b. Flow performance requirements to all levels of assembly and describe the verification method for these requirements/specifications.

The SPVP describes the approach (test, analysis, etc.) that will be utilized to verify each requirement in the SMBA Instrument Performance Requirements Document and the Spacecraft to SMBA ICD. Verification will be to the BOL performance levels in the SMBA Instrument Performance Specification. If verification relies on measurements, tests, or analyses at lower (or other) levels of assembly, this dependence shall be described.

The SPVP shall document plans to verify by test or inspection the proper polarity, orientation, and position of all components (sensors, switches, and mechanisms).

This Plan shall include level of assembly, configuration of item, objectives, test phases, appropriate functional operations, and requirements for procedures and reports. Each analysis activity shall include objectives, a description of the mathematical model, assumptions on which the model will be based, required output, criteria for assessing the acceptability of the results, interaction with related test activity, and requirements for reports.

On-orbit verification tests shall be cross-referenced to the On-Orbit Commissioning Plan (CDRL OO-1).

The Final SPVP due 15 days prior to PER shall incorporate all review comments.

Requirement Verification Matrix (RVM)

The Requirement Verification Matrix (RVM) shall summarize the flow-down of system specification, Interface, Mission Assurance, and calibration/validation requirements verification. The RVM shall stipulate how each requirement (referenced by requirement ID) will be verified, the expected date for closure, the owner of the requirement, and summarizes current status of compliance/non-compliance with requirements. The SPVM shall list a summary description of each requirement, and a summary of the measured/analyzed/demonstrated performance of the system against each requirement. It shall show each SMBA Instrument Performance Requirements Document (as flowed to the SMBA Instrument Performance Specification), Spacecraft to SMBA ICD, IMAR, and Calibration/Validation Plan requirement reference source (to the specific paragraph or line item), the method of compliance, applicable procedure references, report reference numbers, etc. for each requirement set from the SMBA Instrument Performance Requirements Document, and IMAR. It shall show the flow-down of requirements verification through the sub-system (box/board) level.

The RVM shall include the level of assembly at which the requirement is verified and the environments in which the requirements will be verified.

The RVM shall trace requirements backwards to the next level above, i.e., a level 4 requirement shall be traced back to its level 3 parent, etc.

To allow transfer/tracking into DOORS, the RVM shall be prepared and delivered in DOORS compatible (electronic) format (Excel is acceptable), to include, at a minimum, the following content:

- _Source of Requirement (e.g., PRD, Inst to SC Interface Requirements Document (IRD), etc.)
- _Requirement Reference ID (e.g., PRD-1446, SCIRD-0001, etc.)
- _Requirement Text
- _Verification Method (e.g., test, analysis, etc.)
- _Test Plan
- _Test Procedure (number and title)
- _Verification Lead
- _IMS ID
- _Expected Date of Closure (Milestone)
- _Verification Status
- _Artifact report (e.g., Test Report, Analysis Report, etc.) (number and title)

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-5

2. Title:

VERIFICATION REPORTS

3. Reference:SOW 5.7.2
IMAR4. Use:

Provide summary of each integration and test result, conformance, non-conformance, and trend data. A Verification Report for all verification types indicated in the System Performance Verification Plan (Test, Analysis, Inspection, Demonstration) shall be generated. Multiple requirements may be verified in a single report.

5. Preparation Information:

The Contractor shall provide a verification report (s) that includes all instrument requirements.

Verification reports are required for all integration, test and environmental test activities commencing at component level testing through testing at the integrated instrument level.

Contents of these reports shall include, as a minimum:

- A. Summary of the test purpose and results of each activity and an assessment of the quality and acceptability of the item tested, including pass/fail criteria and performance against the criteria
- B. Summary of non-conformances occurring during the test and the resolution and corrective actions taken
- C. Summary of test flow, and test configuration, including GSE/STE.
- D. Trends in the performance of critical components
- E. Actual sequence of these operations including dates and times
- F. For thermal testing, tabulation of test target temperatures and actual test temperatures for all equipment and components along with the location of temperature Instruments used during the test
- G. For thermal balance testing, a tabulation of test prediction and actual temperatures and a tabulation of other pertinent targeted parameters vs. their actual test values, such as heater powers, heater place temperatures, solar intensity, etc.

- H. Across-reference to the test procedure number(s) or analysis used in the verification
- I. The document or a link to the document where the verification is recorded

Contractor format may be used for these reports as long as the required information is included.

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-6

2. Title:INSTRUMENT PERFORMANCE
SPECIFICATION3. Reference:

SOW 2.4, 2.6

4. Use:

To ensure that the Instrument design requirements flow correctly from the Instrument Performance Requirements Document and the other contract requirements documents. Because the requirements in the SMBA Instrument Performance Requirements Document are specified at End of Life (EOL), the Government will use the SMBA Instrument Performance Specification Beginning of Life (BOL) values to determine both conditional acceptance and on-orbit acceptance.

5. Preparation Information:

The SMBA Instrument Performance Specification shall delineate the design requirements for the Instrument that flow from the SMBA Instrument Performance Requirements Document and other contract requirements documents. It shall establish the top-level interface specification(s) placed on the Instrument that are consistent with the Instrument to Spacecraft Interface Control Document.

The lower level Instrument subsystem level specifications shall be directed and controlled by the SMBA Instrument Performance Specification. This specification shall divide and allocate the design responsibilities and interfaces between all of the elements of the Instrument.

The SMBA Instrument Performance Specification shall identify the Beginning of Life and End of Life performance for each requirement in the SMBA Instrument Performance Requirements Document. The SMBA Instrument Performance Specification shall supply the “degradation factor” in the form of an equation or other method, for each performance parameter that is expected to degrade over time.

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-7

2. Title:

ANALYSES AND MODELING

3. Reference:

SOW 2.4, 2.7.1

4. Use:

To provide reports of specific analyses for review.

5. Preparation Information:

All models shall be exchanged in formats that are compatible with those used by the spacecraft Contractor.

Sufficient model documentation shall be provided to describe how the significant thermal and structural characteristics are modeled.

Nodal number ranges and element number ranges should be defined for the individual components that make up the model.

The model will be utilized by the Spacecraft Prime Contractor for system-level modeling.

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-7A

2. Title:STRUCTURAL AND MECHANICAL
ANALYSIS AND MODELS3. Reference:

SOW 2.7.1.1

4. Use:

To document the analyses and data which demonstrates that all structural and mechanical subsystem performance requirements are satisfied. These models are to be used to analyze the structural and dynamic characteristics of each instrument. The models will be used to determine the instrument structural adequacy to withstand transportation, launch, and on-orbit loads. Also, they will be used to predict instrument structural resonant frequencies. The satellite/launch vehicle loads analysis will be used in the development and definition of the instrument/spacecraft interface loads. The model will be utilized by the Spacecraft Prime Contractor for system-level modeling.

5. Preparation Information:

This analysis and report shall contain the following in compliance with applicable requirements as specified in the ICD:

The structural analysis report(s) shall include, but shall not be limited to, the stress analysis of critical structural elements of the chassis, dynamic response of the printed wiring boards and chassis walls containing subassemblies, and deflection analysis. The report should contain, at a minimum: all assumptions and design criteria used in the analysis, descriptions of the analytical models, and results, conclusions, and recommendations. All detail calculations necessary to substantiate results; computer detailed calculations other than standard outputs are exempt. Decompression/venting analysis of the equipment shall also be included.

The Contractor shall verify the accuracy of the models with dynamic test data. The Contractor shall update the models to agree with the structural test results. A structural analysis using maximum equivalent loads shall be conducted on the instrument. The Contractor shall perform a structural analysis on each separately-mounted instrument component.

NASTRAN Finite Element Models (FEM) of the Instrument shall be delivered by the Contractor. A test-verified detailed finite element model shall be provided if the instrument

exceeds 23 kilograms and the fundamental resonant mode frequency is less than 50 Hz, as shown by analysis.

The Contractor shall provide documented verification that all structural and dynamic requirements of the equipment are met. The Contractor shall utilize analytical models and perform the simulations necessary to support and justify the design approach and solution. The structural analysis report(s) shall include, but shall not be limited to, the stress analysis of critical structural elements of the chassis, dynamic response of the printed wiring boards and chassis walls containing subassemblies, and deflection analysis. The analysis report should contain, at a minimum: all assumptions and design criteria used in the analysis, descriptions of the analytical models, and results, conclusions, and recommendations. All detail calculations necessary to substantiate results; computer detailed calculations other than standard outputs are exempt. Decompression/venting analysis of the equipment shall also be included.

The Contractor shall provide analysis and test data that describes the "swept" or deployed volume which accounts for all distortions and misalignments.

STRUCTURAL MODEL VERIFICATION PLAN

In support of the development of the test-verified structural models, the Contractor shall develop a verification plan which includes:

- a. Identification of the modeling techniques and analysis programs to be utilized,
- b. Analytical and testing techniques to be used to verify the analytical models and, where required, plans for revising models and repeating analyses based on verification results,
- c. A description of analyses to be performed, along with the objective, scope, and output of each analysis, and description of testing which will be used for model verification accompanied by associated schedules
- d. A compilation of required loads interface data (e.g., loads to components or deployables, or loads from spacecraft, etc.) and a schedule of need dates.

SPECIAL PREPARATION INSTRUCTIONS

All models, simulations, and/or databases required by this DID shall be delivered on electronic media compatible with the hardware platform on which it is to execute and include everything necessary to provide a fully functioning computer model whose execution requires only commercially available hardware and software:

- Modal results for reduced models shall agree with full-up model as follows:
 - o Fundamental frequency should agree to 3%.
 - o All remaining modes to 100 Hz should have frequencies to within 5%.
 - o Cross orthogonality check with full-up model must have diagonal terms greater than 0.95 and off-diagonal terms less than 0.05.

NASTRAN Model Verification

- If either analysis or a frequency verification test shows any significant mode below 50 Hz, (in a fixed base configuration) these modes shall be verified in accordance with the ICD.
- The analytical frequency predictions shall agree with dynamic test data to within 5 percent for the fundamental mode and to within 10 percent for all remaining significant modes to 50 Hz.

A cross orthogonality check between test and analytical mode shapes should have diagonal terms above 0.9 and off diagonal terms below 0.1 for all significant modes up to 50 Hz.

Instrument alignment and stiffness: This data item shall document the analyses, tests and test data required to verify all instrument alignment and stiffness requirements. The document shall also address instrument mechanical design verification with respect to dynamic interactions with the launch vehicle.

Instrument clearances: This data item shall document all instrument critical clearances and loss of clearances, with the associated analyses. This shall include dynamic loss of clearance between the observatory and the launch vehicle, and between various observatory elements. Clearance loss shall be determined for critical clearances throughout all mission phases, including launch, on-orbit deployments, and on-orbit operations. Components contributing to each clearance loss, such as manufacturing and assembly tolerances, clearance loss due to observatory dynamics and thermal gradients, as well as observatory insulation, harnesses, and grounding provisions, shall be quantified. Analytical and measured data used for clearance verification shall be provided.

Mechanical performance: This data item shall fully document the tests and analyses performed to verify the performance of the instrument mechanisms and deployment devices. Analyses and test data used to verify the instrument structural performance and workmanship shall also be provided. This document shall also address any methods utilized during fabrication and assembly which verify workmanship of the observatory structure and mechanical components.

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-7B

2. Title:

THERMAL ANALYSIS MODELS

3. Reference:

SOW 2.7.1.2

4. Use:

To document the detailed and reduced thermal analyses used in the design of the instrument. Several analysis reports and the instrument thermal models themselves are required to be delivered as noted in the table of CDRL deliverable models and reports. Additionally, any thermal models developed to support instrument environmental testing (e.g., representations of thermal chambers, thermal targets, helium refrigeration systems, etc.), design trade/parametric studies are required to be delivered as they are developed.

The instrument reduced thermal model shall be integrated with the observatory thermal model ('observatory' = spacecraft + instruments). The observatory thermal model will be utilized by the Spacecraft Prime Contractor to characterize observatory thermal performance. The observatory (a.k.a. 'satellite') thermal model will be used to calculate radiation exchange factors, environmental heating rates (solar, albedo, earth infrared) during various phases of the mission including launch, ascent, on-orbit operations, all survival modes, and including instrument calibration maneuvers. The observatory thermal model will be used to characterize the thermal response of the observatory for all spacecraft states and instrument modes over the entire length of the mission. The observatory thermal model will also be used to simulate the thermal environments encountered during the observatory environmental test campaign.

5. Preparation Information:

The analyses shall contain, as a minimum, the following: all pertinent assumptions used for analysis, description of the thermal model with associated boundary conditions, all detailed calculations needed to substantiate the results (if used), the temperature and mechanical effects on the unit in thermal vacuum (if required) versus ambient pressure, and results, conclusions, and recommendations. References/measurements shall be cited to support assumptions about interface conductances, and optical and thermophysical properties.

The detailed thermal analysis shall demonstrate that all internal parts meet established thermal design criteria, based upon program de-rating or electrical performance requirements. The associated reports shall describe in detail the various analysis cases (e.g., launch, cold survival, worst case hot and cold operational, nominal science, etc.). This description shall include:

- a. List of all parameters and assumptions
- b. Relationships between various model, or sub-models, and how boundary conditions are obtained
- c. Detailed calculations and descriptions of computer models and analyses
- d. Results of the final analyses including predicted thermal performance for worst case extremes and design margins

Instrument Thermal Model Requirements (Common to Detailed and Reduced Models):

The Instrument Contractor shall provide both a detailed thermal model and a reduced thermal model using ThermalDesktop (geometry) and SINDA/FLUINT (math model). Thermal models shall not employ SINDA/FLUINT INCLUDE or INSEINSERT files without obtaining Project approval.

The thermal models shall include an adequate level of detail to predict, under worst case hot and cold environments, all critical temperatures, including those that drive operational and survival temperature limits and heater power. Worst case conditions shall include variations in season, orbit selection, orbital time, and environmental flux parameters (seasonal and spatial) and a rational combination of the effects of design tolerances, fabrication uncertainties, material differences and degradation due to aging. See PRD Table 6.4.7.2.1-1 for cold and hot worst case environmental parameters. The instrument thermal models shall also be used to support thermal distortion studies, pointing budget assessments and/or power budget activities as required. For both reduced and detailed thermal models, the geometry used to represent instrument radiators, apertures, optical elements, mounting feet, and any other components identified as 'critical' in the thermal model shall be congruent with the geometry found in the mechanical/CAD model. Instrument thermal models may deviate from their mechanical counterparts to account for soft goods, such as multi-layer insulation blankets, or to intentionally exclude unnecessary features.

For each flight temperature sensor, thermostat, heater, or other active thermal control element (e.g., thermoelectric cooler), a thermal model node(s) shall be used to represent the corresponding hardware location.

For purposes of this DID, instrument 'critical components' are defined as comprising the hardware elements cited in the mechanical/CAD model and the thermal model node(s) shall be used to represent the corresponding hardware location.

Instrument thermal models shall employ closed volumes and/or ThermalDesktop Radiation Analysis Groups to separate radiative exchanges involving interior instrument surfaces from radiative exchanges involving exterior instrument and/or spacecraft surfaces. It is recommended that radiative exchanges between the instrument base plate and the spacecraft, unless intentionally retained, be zeroed by specifying the base plate surface opposite the spacecraft interface as 'Active None' (ThermalDesktop feature).

Models shall use conservative property values for conduction, absorption, emission and MLI effective emittance, and consider contact resistance. Over time, some of these conservative

values are expected to give way to direct measurements and model correlation findings. Analyses should consider beginning of life and end of life properties for external coatings.

All instrument thermal models shall be capable of steady-state and transient analyses. To satisfy the latter portion of this requirement, thermal models must accurately account for the measured/anticipated instrument mass.

Instrument thermal models shall use the following units: centimeters, kilograms, Joules, seconds, and degrees Celsius.

Thermal models shall be updated as necessary to reflect significant design changes during the design phase of the instrument, and analyses used to support such design changes shall be revisited using the test-correlated detailed thermal model on completion of instrument thermal balance testing.

Any thermal models developed to support instrument environmental testing (e.g., representations of thermal chambers, thermal targets, helium refrigeration systems, etc.), design trade/parametric studies are required to be delivered as they are developed (the Instrument Contractor is free to use any simulation software for such analyses).

Instrument Detailed Thermal Model Requirements:

Instrument detailed thermal models, constructed using ThermalDesktop and SINDA/FLUINT, shall be provided by the Instrument Contractor to facilitate assessments/reviews of the design, test, and qualification of the instrument thermal control architecture by Project subject matter experts.

The Instrument Contractor shall correlate the detailed thermal model to instrument level thermal balance tests. The correlated model, when subjected to energy balances simulating instrument thermal balance test conditions to within 5%, shall predict all critical component temperatures to within 3C of values measured testing.

The Spacecraft Prime Contractor will provide the Instrument Contractor either the observatory thermal model, or requisite portions of the observatory thermal model, for integration with the instrument detailed thermal model to support assessments of the integrated on-orbit thermal environments and spacecraft IR backloads. The instrument-to-spacecraft interface conductance will also be provided to the instrument contractor, if the mounting hardware is the responsibility of the Spacecraft Prime Contractor. The Instrument Contractor shall integrate its instrument detailed thermal model with the aforementioned observatory thermal model (or portions thereof) to perform stand-alone thermal analyses to characterize instrument on-orbit thermal performance and to assess IR backloads (a.k.a. 'accommodation loads') from the satellite on the instrument.

Instrument Reduced Thermal Model Requirements:

Instrument reduced thermal models, constructed using ThermalDesktop and SINDA/FLUINT, shall be provided by the Instrument Contractor for use by the Spacecraft Prime Contractor in defining each instrument/spacecraft thermal interface. The reduced thermal model shall be integrated into the observatory thermal model by the Spacecraft Prime Contractor for observatory thermal analyses.

All models shall be fully documented to permit ease of use by other contractors as deemed appropriate by the Project.

The reduced thermal model shall be as small as is reasonably possible. The total number of surfaces is dependent on geometric complexity and the maximum acceptable surface count shall be coordinated with the Spacecraft Prime Contractor and the Project.

All reduced thermal model SINDA/FLUINT entities shall reference SINDA/FLUINT Submodel names agreed to by the Instrument Contractor and the Spacecraft Prime Contractor. Once the instrument detailed thermal model correlation is complete, the reduced thermal model shall be updated to reflect any corrections made to the detailed thermal model. In creating the observatory thermal model, the Spacecraft Prime Contractor will place the instrument reduced thermal models in accordance with the appropriate mechanical interface document (MICD). To support this effort, the Instrument Contractor shall specify the precise relationship between the reduced thermal model geometry and the instrument-to-spacecraft MICD.

When the post-correlation reduced and detailed models are subjected to the same environmental heat loads and internal dissipations, the reduced thermal model shall reproduce detailed thermal model temperature predictions for all critical components to within 3C. When the reduced and detailed models are subjected to the same environmental heat loads and internal dissipations, the reduced thermal model shall reproduce detailed thermal model instrument- spacecraft heat flows to within 5%.

Instrument Thermal Model Documentation:

The following information and data shall be supplied in fulfillment of the requirement of a deliverable instrument thermal model:

- a. Description of the model, any special modeling features used, and the rationale for the modeling methodology.
- b. Illustrations/diagrams of the energy balances, also known as 'heat maps,' simulated by the thermal model.
- c. A summary of the organizational scheme used to group thermal model entities to represent instrument components and/or modules (e.g., an optical module might be identified by using a distinct SINDA/FLUINT submodel, a specific set of node/element IDs, and/or through the use of ThermalDesktop Groups and Layers).
- d. ThermalDesktop verification plots showing the locations of thermal blankets, radiators, apertures, and thermal control hardware.

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- e. A description of any thermal control hardware (e.g., radiators, thermal blankets, heaters, temperature sensors), apertures, optical elements, conductive interfaces (e.g., mounting feet), and any other components identified as ‘critical’ hardware and the key parameters associated with this hardware. A 400 in² aluminum radiator, for instance, might be covered with a 5 mil thick silver Teflon film and have a mass of 1kg. Or, a titanium instrument flexure might have a 0.3 in² minimum section, 0.25 W/K conductance and be secured to the spacecraft mounting plate by 2, M5 bolts. The thermal model correlation effort shall focus on adjusting the detailed thermal model to matching test data for these items.
 - f. Any instrument component represented only by a node(s) (i.e., there is no corresponding surface/solid geometry in the thermal model) shall be identified (e.g., node number 1967), described (e.g., fold mirror) and characterized (i.e., predominate material, mass, dissipation) in the associated thermal model report(s). Such entities might be found in SINDA/FLUINT INCLUDE or INSERT files called by the ThermalDesktop Case Manager.
 - g. A table describing the estimated, or measured, heat dissipated by the instrument for all environmental test or flight modes and physical configuration. If the dissipation for an instrument with a staring telescope is different than it is with scanning telescope, for instance, thermal dissipation estimates are required for both scenarios.
 - h. A table listing beginning-of-life and end-of-life assumed and/or measured optical properties (i.e., solar absorptivity and IR emissivity).
 - i. A table listing thermophysical properties (e.g., density, specific heat, conductivity) of the materials used to create instrument components.
 - j. References shall be provided to support all assumptions regarding optical properties, thermophysical properties, multilayer insulation performance and joint conductances.
 - i. Instrument Thermal Analysis report shall describe in detail the various analysis cases (e.g., launch, transfer orbit, on-orbit, on orbit hot and cold cases, etc.). This description shall include:
 - a) List of all parameters and assumptions used
 - b) Interrelationships between various models and how boundary conditions are obtained
 - c) Detailed calculations and descriptions of computer models and analyses
 - d) Results of final analyses including predicted thermal performance over worst case extremes and design margins
 - e) General outline of the report should include
 - a. Purpose
 - b. Related documentation
 - c. Requirements Verification
 - i. Which requirements are verified through this report
 - d. Thermal Model Description
 - i. Component Models
 - ii. Thermal Optical Properties
 - iii. Thermophysical Properties
 - iv. MLI Effective Emissivity
 - v. Orbits
 - vi. Component Power Dissipations
 - vii. Internal and External Thermal Isolations

- viii. Spacecraft interfaces
- ix. Heater Voltages
- e. Thermal Analysis Results
 - i. Operational Case Temperature Predictions
 - ii. Survival, Decontam, and Safehold Cases Temperature Predictions
 - iii. Temperature Margins
 - iv. Thermal Interface Rate of Change Predictions
 - v. Thermal Gradients (if required)
 - vi. Heater Power Predictions
 - vii. Heat Flow Predictions (Heat Map)
 - viii. Launch Analysis
 - ix. Any other special analysis not covered in sections above

Additionally, for the instrument detailed thermal model:

- a. The Instrument Contractor shall submit to the Project in advance of the instrument level test readiness review:
 - i. A list of critical components to be used in correlating the instrument detailed thermal model.
 - ii. A list of temperature sensors used to guide the instrument thermal vacuum environmental tests.
- b. A report detailing the detailed thermal model correlation to instrument thermal balance tests shall be provided. This correlation report shall include a sequential list of changes made to correlate the detailed thermal model, the reason for each change and the principal impact of each change on model performance.
- c. Flight predictions (e.g., heater duty cycles, temperatures of critical components and performance margins for other active thermal control elements) made using the correlated detailed thermal model shall be reported to the Project to demonstrate the instrument thermal control system shall meet mission performance requirements.

Additionally, for the instrument reduced thermal model:

- a. Sample runs characterizing worst case hot and cold operating environments and worst case hot and cold survival environments shall be provided.
- b. For the cases anticipated by the worst case hot and cold operating environments and worst case hot and cold survival environments, tables comparing detailed and reduced thermal model temperature predictions and heater duty cycles shall be provided.
- c. For each node used to represent a flight temperature sensor a temperature correction shall be provided.

Thermal Model Delivery Schedule:

Delivery	Activity/Event Supported	Need Date
Instrument DTM 1	Instrument Critical Design Review (ICDR)	15 days before ICDR
Instrument RTM 1	Observatory Critical Design Review (OCDR)	90 days before OCDR

Instrument DTM 2 + TVAC GSE	Instrument TVAC Test Readiness Review (ITRR)	90 days before ITRR
Correlated As-tested Instrument DTM 3 + TVAC GSE	Instrument model correlation review	90 days after instrument TVAC thermal balances
Correlated Instrument DTM 4 - Flight Cases - Launch/Ascent/Deploy - Science calibration maneuvers	Pre Ship Review (PSR)	40 days before PSR
Correlated Instrument RTM 2	Pre Ship Review (PSR)	20 days before PSR

Note: Instrument thermal model deliveries are expected to support all physical configurations (e.g., scan drum stowed, cooler door deployed) and/or thermal environment (e.g., worst-case hot flight sun-pointing survival, TVAC cold thermal balance) studied.

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-7C

2. Title:

STRESS ANALYSIS

3. Reference:

SOW 2.7.1.3

4. Use:

It shall serve as the source for design loads, stresses, deformations, margins of safety and information which defines the structural materials such as alloy type, strength, heat treatment, hardness, chemical treatment, finish, and other physical properties that have an influence on structural analysis.

5. Preparation Information:

The Contractor shall provide a detailed stress analysis and margins of safety calculations for all components of the instrument. Margins of Safety shall be summarized in tabular form, providing a description of each structural element, its critical loading condition, including thermal and mechanical loadings, failure mode, margin of safety, and reference to the detailed analysis. Primary load-paths shall be identified, and all simplifying assumptions clearly stated. If separate finite element models are developed for use in analyzing structural components, these models shall also be documented and included in the final stress report. The documentation shall also include any parametric studies and analyses directed toward minimizing component loads and stresses.

All important structural requirements shall be assembled and provided in conjunction with the stress analysis report. It shall serve as the source for design loads, stresses, deformations, margins of safety and information which defines the structural materials such as alloy type, strength, heat treatment, hardness, chemical treatment, finish, and other physical properties that have an influence on structural analysis.

Stress Report Updates

The stress analyses shall be updated to verify margins of safety using the design and final coupled flight loads analysis results. The stress analyses shall be updated and provided using the verification coupled flight loads analysis results, to verify that modifications to the structural models to achieve correlation with modal test data and hardware modifications encountered during observatory manufacturing and integration maintain positive margins of safety.

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-7D

2. Title:RADIATION HARDNESS AND
SURVIVABILITY ANALYSIS3. Reference:

SOW 2.7.1.5

4. Use:

The following analyses demonstrate that the hardware meets the applicable H&S requirements defined in the equipment specification:

(1) Dose shielding Analysis – This analysis shall determine the worst case total ionizing doses at the semiconductors within the unit to a level of detail sufficient to support the Piece Parts Radiation De-rating and Margins Analysis. The equivalent shielding afforded by the host spacecraft and other structure adjacent to the unit will be provided by the Government to the Contractor for use in this analysis.

(2) Piece Part De-ratings and Margins Analysis – This analysis shall use the results of the Dose Shielding Analysis to determine the dose levels to which unit semiconductor part parameters must be de-rated to meet circuit performance requirements with margin. The de-rated parameter values shall be established at a dose level with sufficient margin above the received dose in accordance with the hardness assurance requirements of equipment specification. These radiation de-rated parameters shall be combined with other appropriate de-ratings for age and temperature and used in the worst-case circuit analysis.

(3) Single Event Effects Analysis – This analysis shall verify that the unit meets the single event upset and damage requirements of the specification. Single event upset rates for parts shall be obtained from existing literature or new tests and used to determine the upset rates and modes for the unit. This analysis shall also demonstrate that no catastrophic modes occur as the result of a single event heavy ion or proton hit. Contractor analysis may be limited to SET analysis if the parts used are immune to other SEE's; Contractor shall submit documentation that validated the part's immunity.

5. Preparation Information:

To be provided in Contractor preferred electronic format.

A radiation analysis report shall be provided that includes radiation effects/degradations on electronic performance. This radiation analysis and report shall include radiation

effects/degradations on components, coatings, and on electronic performance, Dose Mapping analysis, Single Event Effects analysis, Displacement Damage analysis parts total dose analysis and test data supporting these analyses. Transient radiation effects that are not distinguishable from noise in the instrument output shall be included as a noise component in signal-to-noise ratio analyses and compliance evaluations. Transient radiation effects on thermistor temperatures used in emissive band calibration shall be analyzed, including impacts on radiometric calibration.

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-7E

2. Title:

SHOCK ANALYSIS

3. Reference:

SOW 2.7.1.3

4. Use:

The analyses demonstrate that the hardware meets the applicable shock requirements defined in the Spacecraft to SMBA ICD.

5. Preparation Information:

To be provided in Contractor preferred electronic format.

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-7F

2. Title:

CONTAMINATION ANALYSIS REPORT

3. Reference:

SOW 2.5

4. Use:

To document contamination analysis, as specified in the Instrument Contamination Control Plan (CDRL MA-30), for Government review.

5. Preparation Information:

The primary focus of the instrument internal mass transport analysis is an understanding of the potential contamination sources within the instrument, as well as a list of surfaces sensitive to contamination depositions. The results should include a list of the deposition onto sensitive surfaces (usually cold surfaces) for the duration of mission life as well as the temperature history of these elements.

A list of the surfaces of concern as contamination sources should also be presented as a result. This allows the engineering team to evaluate the design to see if there are design changes which could limit these effects (lower source temperatures, baffles, sealed areas...). The results can also help determine the necessary levels of bakeout required to insure proper performance of the instrument. In addition, a thorough understanding of the outgassing paths within the instrument can help understand contamination aspects of potential flight problems that may be contamination related.

Instrument level TV test outgassing results will further confirm or refine these models.

Information required for NEON to perform analysis:

If the instrument vendor is unable to perform the necessary mass transport analysis, the NEON mission may be able to provide support. If an agreement can be reached to allow this assistance, the following information is required for analysis to be performed. If this information is not provided, NEON will not be able to adequately perform the analysis.

A STEP and/or ThermalDesktop file for geometry is required. If these are not available, up to date drawings may be able to be used to develop this model however this will delay the performance of the analysis. In addition, the material used to build the hardware and the

materials properties are also required. Materials selection and location is a key component in determining where contaminants will move.

A file of the temperature predicts for a typical hot and a cold case for each node including bus, instruments, and other exposed hardware. Each component must specify radiator location(s) and vent size / location for all hardware. The maximum amount of contaminants allowed onto sensitive surfaces and apertures for the duration of mission life in order to satisfy throughput requirements for optics is also required.

For internal hardware analysis, a reasonably detailed geometry file for internal layout of the component (how many cavities inside the instrument, baffles, venting path, and vent location to outside, identify contamination sensitive surfaces), typical operating temperatures for all of the nodes, and outgassing rates of each component are all required.

Requirements for chamber bakeout performance analysis:

If the analysis is to determine the bakeout performance criteria for a hardware outgassing certification, even more information is required. In addition to all of the above data, the layout and operating parameters of vacuum chamber are also required. A detailed layout of the arrangement of the hardware within the chamber is essential. Temperatures of all chamber components, and locations of all vents and cold traps are also required. The location and operating temperature of the Thermal Quartz Crystal Microbalances (TQCM) used to monitor the bakeout/certification must also be specified. In addition, the chamber pumping efficiency must also be determined by the vendor.

The chamber pumping efficiency is a function correlating the amount of contaminants which are pulled out of the chamber by the pump and/or cold traps versus the amount of contaminants which will be deposited on the TQCM head. This function is usually calculated by performing a chamber background bake under two different conditions. For the first portion of the test, the background should be taken with no cold traps operating. Once at least 8 hours have passed under these conditions, a cold trap of well-known temperature and area should be run for at least 8 hours. This data is required to then calculate the pumping efficiency for the chamber analysis.

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-7G

2. Title:ON-ORBIT POINTING & JITTER ANALYSIS

3. Reference:SOW 2.7.1.3

4. Use:

Error Sources:

1) Instrument Thermal Distortion and Jitter

Provide an instrument level model and show the expected angular jitter and linear acceleration disturbances created by any Instrument mechanisms or moving parts as a function of frequency, based on either component specifications or direct measurements, and mechanical analysis, inclusive of any instrument borne disturbance sources with accompanying descriptions of application (i.e., model location/source application points, grid point, apply torque/force).

2) Line-of-Sight Error

a) Line-of-Sight Sensitivity

Provide an instrument level model and analysis of the combined effects of mechanical jitter from all spacecraft and Instrument sources on the instrument line of sight. Include the effects of any jitter suppressing mounting fixtures. Spacecraft jitter sources will be provided by the Government. The model shall also be capable of utilization for assessment of all pertinent thermally or mechanically induced distortion effects such as, but not limited to: worse case temperature profile distribution, boundary distortion effects, hydroscopic desorption, etc.

The model will be utilized by the Spacecraft Prime Contractor for system-level modeling.

5. Preparation Information:

Pointing & Jitter Performance

1) Instrument Jitter

Show the expected angular jitter and linear acceleration disturbances created by any Instrument mechanisms or moving parts as a function of frequency, based on either component specifications or direct measurements, and mechanical analysis.

2) Line-of-Sight Pointing

Provide an analysis of the combined effects of distortion due to all mechanical and thermal error sources.

3) Line-of-Sight Jitter

a) Line-of-Sight Sensitivity

Provide an analysis of the combined effects of mechanical jitter from all spacecraft and Instrument sources on the instrument line of sight. Include the effects of any jitter suppressing mounting fixtures. Spacecraft jitter sources will be provided by the Government.

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-7H

2. Title:

EMI/EMC ANALYSIS REPORT

3. Reference:

SOW 2.7.1.4

PRD 5.7.5, 6.2, 7.3

MIL-STD-461G

4. Use:

To document the EMI/EMC analyses used in the design of the instrument, the Contractor shall perform EMC/EMI analysis and/or test in accordance with MIL-STD-461G (as defined in the Instrument Performance Requirements Document) to verify the hardware meets the performance requirements of the equipment specification.

5. Preparation Information:

To be provided in Contractor preferred electronic format.

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-7I

2. Title:HYPERSENSPECTRAL CAPABILITY ANALYSES

3. Reference:SOW 2.7.1.8

4. Use:

To document the characteristics and performance of the SMBA instrument's hyperspectral capability.

5. Preparation Information:

The delivery shall include:

1. A description of the hyperspectral frequency configuration:
 - a. Identification of which bands and frequency ranges have hyperspectral capability
 - b. The number of channels for each hyperspectral band
 - c. The start and stop frequencies, the center frequency and bandwidth of each hyperspectral channel
 - d. Performance trades and studies used to select the hyperspectral configuration, including rationale on the selected configuration
2. The performance characteristics of the hyperspectral channels
 - a. The SRF of each channel, including a description of which passbands are combined to form the channel (a reference to CV-10 SRFs is acceptable).
 - b. The raw data used to derive the SRF along with the frequencies associated
 - c. The description of the SRF measurement test that provides details about the test procedure (set up and instrumentation), the environmental conditions, the data generated, the data analysis and the results obtained.
 - d. A description of the algorithms that combine the channels including rationale; limitations and specifications; mathematical description; a performance analysis; and assumptions such as the data sets used for simulating the environmental conditions
 - e. The start and stop frequencies, the center frequency stability, calibration accuracy, and NEDT of each channel
3. Observatory- and Mission-level Considerations
 - a. Flexibility in the frequency plan, including capabilities to update the frequencies pre- and post-launch
 - b. Power consumption impacts

- c. Thermal impacts
 - d. Data rate impacts on downlink
 - e. Data storage impacts
- 4. Hyperspectral subject matter expert inputs
 - a. Description of the signal flow and digitization, including Fast Fourier Transforms, binning, windowing, and similar functions as applicable
 - b. A description of the key contributing hardware elements and their functions
 - c. A description of the software
 - d. Digitization errors, including how closely a digitized signal matches the analog signal

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-7J

2. Title:RADIO FREQUENCY INTERFERENCE
DETECTION CAPABILITY ANALYSIS3. Reference:SOW 5.2, 5.3
MIL-STD-461F4. Use:

To document the capability for the SMBA instrument to detect radio frequency interference from anthropogenic RF sources in the SMBA detector frequency range. Because the frequency range of the instrument detector is near the range used for RF communications it is important for the instrument to be able to identify and flag RF interference that may contaminate science data measurements. This analysis documents the methods and capabilities the SMBA instrument will use to identify RF interference.

5. Preparation Information:

1. The delivery shall consist of the description of the Radio Frequency detection capability of the SMBA instrument covering both the hardware and software aspects for each of the bands identified in PRD Table 4.2.1. It shall include:

- The number of channels for each band
- The bandwidth of each channel
- The Radio Frequency Interference (RFI) detection algorithms chosen for each band
- The implementation of the RFI detection algorithms

For each of the selected RFI detection algorithm, the delivery shall consist of:

- An overview of each algorithm including the objective or purpose of the algorithm
- The rationale for each algorithm and identification of any specifications and limitations
- A description of each algorithm's inputs and outputs including any external ancillary data (i.e., thresholds) used by the algorithm
- A mathematical description of each algorithm
- An implementation of the algorithm in source code and identification of the Operating System that the source code was built on.

- A performance analysis of the algorithm that provides metrics on the algorithm including its detection sensitivity, its probability of detection and its false alarm rate as function of RFI characteristics
- A description of the methods used to validate each algorithm and description of the validation results

A test report that describes the procedure used to test the SMBA instrument's RFI detection capability for each of the bands identified in PRD Table 4.2.1 shall be delivered. This test report shall include a description of:

- The RFI test set up and instrumentation
- The RFI characteristics tested
- The RFI detection algorithms tested
- The data sets generated by the test of the RFI detection algorithms
- Test Objectives
- Test Methods
- Assumptions
- Test Equipment Identification
- Environmental and/or other conditions to be maintained
- Test durations
- Test configurations
- Sequence of events
- Step-by-step instructions
- Data Recording/Output Format Requirements
- Data Recording Forms and Tables
- Analysis techniques
- Expected results
- Pass/Fail Criteria
- Test Termination Procedure

A test report that describes the procedure used for the RFI test cases reproducing what is currently observed from both terrestrial and space-based RFI sources shall be delivered. This test report shall include a description of:

- The RFI test set up and instrumentation
- The RFI characteristics tested
- The RFI detection algorithms tested
- The data sets generated for the test of the RFI detection algorithms
- Test Objectives
- Test Methods
- Assumptions
- Test Equipment Identification
- Environmental and/or other conditions to be maintained
- Test durations
- Test configurations
- Sequence of events

-
- Step-by-step instructions
 - Data Recording/Output Format Requirements
 - Data Recording Forms and Tables
 - Analysis techniques
 - Expected results
 - Pass/Fail Criteria
 - Test Termination Procedure

2. For each analysis required to validate the RFI detection capability tests, the procedure for each analysis shall include:

- Analysis Objectives, i.e., short description of the item or items being validated by this analysis
- Analysis Methods
- Assumptions
- Applicable Documents and Software
- Associated algorithms and software tools
- List of input data that will be used and related test IDs that were used to acquire the data
- List of external reference sources of input data and/or model used in analysis
- List of fixed analysis parameters and set points
- List of adjustable analysis parameters and set points that may vary between runs.
- Sequence of events
- Step-by-step description of the data analysis procedure/algorithm
- Data Recording/Output Format Requirements
- Data Recording Forms and Tables
- Expected results
- Pass/Fail Criteria

Raw data generated to assess the SMBA instrument's RFI detection capability for each test performed shall be delivered along with the test report and the analysis report. The raw data delivery shall include:

- The ancillary data used for the test
 - Raw data
 - Frequency channels, their bandwidth
 - Implementation of the RFI detection algorithms tested
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DATA ITEM DESCRIPTION

1. CDRL No.:

SE-8

2. Title:INSTRUMENT CONCEPT OF OPERATIONS
DOCUMENT3. Reference:

SOW 2.4

4. Use:

To provide information on the Instrument Concept of Operations derived from unique aspects of the Contractor's design.

5. Preparation Information:

The Instrument Concept of Operations Document shall describe the methodology required to operate the instrument. Relative to the Contractors design (including any and all instrument unique operational features) the operations concepts shall address:

- a) Instrument operational environment and limitations
- b) Operational requirements over and above existing instrument requirements necessary to accommodate design unique factors
- c) Unique ground command and data handling operations
- d) User support operations
- e) Contingency and emergency instrument unique operational scenarios, including flight software changes.
- f) RFI detection related activities

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-9

2. Title:Inputs to the Spacecraft to Instrument Interface
Control Document3. Reference:

SOW 2.4.2.1

4. Use:

To establish interface definition and control between SMBA and the spacecraft.

5. Preparation Information:

This document shall provide detailed technical information regarding the interface of the SMBA to the spacecraft. The inputs to the Spacecraft to Instrument ICD shall include interface requirements for the SMBA. Interface inputs shall consist of the following:

- ICD Agreements/Understanding
 - Flight or test hardware supplied by one organization to the other organization (e.g., matching connectors, test fixtures, GSE)
 - Services and materials provided by one organization to the other organization (e.g., purge gas and fittings)
 - Flight hardware design and fabrication responsibility for interfaces (e.g., flight harness segments, ground straps, fasteners)
 - Pointing budgets and allocations, including estimated required spacecraft allocations
- ICD Design
 - First circuit schematics for all power and signal interfaces
 - Connector I/O, including connector designators, connector part numbers, signal names, pin numbers and harness wiring (e.g., twisted-shielded pair; 100 ohm controlled impedance)
 - CBE or measured orbit average power, by mode
 - CBE or measured Main Bus peak current
 - Modeled or measured Main Bus inrush transients
 - Modeled Main Bus impedance for source and load
 - Main Bus power service fault protection implementation (e.g., overcurrent set point)
 - Data – Rates (peak and orbit average), volume, compression, format, high speed data interface

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- Heater circuit schematic with equivalent resistance
 - Design implementation of data bus and command/telemetry bus timing and protocols
 - Baseplate temperature range
 - Instrument calibration spacecraft requirements
 - The type, location, and orientation of the Instrument Alignment References
 - Mechanical
 - Mass, CG, MOI
 - All instrument fields of view, including, where applicable, optical, glint, thermal cryo-radiator, electronics radiator, and solar diffuser fields of view.
 - Static and dynamic envelopes/volumes and keep out zones, including thermal blankets, instrument connectors, mounting feet, instrument-supplied harnesses, and unique mounts.
 - Information on field-of-view-related issues such as clearances, multi-path tolerances, indirect effects such as glint and allowable secondary radiation, and percent interference allowed.
 - Engineering drawings with a set of "not to exceed" dimensions for the launch, deployed on-orbit, and installation envelopes.
 - Specific requirements for pinning the instrument.
 - The spacecraft mounting interface requirements for each separately-mountable instrument component.
 - Mounting hole pattern
 - Mounting footprint
 - Mounting fasteners
 - Mounting type and materials (related to thermal and electrical bonding)
 - Volume/envelop with dimensions
 - Ground strap attach points, if any
 - Connector location with connector designators
 - Fields of View
 - Harness and GSE restraints, support, mounting interfaces
 - Purge gas interface
 - Location and proximity of other hardware mounted on the same panel
 - Thermal
 - Interface temperatures, radiator FOV
 - Thermal Control System
 - Flight Software (FSW)
 - Protocol, rates, and handshaking (commanding acknowledgment)
 - Continuous, burst, etc.
 - FSW loads/patches for software updates
 - SMBA FSW updates
 - Ground Support Equipment used during observatory
 - NEdT I&T
 - Mechanical Fixtures
 - Lifting Slings
 - Simulators & Stimulators
 - Contamination

- Allocation of contaminant contributions from Observatory sources
- Observatory I&T and launch site facility constraints
- MLI vent locations
- Observatory level instrument testing
 - Instrument simulators

The inputs to the ICD shall include the special sequences required to re-configure from sudden entry into OFF mode.

The inputs to the ICD shall include what activation sequences/events are required, where they are necessary, and specific spacecraft control requirements, including appropriate algorithms to be implemented.

The inputs shall include a list of items that may need to be accessible at observatory I&T, as well as a list of items to remove/install prior to flight, venting location, size, path etc.

The Instrument contractor shall provide the location, size, path, venting forces and impulses, direction of net thrust, and operation time of vents in the instrument.

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-10

2. Title:Inputs to the Mechanical Instrument Interface
Control Document3. Reference:

SOW 2.4.2.1

4. Use:

To establish interface the mechanical interface definition and control between SMBA and the spacecraft.

5. Preparation Information:

The Spacecraft to Instrument MICD shall contain, at a minimum, the following information:

- a. Instrument placement on Satellite
- b. Footprints
- c. Coordinate system
- d. Mass and center-of-gravity
- e. Volume allocation and clearances
- f. I&T access/clearances
- g. Alignment features
- h. Flight configuration (stowed for launch), to include with respect to fields of view and surrounding structure, components, and other hardware
- i. On-orbit configurations (each state and intermediate positions of moving mechanical assemblies and deployables), to include with respect to fields of view and surrounding structure, components, and other hardware
- j. Hardpoints and mounting interfaces for mechanical ground support equipment
- k. Fields-of-view (optical, thermal, glint, RF), including tolerances
- l. Electrical connector designators (e.g., J1, J2)
- m. Ground straps
- n. Mounting hardware (and provider), torque specifications

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-11

2. Title:ELECTRICAL SYSTEMS REQUIREMENTS
DOCUMENT3. Reference:

SOW 2.6

4. Use:

To document allowable methods and practices for electrical harnessing, grounding, shielding, and interfaces on SMBA.

5. Preparation Information:

The Electrical System Requirements Document (ESRD) shall include, as a minimum, the following information as requirements on the instrument to guide the design of and interfaces between components on the instrument:

- System electrical block diagram
- System grounding and bonding diagram
- Grounding practices for components, interconnect harnesses, blankets
- Electromagnetic Interference (EMI)/EMC control, (i.e., Direct Current [DC]/DC converter frequency synchronization, digital/analog circuit separation, EMI gaskets)
- System electrical fault protection
- Intra-instrument harness shielding and termination
- Power quality requirements including filtering, steady state voltage and current, peak current, turn-on inrush and operational transients
- Instrument charging requirements and design practices – thermal (paint, anodize) & electrical coatings • Electrical isolation of power services, returns, and chassis
- General electrical interface requirements
- Breakout/Test Points
- Non-flight interface (e.g., for FSW changes, de-bug terminal monitoring etc.)
- Connector and backshell requirements

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-12

2. Title:

SPARE PARTS PLAN AND LIST

3. Reference:

SOW 4.1, 4.3.2.3

4. Use:

To review Contractor's spares plan and parts list.

5. Preparation Information:

The Spares Parts Plan shall define and justify the Contractor's position for the spares proposed for the Instrument build, preparation, and test. This plan shall also present the schedule and method for obtaining the spares. The Plan shall provide a listing of Spare Parts. For the purposes of this CDRL, the Contractor should concentrate on parts that are not commonly available, or may cause schedule problems if out of stock.

The Contractor shall develop, deliver, and implement an approved spare parts program plan for critical flight spares and for critical Ground Support Equipment (GSE) spares built by the Contractor (e.g., Calibration Test Equipment, target assemblies and electronics). The plan shall consider the judicious definition, procurement, and stock of a set of these critical spares to help minimize potential schedule impact created by subsystem failures, by contamination, or by other plausible events or conditions.

Once the plan is approved by the Government, the Contractor shall procure sufficient spare parts to account for typical attrition during the manufacturing process due to handling, random failure, programming, etc. The Contractor shall provide a spare kit of electrical, electronic, electro-mechanical (EEE) parts and spare boards. The spare boards shall have coupons evaluated by the Government as required in the IMAR. The spares may be retained as fully assembled, partially assembled or piece part spares. Since the need to use a spare may be a failure condition in the instrument and, depending on the specifics of the fault, the spare itself may need rework or repair, simple substitution of a spare may not be a viable approach. With this consideration in mind, the judgment of the Contractor shall establish the assembly level of sparing.

In defining the spare parts program, the Contractor shall consider the reliability, handling, and environment conditions of subsystems, components, and parts under consideration for sparing, including the likelihood that these items would need to be replaced. The Contractor should provide enough spares that they could recover from a failure within three months. The spare

parts shall be qualified, tested, and calibrated to the same standards as the flight instrument articles or primary GSE article consistent with their spared level of assembly.

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-13

2. Title:INSTRUMENT SIMULATOR SPECIFICATION

3. Reference:SOW 4.2.11

4. Use:

The Instrument Simulator Specification establishes the design parameters required to meet the functionality required of the Instrument Simulator.

5. Preparation Information:

The Instrument Simulator Specification shall contain at a minimum the following items:

Design parameters necessary to meet the Instrument Simulator requirements established in the Flight Vehicle Test Suite (FVTS) Requirements Specification including:

Part 1:

- Minimum hardware platform and operating system requirements
- Simulator external interfaces to the SMBA spacecraft simulator – electrical, command/telemetry, data rates, protocol, packet structure/definition
- Simulator internal interfaces
- Time maintenance and correlation
- Fault detection and handling
- Modes of operation of the Simulator
- Simulator displays
- Allowable data types and data conversions
- Simulator alerts and warnings
- Access control requirements.

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-14

2. Title:INSTRUMENT SIMULATOR INTERFACE
VERIFICATION REPORT3. Reference:

SOW 4.2.11

4. Use:

To document the verification of how each interface requirement of the Instrument to Spacecraft ICD is simulated by the Instrument Simulator.

5. Preparation Information:

The Instrument Simulator Interface Verification Report shall document how each interface of the Instrument to Spacecraft ICD is simulated by the Instrument Simulator and the results of the verification of the simulation of each of these interfaces.

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-15

2. Title:Field Programmable Gate Array (FPGA)
Development Plan3. Reference:

SOW 4.2

4. Use:

Document the vendor's systematic approach to, and processes used in, the management, design, development, testing (verification, validation, and qualification), documentation, configuration management, review, and quality assurance of any Field Programmable Gate Arrays (FPGA). The FPGA Development Plan shall apply to all FPGAs used by the Contractor for the SMBA instrument.

Reference GSFC 500-PG-8700.2.8, Field Programmable Gate Array (FPGA) Development Methodology.

5. Preparation Information:

The FPGA Development Plan shall apply to all Advanced Orbiting System FPGAs (heritage, modified, or newly developed).

The FPGFA Development Plan shall address:

1. Required documentation (requirements, specification)
2. FPGA development flow diagram
3. Test and flight FPGA part types, selection, justification, and usage
4. Use of 3rd party or Government-developed FPGA designs (design-reuse) and/or intellectual property code (includes core selection, management, and verification)
5. FPGA design capture process
6. FPGA verification plan, including:
 - a. Simulation
 - b. Any other analysis methods
 - c. Non-flight hardware test configuration(s)
 - d. Flight test configuration(s)
7. FPGA computer-aided engineering (CAE) tools including version number

8. FPGA utilization/margin guidelines at specified stages of development for I/O, combinatorial logic, sequential logic, RAMs, cores, etc. as well as internal and interface timing
9. Design Review process – Reviews specified may be combined with FPGA in-house, Engineering Peer and Inherited Items reviews.
10. Quality Assurance involvement/monitoring/oversight/audit of FPGA development processes.
11. Revision control, configuration management, daily backups and archive processes used at specified stages of development.

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-16

2. Title:Field Programmable Gate Array (FPGA) Design
Data Package.3. Reference:

SOW 4.2

4. Use:

An FPGA Design and Analysis Report is required for each FPGA design to allow Government review of Contractor FPGA designs.

5. Preparation Information:

An FPGA Design Data Package shall be delivered for all FPGA designs (heritage, modified, or newly developed). If an FPGA is revised after its Design Data Package was delivered, then the Contractor shall deliver an updated Design Data Package so that each FPGA's final Design Data Package contains the flight design.

NOTE: For heritage or inherited FPGA designs, it is expected that items (a) through (g) below will be delivered to the extent that these artifacts exist. It is not intended for the Contractor to create new artifacts for heritage or inherited designs simply for submission with this CDRL/DID.

The FPGA Design Data Package shall include:

- a. Design (place and route) database
- b. Synthesis report files
- c. Circuit board schematics containing this gate array
- d. Circuit board netlist(s) (any ASCII format such as PADS, MGC, Allegro)
- e. Requirements, specifications, and verification document(s) as well as any supporting material (e.g., block diagrams, presentation material) relevant to the gate array
- f. Source code (e.g., VHDL, Verilog, state machines, state tables Portable Document Format (PDF) of schematics) including any 3rd party intellectual property code and cores
- g. Responses to the GSFC 500-PG-8700-2.7B, Design of Space Flight Field Programmable Gate Arrays checklist or equivalent including:
 - a. Timing analyses for external inputs and outputs, internal domain(s), etc.
 - b. Disposition of all clock domain crossings.

- c. Simulation analysis summary (e.g., functional coverage, code coverage) and simulation test bench/script code.
- d. Signal integrity analyses relevant to this gate array
- e. Power integrity analyses relevant to this gate array

To facilitate reviewing the FPGA design, the FPGA Design Data Package may also include:

- a. System, box, and circuit board requirements, specifications, presentations, and verification document(s) relevant to the gate array and its role in the system, box, and circuit board
- b. Board part list (any ASCII or common spreadsheet format)
- c. PDF of the board layout, such as an assembly drawing
- d. Any other files relevant to the review of the gate array

For any heritage, inherited, or build-to-print FPGA designs, the FPGA Design Data Package shall also include a “was/is” table consisting of:

- a. Full manufacturer part number
- b. Lot/Date code(s)
- c. Vendor part number (e.g., Altered Item Drawing number, etc.)
- d. Logic (e.g., any changes in the hardware description language code, etc.)
- e. Synthesis (e.g., not re-synthesized, re-synthesized to preclude Government Industry Data Exchange Program (GIDEP) xyz, etc.)
- f. Layout (e.g., not re-routed, re-routed per GIDEP xyz, etc.)
- g. Verifications
 - a. Simulations (e.g., not re-run, new simulations added, etc.)
 - b. Analyses (e.g., not updated, timing analysis updated with values exported from latest software version that supports this FPGA, etc.)
 - c. Tests
- h. Environments
 - a. Operating temperature range
 - b. Operating voltage range(s)
 - c. Radiation

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-17

2. Title:

MASS PROPERTIES REPORT

3. Reference:

SOW 2.4

4. Use:

The purpose of the Mass Properties Report is to document all mass properties of the Instrument, its subsystems, and components from preliminary design through final assembly and observatory integration, and to satisfy the mass properties reporting requirements of the launch services provider.

5. Preparation Information:

This document shall provide a mass properties database for Instrument. Mass properties shall include mass, center of gravity, moments of inertia, products of inertia, uncompensated momentum, principal axis misalignment, relevant alignment measurements for each separately-mounted component of the instrument, for each delivered unit, as well a composite total, and physical dimensions, where applicable. The report shall be initially based upon calculated values and shall be updated as calculations are revised and actual measured data becomes available. Following environmental testing and prior to the Instrument shipment to the observatory integrator, the report shall contain a complete mass properties summary of the final Instrument mass properties as measured. The report shall also include the appropriate mass contingency for the current stage of hardware development, along with the allocated mass allowables. If the instrument contains movable masses, expendable masses, or deployables, the respective variations shall be documented in the report.

The mass properties report shall contain the following:

- a. An overall Instrument mass summary. In addition provide: (a) An Instrument mass properties summary for the various phases of the mission through Observatory integration. (b) A detailed mass properties summary of all Instrument hardware organized by subsystem. (c) A summary of all mass properties changes incorporated into the Instrument mass properties database since the last report.

The deliveries at L-54 weeks and L-20 weeks shall satisfy launch services provider requirements as they apply to Instrument, and shall include nominal and 3-sigma uncertainties for mass, centers of gravity, moments of inertia, products of inertia, and principal axis misalignment.

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-18

2. Title:INSTRUMENT SYSTEMS ENGINEERING
MANAGEMENT PLAN

3. Reference:

SOW 2.0

4. Use:

To document the implementation plan for the performance of systems engineering functions and the development of systems engineering products. This plan identifies what, when, where, by whom, and how the functions are performed. It specifies the schedule for the development, and the resources required.

5. Preparation Information:

If the vendor chooses to use Model Based Systems Engineering (MBSE), the SEMP shall describe the plan for which MBSE will be used to produce the CDRL items listed below to be included in the SEMP. Additionally, the vendor may propose to the Government any alternate forms of delivery of CDRLs that are natural products of the MBSE effort (ex. delivering a model in lieu of a document). If MBSE is proposed by the vendor, all models shall be made available to the Government for inspection remotely/electronically. Deliveries of models for CDRLs shall be in a format acceptable to the Government.

The SEMP shall include the following for the areas and activities within the Contractor's sphere of control:

- Describe the overall lifecycle including the major systems engineering activities for each phase.
- Describe critical decisions and activities. Include approach for performing the system engineering activities especially where subcontracts are planned.
- Describe methods utilized for communicating systems engineering activities, progress, status, and results. Include communication methods such as meeting makers, tracking tools, email, websites, etc. that are planned.
- Define who develops the Architecture and Design, what format is planned and when it is due.
- Define who develops the requirements hierarchy, who is responsible for each part of the hierarchy, processes for requirements traceability (up/down), and who identifies and is responsible for the crosscutting requirements.

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- Define what format is planned and what tools if any are to be used for documenting and tracking the requirements.
 - Define the requirements identification process and the formal requirements configuration control process, and the process for requirement change.
 - Define who is responsible for the validation activities and how this is accomplished, what analysis or performance predictions are planned, who performs each, and how they will be accomplished.
 - Define who is responsible for performing the verification activity and tracking the progress for each level within the requirements hierarchy.
 - Define who has technical approval authority for verification at each level within the requirements hierarchy.
 - Define what tools if any are planned to track verification status.
 - Define the due dates for showing requirements compliance.
 - List the resource budgets Systems Engineering will track, the margin philosophy (e.g., the BOL vs. EOL performance margins), who will collect the inputs, how often they will be collected, and when allocation of the budgets are due and when they will be placed under formal configuration management.
 - Document the process for waivers and deviations
 - Document the process for performing trade studies
 - Define which system engineering reviews are planned, who is responsible for organizing them and their content.
 - Define the Systems Engineering Organization Chart and Job Responsibilities. If the responsibility includes subcontractor work, define the scope of the work.

Appendix A: GOLD Rules Alignment Matrix

Project Name: _____						
Initial Issue Date: _____ Revision Letter: _____ Revision Issue Date: _____						
Approval Date of Revision _____						
Project Manager's Signature & Date: _____						
GOLD Rules			Assessment			
Rule No.	Title	Principle	Aligned (Yes = Y; No = N; TBD = T) [Note 1*]	Supporting Documentation [Note 2*]	Rationale [Note 3*]	Waiver Request ID & Status [Note 4*]
1.01	RESERVED					
1.02	RESERVED					
1.03	RESERVED					
1.04	RESERVED					
1.05	Single Point Failures	Single point failures that prevent the ability to fully meet Mission success requirements shall be identified, and the risk associated with each shall be characterized, managed, and tracked and the system trades necessary to determine the need and effectiveness of mitigation efforts (e.g., redundancy, selection of robust parts, etc.) commensurate with mission class shall be conducted and documented.				
1.06	Resource Margins	Mission-level resource margins shall be met in accordance with Table 1.06-1.				
1.07	End-to-End GN&C Phasing	All hardware whose performance is affected by polarity, orientation or position (sensors, switches, mechanisms and actuators) shall be verified by test and inspection for the proper polarity, orientation and position of all components. This includes all GN&C sensors and actuators which shall also undergo full end-to-end (i.e., from sensor stimulus to actuator response) phasing/polarity testing after spacecraft integration in the final flight configuration (hardware and software). All hardware that cannot be fully verified end-to-end in flight configuration shall have flight software mitigations to efficiently correct phasing/polarity errors. The test methodology and results for all polarity/phasing testing shall be independently reviewed.				
1.08	End-to-End Testing	System end-to-end testing shall be performed in the final flight configuration, hardware and software. End-to-end testing shall be from instrument(s) sensor input, through the spacecraft, to a command and telemetry ground system.				
1.09	Test as You Fly	All GSFC missions shall follow a "Test as You Fly (TAYF) - Fly as You Test" approach, throughout all applicable life cycle phases. Each deviation to this approach, along with the rationale for the deviation, shall be documented and a waiver submitted. A project may elect to hold an engineering peer review (EPR), chaired by the rule owner or their designate, in the CDR time frame to discuss the expected TAYF exceptions, rationale and associated risks. This EPR can serve as an expedited method for achieving rule owner acceptance of the project's TAYF waivers. The TAYF exceptions list, along with the EPR minutes shall be maintained by the project configuration management systems. Deltas to the list shall be discussed at subsequent milestone reviews. Note: A waiver or exception to this rule will be based only on the specific elements that appear and are approved in the request and is not a global approval to waive TAYF for all elements.				
1.10	RESERVED					

1.11	Qualification of Heritage Flight Hardware	All heritage flight hardware shall be fully qualified and verified for use in its new application. This qualification shall take into consideration necessary design modifications, changes to expected environments, and differences in operational use.				
1.12	RESERVED					
1.13	RESERVED					
1.14	Mission Critical Telemetry and Command Capability	During time-sensitive events where failure to execute could result in failure to meet mission objectives, event telemetry shall be monitored and downlinked in real-time and/or recorded for downlink later. Mission-critical events include separation from the launch vehicle; power-up of major components or subsystems; deployment of mechanisms and/or mission-critical appendages; initial thruster firings and all planned propulsive maneuvers required to establish mission orbit and/or achieve safe attitude. Following launch vehicle separation, critical deployments, and initial orbit attitude acquisition, continuous command coverage shall be maintained during all subsequent mission-critical events.				
1.15	RESERVED					
1.16	RESERVED					
1.17	Safe Hold Mode	All spacecraft shall have a power-positive, thermally safe, control mode (Safe Hold) to be entered in spacecraft emergencies. Safe Hold Mode shall have the following characteristics: (1) its safety shall not be compromised by the same credible fault that led to Safe Hold activation and (2) it shall employ the minimum hardware set required to maintain a safe attitude.				
1.18	RESERVED					
1.19	Initial Thruster Firing Limitations	If alternate actuators (e.g., reaction wheels) are present, the momentum induced by initial thruster firings shall be within the alternate actuators' capability to execute safe recovery of the spacecraft. For subsequent firings, the Attitude Control System and Failure Detection and Correction shall be designed to limit excessive momentum beyond the capacity of the RWAs.				
1.20	Wetted Joints of Hazardous Propellants	All joints in the propellant lines between the propellant supply tank and the first isolation valve shall be NDE-verified welds.				
1.21	Over pressurization Protection in Liquid Propulsion Systems	The propulsion system design and operations shall preclude damage due to pressure surges ("water hammer"). (Note: See also rule 1.28 "Unintended Propellant Vapor Ignition.")				
1.22	Purging of Residual Test Fluids	Propulsion system design and the assembly & test plans shall preclude entrapment of test fluids that are reactive with wetted material or propellant.				
1.23	Spacecraft 'OFF' Command	No single command shall result in Spacecraft "OFF." This includes both the single string spacecraft case and the redundant spacecraft with one side failed case.				
1.24	Propulsion System Safety Electrical Disconnect	An electrical disconnect "plug" and/or set of restrictive commands shall be provided to preclude inadvertent operation of propulsion system components.				
1.25	Redundant Systems	When redundant systems or functions are implemented, the redundant components, or functional command paths, shall be independent, such that the failure of one component or command path does not affect the other component or command path. The design shall avoid routing of redundant power/signals through a single connector, relay, integrated circuit or other common interface.				
1.26	DELETED					
1.27	Propulsion System Overtemp Fuse	Flight fuses (or other over-current protection devices) for wetted propulsion system components shall be derated per Table 4 of Section F3 of EEEINST-002 relative to the current at which overheating of propellant will occur. (Note: See also rule 2.06 "System Fusing Architecture.")				
1.28	Unintended Propellant Vapor Ignition	Propulsion system design and operations shall preclude ignition of propellants in the feed system.				
1.29	RESERVED					
1.30	Controller Stability Margins	The Attitude Control System (ACS) shall have stability margins of at least 6db for rigid body stability with 30 degrees phase margin. The magnitude of the flexible modes transmission in the open-loop transfer function shall be less than minus 12dB.				

1.31	Actuator Sizing Margins	The Attitude Control System (ACS) actuator sizing shall reflect specified allowances for mass properties growth.				
1.32	Thruster and Venting Impingement	Thruster or external venting plume impingement shall be analyzed and demonstrated to meet mission requirements.				
1.33	DELETED					
1.34	DELETED					
1.35	DELETED					
1.36	RESERVED					
1.37	Stowage Configuration	When a spacecraft is in its stowed (launch) configuration, it shall not obscure visibility of any attitude sensors required for acquisition and shall not block any antenna required for command and telemetry.				
1.38	RESERVED					
1.39	Propellant Sampling in Liquid Propulsion Systems	Liquid propellant quality shall be verified by sampling at point of use prior to loading spacecraft propulsion system.				
1.40	Maintaining Command Authority of a Passive Spacecraft	All spacecraft shall be designed to prevent loss of command authority and command integrity.				
1.41	GSE Use At Launch Site	All testing of flight systems at the launch site shall only use GSE and test configurations that have been previously demonstrated with the flight hardware. Proper operation of the spacecraft with umbilical length equal to or with similar impedance and circuit characteristics to that expected at the launch site shall be demonstrated. Note: Does not apply to launch site resident GSE.				
1.42	Powering Off RF Command Receiver	The spacecraft RF Command Receiver shall not be powered off during nominal flight operations.				
1.43	Flight Software Update Demonstration	There shall be a pre-flight, end-to-end demonstration of code change, using the MOC and flight observatory, for any software which can be changed in flight. (If the system contains FPGAs that can be reprogrammed in-flight, the ability to do so from the MOC shall also be demonstrated.)				
1.44	Early Interface Testing	Spacecraft-to-payload electrical interfaces, including protocol and software compatibility, shall be tested with breadboard or engineering unit hardware, as soon as the hardware is available, preferably before the instrument (or component) CDR.				
1.45	System Alignments	System alignment verifications shall be performed before and after exposure to system environmental testing to demonstrate alignment stability.				
1.46	Use of Micro-Switches	Micro-switches shall be used for information only and shall not be used as the single means to initiate on-board autonomous activity or as an on-board interlock.				
1.47	Design Deployables For Test	Whenever practical, appendages and other deployables shall be capable of deployment under 1G conditions without the use of g-negation ground support equipment. When it is not practical to design for unassisted 1G deployment, the design shall have provisions for interfacing to gravity off-load GSE.				
1.48	Space Data Systems Standards	Data systems standards (e.g., CCSDS, OMG, commercial, international) shall be utilized by missions and implemented in all space communication systems.				
2.01	Flight Electronic Hardware Operating Time	One thousand (1000) hours of operating/power-on time shall be accumulated on all flight electronic hardware (including all redundant hardware) prior to launch. The last 350 hours of operating/power-on time shall be failure-free, of which at least 200 hours shall be in vacuum. For Class D and below, only the failure-free and vacuum requirements shall apply. For hardware expected to operate for less than 100 hours in-flight, proposed pre-launch operating hours shall be discussed with the rule owner.				
2.02	DELETED					
2.03	DELETED					
2.04	RESERVED					
2.05	System Grounding Architecture	For all missions, a system grounding design shall be developed and documented for flight and GSE test configurations. Except for coaxial interfaces, structure or shields shall not be used for the primary circuit current return path. A dedicated conductor shall be included to provide the current return path with the smallest loop area possible.				
2.06	System Fusing Architecture	A system fusing architecture shall be developed and documented for all missions, including the payloads. All				

		circuit breakers that can't be reset by command (i.e., fuses) should be easily accessible for replacement and/or for integrity verification at any time prior to launch vehicle integration.				
2.07	DELETED					
2.08	RESERVED					
2.09	RESERVED					
2.10	RESERVED					
2.11	RESERVED					
2.12	DELETED					
2.13	Electrical Connector Mating	All flight connectors where mating cannot be verified via ground tests, shall be clearly labeled and keyed uniquely, and mating of these connectors shall be verified visually to prevent incorrect mating. The design shall not use connectors that require a blind mating in system-level integration, test and launch operations.				
2.14	Protection of Avionics Enclosures External Connectors Against ESD	All avionics enclosures shall be protected from ESD. All external connectors must be fitted with shorting plugs or appropriate caps during transportation between locations. Additionally, all test points and plugs must be capped or protected from discharge for flight.				
2.15	DELETED					
2.16	RESERVED					
2.17	RESERVED					
2.18	DELETED					
2.19	RESERVED					
2.20	RESERVED					
2.21	RESERVED					
2.22	Corona Region Testing of High Voltage Equipment	Assemblies containing a High Voltage (>150V) supply that is not tested through the Corona region shall undergo venting / outgassing analysis to determine when it is safe to turn on and operate after launch.				
2.23	RF Component Testing for Multipaction and Corona	Multipactor and corona margins for component of spacecraft RF communications subsystems shall be maintained at the mission frequencies. All components shall be vented. 1. If the RF transmitter is on during launch and ascent, all flight components in the transmit path shall be verified as corona free at all pressures from sea level to 1E-4 Torr. 2. Resonant passive flight components shall be verified as multipactor free by test on all units. 3. Non-resonant passive flight components shall be verified as multipactor free by test or analysis. The test setup shall be verified with a known breakdown device. 4. Multipactor analysis shall show a 10dB margin. 5. Multipactor test level for the passive components shall be at least 6dB above the nominal power level in vacuum (<1E-5 Torr) during unit acceptance testing.				
2.24	Solar Arrays	Solar arrays shall incorporate solar cells that have been qualified per AIAA-S-111A-2014, "Qualification and Quality Requirements for Space Solar Cells." If a later revision of AIAA-S-111 has been released by the time of contract award for the mission, the later revision shall govern. b. Solar panels shall be qualified to the mission environment via qualification panels per AIAA-S-112A-2013 (or equivalent), "Qualification and Quality Requirements for Electrical Components on Space Solar Panels." If a later revision of AIAA-S-112 has been released by the time of contract award for the mission, the later revision shall govern. c. Qualification and flight solar panels shall be tested at ambient temperature and at their highest predicted operating temperature including calibrated I-V curves (where practical) before and after panel-level environmental testing. d. Flight solar arrays shall be tested at wing level or array level at ambient temperature including calibrated I-V curves after all environmental testing (integrated to the spacecraft or not) is complete. Should the flight solar				

		array be stored for a period of more than two years after the postenvironmental array testing is complete, the calibrated I-V curve measurements at ambient temperature shall be repeated prior to launch.				
2.25	Electrical Interface Verification	Electrical Interface (i.e., copper-path) Verification Test (IVT) shall be performed on all flight connectors following final flight mating. This may be performed via powered testing and/or physical (e.g., resistance) measurements.				
2.26	Power-On Reset Visibility	A power-on reset occurrence shall be unambiguously identifiable via telemetry. Note: This does not imply real-time telemetry as the reset is occurring.				
2.27	Spacecraft Strip-Charting Capability	A minimal set of hard-line spacecraft parameters, sufficient to establish spacecraft health and safety, shall be monitored and captured (stored), independent of the spacecraft telemetry system, by the EGSE whenever the spacecraft is powered. This data should be sampled at a rate sufficiently high to aid in diagnosis of abnormal power events.				
3.01	Verification and Validation Program for Mission Software Systems	A thorough verification and validation process shall be applied to all mission software systems. This process shall trace customer/mission operations concepts and science requirements to implementation requirements and system design and shall include requirements-based testing of all mission elements, and end-to-end system operations scenario testing.				
3.02	Elimination of Unreachable Software	An analysis of unreachable code, as defined per Table 3.02-1, shall be performed on the intended software that is associated with space flight operation. The analysis shall identify all instances (areas) of unreachable code, the general functionality associated with the code, the reason each is intended to be left within the software, and the justification (e.g., mitigating action) that explains why the included code does not provide a risk to the mission. The focus is on technical risk to the long-term mission, not cost.				
3.03	High Fidelity Interface Simulation Capabilities	A high-fidelity software simulation capability for each external interface to FSW shall be provided in the FSW development/maintenance environments. Both nominal and anomalous data inputs to FSW shall be configurable in real-time using the procedure language of the FSW test workstation. The organization building the flight article being simulated shall also be responsible for the simulator. If this is not feasible, then the developing organization shall provide inputs for and participate in the requirements, design and development of the simulator.				
3.04	Independent Software Testing	Software build testing, system testing, and acceptance testing shall be performed by qualified testers that are independent of the software designers and developers. NOTE: Members of the same development team can perform independent testing as long as the assigned testers have not been involved in any part of the design and development of the software components being tested.				
3.05	Ground System/Operations Testing and Operations Team Readiness	Access to flight system interface and functional capabilities, provided either by the spacecraft or by spacecraft simulators, shall be negotiated with all stakeholders, including the ground system and operations teams. Schedules and agreements should address the spacecraft/spacecraft simulators/instrument(s)/instrument simulator(s) at all levels of fidelity.				
3.06	Dedicated Hardware Computing Platform Testbed for Flight Software and Reconfigurable FPGA Lifecycle Development	A data processing system testbed(s), representative of the flight hardware, shall be dedicated to FSW/FPGA product development teams specifically for development, integration and test. The quantity of f data system testbed units shall be sufficient to support the FSW/FPGA development schedule and the overall mission schedule.				
3.07	Flight Software Margins	Flight software resource margins shall be maintained in accordance with Table 3.07-1 and presented at Key Decision Point (KDP) milestone reviews.				
3.08	RESERVED					
3.09	RESERVED					
3.10	Flight Operations Preparations and Team Development	Experienced operations personnel shall participate as early as possible during mission development, preferably during the mission operations concept phase and the development of specifications for the spacecraft and/or instruments which impact operations. Ideally, the Flight Operations Team (FOT) will supply Test Conductors to support Observatory I&T, which will serve to prepare and				

		train the FOT. As a minimum, the FOT shall participate in flight operations readiness tests that are specified in Table 3.10. Note that these serve as guidelines and are not intended to be prescriptive.				
3.11	Long Duration and Failure Free System Level Test of Flight and Ground System Software	Ground test of the fully integrated FSW and ground system shall include demonstration of error free operations-like scenarios over an extended time period. The minimum duration of uninterrupted FSW system-level test (on the highest fidelity FSW testbed) and ground system operations is 72 hours for Class A and B missions; 48 hours for Class C missions; and, 36 hours for Class D missions, respectively. Planetary missions should consider test durations longer than the above guidance commensurate with the planned ConOps.				
3.12	RESERVED					
3.13	Maintaining Adequate Resources for Mission Critical Components	The updating of mission critical components during the mission operations phase (including any combination of hardware platforms, hardware devices, and software code) shall not compromise the capability of the system to meet mission requirements. Missions shall provide sufficient quantities of flight and ground resources to allow development, test, and operations activities to be conducted without compromising mission availability requirements.				
3.14	Command Procedure Changes	Command procedures and/or scripts, and mission databases (onboard and ground) shall be controlled (treated with the same rigor as changes to flight critical software). This includes formal configuration management, peer review by knowledgeable technical personnel, and full verification with up-to-date simulations wherever possible. (Routine command loads to perform nominal operations may require less test rigor based on experience of senior engineers.)				
3.15	RESERVED					
4.01	Contamination Control, Planning, and Execution	Specific contamination control requirements and processes (such as analytical modeling, laboratory investigations, and contamination protection and avoidance plans) that support mission objectives shall be identified.				
4.02	RESERVED					
4.03	Factors of Safety for Structural Analysis and Design, and Mechanical Test Factors & Durations	Structural analysis and design factors of safety shall apply to all systems in accordance with GEVS Section 2.2.5. The project shall employ the mechanical test factors and durations in accordance with GEVS Section 2.2.4.				
4.04	RESERVED					
4.05	RESERVED					
4.06	Validation of Thermal Coatings Properties	All thermal coatings properties that drive thermally significant performance shall be determined, measured and validated to be accurate for materials and mission flight parameters over the lifecycle of the mission. All thermal analysis shall employ these properties. The GSFC Coatings Committee (chaired by Code 546) shall review and approve the coatings properties.				
4.07	DELETED					
4.08	DELETED					
4.09	RESERVED					
4.10	Minimum Workmanship	All electrical, electronic, and electro-mechanical components shall be subjected to minimum workmanship test levels as specified in GEVS Section 2.4.2.5.				
4.11	Test In Flight Configuration	Mechanical environmental testing (sine, random, & acoustic, shock, etc.) of flight hardware shall be performed with the test article in the flight like configuration. Mechanisms shall be configured for flight, and the flight (or flight like) blankets and harness shall be present for test. The flight optical system shall also be present for the test and configured for flight.				

4.12	Structural Proof Testing	Primary and secondary structures fabricated from nonmetallic composites, beryllium, or containing bonded joints, bonded inserts, or critical welds shall be proof tested in accordance with GSFC-Std-7000 Section 2.4.1.4.1. The following definitions should be used to interpret this GOLD Rule: Primary Structure – Structure in the primary load path that carries the operational or test loads of the system to the structural boundary and whose failure would result in loss of structural integrity. Secondary Structure – Structure that is not in the primary load path and whose failure would not result in loss of structural integrity but would result in an unacceptable loss of capability for the system to meet functional requirements. Secondary structure includes structure whose failure could result in damage to other hardware critical to meeting the functional requirements of the system. Tertiary Structure – Structure not in the primary load path whose failure would not affect structural integrity or the ability of the system to meet functional requirements. Note: Classification of structures should be evaluated at each level of assembly as defined in GEVS (system, subsystem, component).				
4.13	RESERVED					
4.14	Structural and Mechanical Test Verification	Structural and Mechanical Test Verification program shall comply with GEVS-Table 2.4-1, Structural and Mechanical Verification Test Requirements.				
4.15	Torque/Force Margin	The Torque/Force Margin (TM) requirement defined in NASA-STD-5017, Section 4.3 shall apply to all mechanical functions, those driven by motors as well as springs, etc. at beginning of life (BOL). End of Life (EOL) mechanism performance shall be determined by life testing, and/or by analysis; however, all torque increases due to life test results and/or analysis shall be included in the final TM calculation and verification. Margins shall include all flight drive electronics effects and limitations. Note: use higher safety factors as appropriate for immature mechanism designs with no engineering test data to significantly substantiate resistive torque/force loads. See GEVS Section 2.4.5.3 for suggested factors to be considered by mission phase.				
4.16	RESERVED					
4.17	RESERVED					
4.18	Deployment and Articulation Verification	All flight deployables, movable appendages, and mechanisms shall demonstrate full range of motion and articulation under worst-case conditions, when being driven by the flight avionics (i.e., not EGSE) prior to flight.				
4.19	RESERVED					
4.20	Fastener Locking	All threaded fasteners shall employ a minimum of one locking feature that does not depend on fastener preload to function. Exception: Swagelock compression fittings are not required to have a locking feature, but it is recommended. See Code 543 for best practices/approaches for adding a secondary locking feature.				
4.21	Brush-type Motor Use Avoidance	Designs shall avoid brush-type motors for critical applications with very low relative humidity or vacuum operations. Intentionally excluded from this rule are contacting sensory and signal power transfer devices such as potentiometers and electrical contact ring assemblies (slip rings, roll rings), etc.				
4.22	Precision Component Assembly	When precise location of a component is required, the design shall use a stable, positive location system (not relying on friction) as the primary means of attachment.				
4.23	Life Test	Once requirements and design are stabilized to a high degree of certainty, a life test shall be conducted, within representative operational environments, to at least 2x expected life for all repetitive motion devices with a goal of completing 1x expected life by CDR. The differences between the life-test drive electronics and the flight drive electronics (e.g., voltage, current, duty cycle, etc.) could affect mechanism operating life and should be considered in the life-test.				
4.24	Mechanical Clearance Verification	Verification of mechanical clearances and margins (e.g., potential reduced clearances after blanket expansion) shall be performed on the final as-built hardware.				

4.25	Thermal Design Margins	Thermal design shall provide adequate margin between stacked worst-case flight predictions and component allowable flight temperature limits per GEVS 2.6 Note: This applies to normal operations and planned contingency modes. This does not apply to cryogenic systems.				
4.26	RESERVED					
4.27	Test Temperature Margins	Components and systems shall be tested beyond allowable flight temperature limits, to proto-flight or acceptance test levels as specified in GEVS section 2.6.3.2 Note that at levels of assembly above component, full specified margins may not always be achievable for all components due to test setup limitations. In these cases, the expected test levels shall be approved by the GSFC Project, and shall be presented at the earliest possible formal review, no later than PER.				
4.28	Thermal Design Verification	All subsystems/systems having a thermal design with identifiable thermal design margins shall be subject to a Thermal Balance Test at the appropriate assembly level per GEVS Section 2.6.4.				
4.29	Thermal-Vacuum Cycling	All systems flying in unpressurized areas shall have been subjected to a minimum of eight (8) thermal-vacuum test cycles prior to installation on a spacecraft. For an instrument, a minimum of four (4) of these eight (8) Thermal Vacuum cycles shall be performed at the instrument level of assembly. For units where there is an institutional or organizational delivery to an interim level of assembly, pre-delivery testing should include a minimum of 4 cycles.				
4.30	Materials Engineering Implementation	Materials and processes intended for use in flight designs shall be validated by Materials Engineering to be appropriate for the flight configuration, from concept through delivery of hardware. Materials properties testing and verification needed to inform engineering analyses as well as Non-Destructive Evaluation (NDE) of hardware, shall be identified in the early stages of the project.				
5.01	RESERVED					
5.02	RESERVED					
5.03	RESERVED					
5.04	Instrument Testing for Multipaction	Active RF components, such as radars, that develop significant RF power shall be designed and tested for immunity to multipaction. If multipaction immunity is demonstrated by test alone, the test shall be performed at least 6dB above the nominal power level. If satisfied by analysis and test, the analysis shall show at least 10dB of margin above the nominal power level and the test shall be performed at least 3dB above the nominal power level. Due to the inherent uncertainty in the analysis at these power levels, satisfaction by analysis alone is not allowed.				
5.05	Fluid Systems GSE	Fluid systems GSE used to pressurize flight systems shall be compliant with the fault tolerance requirements of Rule 1.26.				
5.06	Flight Instrument Detector Characterization Standard	Instrument detector systems (and associated components) shall demonstrate performance via test over the expected operating temperature range before the Pre-Environmental Review (PER) to establish a performance baseline and provide a provisional verification of performance prior to exposure to non-operational environments, such as vibration, acoustics, non-operational temperatures, or other conditions required to demonstrate survival. At the conclusion of environmental testing, performance shall again be characterized via test and the results compared to the baseline results.				
5.07	RESERVED					
5.08	DELETED					
5.09	DELETED					

5.10	Early Demonstration of Instrument Opto-Mechanical System Alignment and Test	For instrument opto-mechanical systems without significant flight heritage, an early demonstration of the capability to fabricate, assemble, align, and test the opto-mechanical system shall be performed. Optics, mechanisms, structures, and other components relevant to the instrument system, including all opto-mechanical features and interfaces, using components of the approximate fit, form, and function of the flight hardware should be part of the early demonstration. The hardware configuration for the demonstration shall be agreed to by all stakeholders and phased with the flight unit to ensure that demonstration occurs early enough to be valuable.				
5.11	Instrument System Performance Margins	Instrument performance budgets shall be developed for instrument systems and their sub-systems. The performance budgets shall account for uncertainties including, but not limited to, fabrication, assembly, stability and test/verification. The project must have justification for the adequacy of their margins; test demonstration of predicted on-orbit performance with margins against the performance budgets is the preferred justification.				
5.12	Instrument Alignment, Integration and Test	Instruments containing optical systems shall develop an alignment plan in Phase A which will be refined and tracked throughout the project life cycle. The alignment plan should address such considerations as: alignment philosophy including the number of datasets required for appropriate statistics to verify requirements; cross-checks for critical data; leveling the instrument to gravity during metrology as appropriate; fiducials and other references; and authority to proceed before breaking an alignment configuration. In addition, consideration must be given to likely failure modes during testing to ensure that the hardware and test design is adequate to determine test failure causes and corrective action.				
5.13	Laser Life Testing	There shall be a project-approved and peer-reviewed plan, consistent with the mission risk profile, for life-testing a laser prototype to a minimum of 1x of the mission lifetime requirement. The life-test unit should be a high-fidelity representation of the flight laser and any differences between the life test unit and the flight laser should be delineated in the plan. The plan should include system and component-level testing and/or analysis. Any components that have a wear-out or failure mechanism need to be addressed in the plan either by testing or with justification for why testing is unnecessary. Accelerated tests are permitted (and even encouraged) if the acceleration factors are understood and justified. The plan should include technical, budget, schedule and resource assumptions upon which the plan is based.				
5.14	Cryogenic Thermal Margins	The Cryogenic Thermal Design shall provide adequate margin to account for increased heat load or decreased cooling capability from conceptual design to implementation. This is applicable to passive systems operating below 120K and actively cooled systems below 200K.				
5.15	Stray Light Modeling and Mitigation	All optical systems shall have an end-to-end stray light modeling and test campaign performed at the system level to identify background due to stray light effects and develop appropriate mitigation strategies to keep stray light effects within documented requirements. Throughout the life cycle, the model and test configuration shall be continually updated to reflect the current state of the design, ultimately accurately capturing the as-built flight hardware*. "End-to-end" is defined as the entire path from the observed target to the detecting surface. "Optical systems" include, but are not necessarily limited to scientific instruments, guiders, cameras or other vision-type systems, lidar instruments, star trackers, and sun sensors.				
5.13	Laser Life Testing	There shall be a project-				
*Notes						
1. Include a completed <u>compliance assessment</u> per GPR 8070.4, Section 2.0.						
2. Include reference(s) to relevant documentation for each non-compliance.						
3. Supply comments conveying the rationale for the project's assessment of each non-compliance and related future intent (i.e., waiver request, redesign to compliance, etc.).						

4. Supply the waiver ID# (if known) and status of each non-compliance (i.e., approved, disapproved, submitted, pending, etc.); Refer to Section 3.0 of GPR 8070.4.
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Appendix B: Abbreviations and Acronyms

ΔG/G	Gain Stability
A	Approval
ACS	Attitude Control System
ADB	Ada body file extension
AFM	Additional Flight Models (i.e., in addition to FM1)
AM	Additive Manufacturing
AMCP	Additive Manufacturing Control Plan
ANSI	American National Standards Institute
APID	Application Process Identifier
APPLIC	Applicability
ASCII	American Code Standard for Information Interchange
ATMS	Advanced Technology Microwave Sounder
ATP	Ambient Temperature and Pressure
ATP	Authority To Proceed
AVI	Audio Video Interleave
BAT	Bench Acceptance Test
BCR	Baseline Change Request
BOL	Beginning of Life
C&DH	Command and Data Handling
C&T	Command and Telemetry
CADRe	Cost Analysis Data Requirement
CAD	Computer-Aided Design
CAE	Computer-Aided Engineering
Cal/Val	Calibration/Validation
CAT	Category
CBE	Current Best Estimate
CCB	Configuration Control Board
CCP	Contamination Control Plan
CCR	Configuration Change Request
CDR	Critical Design Review
CDRL	Contract Data Requirements List
CG	Center of Gravity
CIIL	Configuration Item Identification List
CIL	Critical Items List
CM	Configuration Management
CMO	Configuration Management Office
CMP	Configuration Management Plan
CO	Contracting Officer
COR	Contracting Officer's Representative
COTS	Commercial-off-the-shelf
CPR	Contract Performance Report
CPT	Comprehensive Performance Test
CPU	Central Processing Unit

CRC	Cyclic Redundancy Check
CrIS	Cross-track Infrared Sounder
CSC	Computer Software Components
CSCI	Computer Software Configuration Item
CSU	Computer Software Unit
csv	Comma Separated Values file extension
CTB	Consent to Break
CTDB	Command and Telemetry Database
CTI	Consent to Integrate
CTP	Consent to Proceed
CV	Calibration/Validation
D	Definitization
DACA	Days After Contract Award
DC	Direct Current
dCDR	Delta Critical Design Review
DD	Department of Defense (forms only)
Del	Deliverable
DFMR	Designed For Minimum Risk
DID	Data Item Description
DM	Data manager
DMR	Design Modification Review
DPA	Destructive Physical Analysis
DOORS	Dynamic Object-Oriented Requirements System
DTD	Document Type Definition
DTM	Detailed Thermal Model
DWF	Design Web Format
E	Electronic Format
EAC	Estimate at Completion
EAR	Export Administration Regulations
EDR	Environmental Data Records
EDU	Engineering Development Unit
EEE	Electrical, Electronic, and Electromechanical
EEO	Early Engineering Opportunity
EGSE	Electrical Ground Support Equipment
EIDP	End Item Data Package
EM	Electromagnetic
EMC	Electromagnetic Compatibility
EMI	Electromagnetic Interference
EOL	End of Life
EOMP	End of Mission Plan
EPR	Engineering Peer Review
ESD	Electrostatic Discharge
ESRD	Electrical System Requirements Document
ETU	Engineering Test Unit
EVMIG	Earned Value Management Implementation Guide

EVM(S)	Earned Value Management (System)
EVP	Environmental Verification Plan
FAR	Federal Acquisition Regulation
FCA	Functional Configuration Audit
FEM	Finite Element Model
FM	Flight Model
FMEA	Failure Modes and Effects Analysis
FMECA	Failure Modes and Effects Criticality Analysis
FOV	Field of View
FPGA	Field Programmable Gate Array
FQT	Formal Qualification Test
FRB	Failure Review Board
FRR	Flight Readiness Review
FSTB	Flight Software Test Bed
FSW	Flight Software
FVTS	Flight Vehicle Test Suite
G-R	Giver-Receiver
GFE	Government Furnished Equipment
GIDEP	Government Industry Data Exchange Program
GOLD	GSFC Open Learning Design
GOTS	Government-off-the-shelf
GPR	Goddard Procedural Requirement
GSE	Ground Support Equipment
GSFC	Goddard Space Flight Center
H	Hardcopy
H&S	Hardness and Survivability
Hz	Hertz
I	Information
I/O	Input/Output
I&T	Integration and Test
ICDR	Instrument CDR
ID	Identification
IBR	Integrated Baseline Review
ICD	Interface Control Document
ICDR	Instrument Critical Design Review
iDel	Internal Deliverable
IEEE	Institute of Electrical and Electronics Engineers
IF	Intermediate Frequency
IMAR	Instrument Mission Assurance Requirements
IMS	Integrated Master Schedule
IMT	Integrated Mission Timeline
IOC	Initial Operating Capability
IPMR	Integrated Program Management Report
IPT	Integrated Product Team
IR	Infrared

IRD	Interface Requirement Document
iRec	Internal Receivable
ISAR	Instrument Safety Assessment Report
ISRR	Instrument System Readiness Review
IT	Integration and Test
ITAR	International Traffic in Arms Regulations
ITRR	Instrument TVAC Test Readiness Review
IV&V	Independent Verification and Validation
JCT	Joint Compatibility Test
JPEG	Joint Photographic Experts Group
JPSS	Joint Polar Satellite System
K	Kelvin
KDP	Key Decision Point
Kg	kilograms
L	Launch
LEO	Low Earth Orbit
LEO&A	Launch, Early Orbit, and Activation
LOA	Letter of Agreement
LPD	Low Earth Orbit Programs Division
LPT	Limited Performance Test
LRD	Launch Readiness Date
LRR	Launch Readiness Review
LTS	Lower Tier Supplier
M&P	Materials and Processes
MA	Mission Assurance
MATLAB	Matrix Laboratory
Mbps	Mega bits per second
MBSE	Model Based Systems Engineering
MDR	Mission Definition Review
MGC	ASCII file format extension
MGSE	Mechanical Ground Support Equipment
MICD	Mechanical Interface Control Document
MIUL	Materials Identification and Usage List
MIP	Mandatory Inspection Point
MLI	Multi-Layer Insulation
MOC	Mission Operations Center
MOI	Moment of Inertia
MOR	Mission Operations Review
MOT	Mission Operations Team
MOTS	Modified-off-the-shelf
MOU	Memorandum of Understanding
MPEG	Moving Picture Experts Group
MPSCIP	Materials and Processes Selection, Control, and Implementation Plan
MR	Mission Rehearsal
MRB	Material Review Board

MRR	Manufacturing Readiness Review or Mission Readiness Review
MST	Mission Support Team
MUA	Materials Usage Agreement
NASA	National Aeronautics and Space Administration
NASTRAN	NASA Structural Analysis
NEAT or NEdT	Noise Equivalent Delta Temperature
NEON	Near Earth Orbit Network
NF	Noise Factor
NFS	NASA FAR Supplement
NGS	NOAA Ground System
NIST	National Institute of Standards and Technology
NLT	No Later Than
NOAA	National Oceanic and Atmospheric Administration
NPR	NASA Procedural Requirement
NS1	NEON Series-1 Project
NSOF	NOAA Satellite Operations Facility
OCDR	Observatory Critical Design Review
ODAR	Orbital Debris Assessment Report
OHA	Operations Hazard Analysis
OHR	Operations Handover Review
OO	On Orbit
P _{CS}	Probability of not meeting cost/schedule allocations
P _S	Probability of safety event occurrences
P _T	Probability of not meeting technical performance requirements
PADS	ASCII file format from Mentor Graphics
PCA	Physical Configuration Audit
PCB	Printed Circuit Board
PDF	Portable Data Format
PDR	Preliminary Design Review
PDS	Post-Delivery Support
PER	Pre-Environmental Review
PFM	Proto-Flight Model
PHS&T	Packaging, Handling, Storage, and Transportation
PLAR	Post-Launch Assessment Review
PLO	Phase-Locked Oscillator
PMB	Performance Measurement Baseline
PM&P	Parts, Materials and Processes
PMP	Project Management Plan
PMPCA	Parts, Materials and Process Control Authority
PMR	Program Management Review
PPP	Part Production Plan
PRD	Performance Requirements Document
PROC	Procedures
PROM	Programmable Only Memory

PRT	Platinum Resistance Thermometer
PSR	Instrument Pre-Ship Review
PSR	Program Status Review
QA	Quality Assurance
QPMR	Quarterly Program Management Review
QTRR	Qualification Test Readiness Review
R	For Review
RAM	Random Access Memory
RDR	Raw Data Record
RE	Review
Rec	Receivable
Rec-Del	Receive-Deliver
RF	Radio Frequency
RFE	Receiver Front End
RFA	Request for Action
RFI	Radio Frequency Interference
RHA	Radiation Hardness Assurance
RMM	Radiometric Math Model
RMP	Risk Management Plan
ROP	Recommended Operating Procedures
RPS	Receiver Power Supply
RTM	Reduced Thermal Model
RVM	Requirement Verification Matrix
SC	Spacecraft
SCCDR	Spacecraft CDR
SCPDR	Spacecraft PDR
SCIRD	Spacecraft IRD
SDD	Software Detailed Design
SDE	Scan Drive Electronics
SDP	Software Development Plan
SDR	Sensor Data Record
SDR	System Definition Review
SE	Systems Engine
SEE	Single Event Effect
SEMP	Systems Engineering Management Plan
SEN	Software Engineering Notebook
SET	Single Event Transient
SFTP	Secure File Transfer Protocol
SIR	System Integration Review
SMBA	Sounder for Microwave-Based Applications
SOE	Satellite Operations Exercise
SOW	Statement of Work
SPA	Signal Processor Assembly
SPMM	Sensor Performance Math Model
SPVM	System Performance Verification Matrix

SPVP	System Performance Verification Plan
SRF	Spectral Response Function
SRR	Systems Requirements Review
SRR	Scene Release Reconstruction data file extension
SRS	Software Requirements Specification
STE	System Test Equipment
STEP	Standard for the Exchange of Product Data
SW	Software
SWAR	Software Acceptance Review
SWE	Software Engineering Requirement
SWTRR	Software Test Readiness Review
T	Time of the launch date
TAA	Technical Assistance Agreement
TBD	To Be Determined
TBR	To Be Resolved
TBS	To Be Scheduled
TDA	Test Data Analysis
TDR	Temperature Data Record
TM	Torque Margin
TMM	Thermal Math Model
TPM	Technical Performance Metric
TQCM	Thermal Quartz Crystal Microbalance
TRL	Technology Readiness Level
TRR	Test Readiness Review
TVAC	Thermal Vacuum
TYF	Test as You Fly
UAR	Update As Required
UID	User Identification
U.S.	United States
VDD	Version Description Document
VHDL	Very High-Speed Integrated Circuits Hardware Description Language
VTL	Verification Tracking Log
W	Watt
WBS	Work Breakdown Structure
WGVC	Working Group on Calibration and Validation
XML	Extensible Markup Language